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TOPICAL REVIEW

The *B*-spline *R*-matrix method for atomic processes: application to atomic structure, electron collisions and photoionization

Oleg Zatsarinny and Klaus Bartschat

Department of Physics and Astronomy, Drake University, Des Moines, IA 50311, USA

E-mail: oleg.zatsarinny@drake.edu and klaus.bartschat@drake.edu

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Abstract

The basic ideas of the *B*-spline *R*-matrix (BSR) approach are reviewed, and the use of the method is illustrated with a variety of applications to atomic structure, electron–atom collisions and photo-induced processes. Special emphasis is placed on complex, open-shell targets, for which the method has proven very successful in reproducing, for example, a wealth of near-threshold resonance structures. Recent extensions to a fully relativistic framework and intermediate energies have allowed for an accurate treatment of heavy targets as well as a fully nonperturbative scheme for electron-impact ionization. Finally, field-free BSR Hamiltonian and electric dipole matrices can be employed in the time-dependent treatment of intense short-pulse laser–atom interactions.

(Some figures may appear in colour only in the online journal)

1. Introduction

Over the past decades, a number of general computer codes have been developed to generate accurate data for atomic and molecular structure, as well as electron and photon collisions with atoms, ions and molecules. Theoretical and computational work in this area is supporting many ongoing experiments, providing not only insight at the fundamental level of submicroscopic, often highly correlated quantum mechanical few-body systems, but also data needed in many practical applications, such as modelling the behaviour of various plasmas and discharges as well as diagnosing their properties. Numerous calculations have been performed over the past decades, with rapidly increasing complexity in the light of the fact that computational resources have become abundant. Comparison with experimental benchmark data, of course, remains necessary to validate the theoretical predictions, thereby providing some confidence in the use of theoretical numbers, especially when theory is the only source of sufficiently complete datasets.

This topical review is devoted to one recently (since about 2000) developed method, namely the *B*-Spline *R*-matrix (BSR) approach. It is one of many that can, in principle, be used to solve the so-called close-coupling equations. As will be illustrated below, the BSR method has been employed very successfully for a number of atomic structure calculations, but especially for the treatment of electron-induced processes (elastic scattering, excitation, de-excitation, ionization), and the interaction of weak (generally continuous) and strong (usually pulsed) electromagnetic fields with atoms and atomic ions (positive and negative). All these processes can be handled with the same basic over-arching approach, with the appropriate details being implemented through the boundary conditions for the close-coupling equations and the extent to which various physical phenomena (e.g., electron correlations, relativistic effects, channel coupling) are accounted for. While the method can be further developed towards molecules, we will exclusively deal with atomic targets here.

A summary of BSR-related publications is provided in tables 1–4. Table 1 lists a number of write-ups and methodology papers, including the publicly available suite of

Table 1. List of BSR programs and write-ups (2000–present).

Topic	Remarks	Reference
Write-up	Nonrelativistic + Breit–Pauli	[1]
(Some programs)	Structure calculations	[2]
	<i>B</i> -splines for DBSR	[3]
	Angular integrals in nonorthogonal basis	[4]
	Time-dependent BSR approach	[5]

Table 2. List of BSR publications for atomic structure (2000–present).

Topic	Target	Reference
Rydberg series	C	[6]
Oscillator strengths	Ne	[7]
	S	[8]
	S II	[9]
	Ar	[7, 10]
	Kr	[7]
	Xe	[7, 11]
Polarizabilities	F	[12]
	Cl	[13]

codes [1] that was used in many of the early works listed in table 2 for atomic structure calculations, table 3 for electron collisions and table 4 for radiative processes. Over the past few years, however, significant further progress has been achieved, including the development of a fully relativistic Dirac-based (DBSR) version, the inclusion of a large number of pseudo-states that allow for coupling to the ionization continuum and, ultimately, the fully nonperturbative treatment of ionization processes, and the use of field-free Hamiltonian and dipole matrices from the BSR complex in the solution of the time-dependent Schrödinger equation (TDSE) for intense short-pulse laser–atom interactions.

As seen from the tables, the (D)BSR method is very versatile. While many test calculations were performed for relatively simple quasi-one-electron (hydrogen-like) and quasi-two-electron (helium-like) targets, the method is particularly suitable when the complexity of the target is increased, and especially when the coupling of another electron is required to properly describe an electron collision process. Consequently, it has enjoyed the largest success in dealing with complex open-shell targets.

The two most significant innovations of the BSR approach compared to other methods and the accompanying computer codes are the following:

- (i) Different sets of nonorthogonal orbitals can be used to represent both the bound and continuum one-electron orbitals.
- (ii) A set of *B*-splines defines the *R*-matrix basis functions.

The use of nonorthogonal bound orbital sets allows for a much higher accuracy in the description of the target states than what is typically achieved when orthogonality is enforced. Since the orbitals are optimized in separate calculations for individual terms, a high level of accuracy can be obtained with compact configuration interaction (CI) expansions. Regarding the close-coupling expansion of the

Table 3. List of BSR publications for electron collisions (2000–present).

Topic	Target	Remarks	Reference
e–ion collisions	Fe II		[14]
	Fe VIII		[15]
	S II		[9]
	K II		[16, 17]
e–neutral collisions (Discrete states, excitation)	He		[18–20]
	C		[21]
	N		[22]
	O		[23–26]
	S		[27, 28]
	Cl		[29]
	Ne		[30–33]
	Ne	Plasma modelling	[34, 35]
	Na	Autoionizing levels	[36, 37]
	Mg		[38]
	Si		[39]
	Ar		[40, 41]
	Ar	Plasma modelling	[42–44]
	K	Autoionizing levels	[45, 46]
	Ca		[47, 48]
	Cu		[49, 50]
	Zn		[51–53]
	I		[54]
	Kr		[55–57, 44]
	Rb	Autoionizing levels	[58]
	Xe		[41]
	Xe	Plasma modelling	[59]
	Cs	Autoionizing levels	[60]
	Kr	DBSR	[61]
	Xe	DBSR	[61]
	Cs	DBSR	[62]
	Au	DBSR	[63, 64]
	Hg	DBSR	[65–67]
	Pb	DBSR	[68, 69]
e–neutral collisions (Large RMPS, incl. ionization)	He		[70, 71]
	C		[72]
	Ne		[73–75]
	Ar		[76, 77]

Table 4. List of BSR publications for photo-induced processes (2000–present).

Topic	Target	Remarks	Reference
Photodetachment	He [−] (1s2s2p) ⁴ P ^o		[78]
	Li [−]		[79]
	B [−]		[80]
	O [−]		[81]
	Ca [−]		[48]
Weak-field photoionization	Li		[25]
	K		[82]
	Zn		[83]
Strong-field photoionization	He	Double ionization	[84]
	Ne	Single ionization	[85]
	Ar	Single ionization	[86]

total wavefunction for collision problems, certain $(N + 1)$ -electron bound configurations must often be included to compensate for orthogonality constraints imposed on the continuum orbitals. However, it can be difficult to keep the expansion fully consistent, and any inconsistency may lead to a pseudo-resonance structure. Using nonorthogonal continuum orbitals, on the other hand, avoids the introduction of these

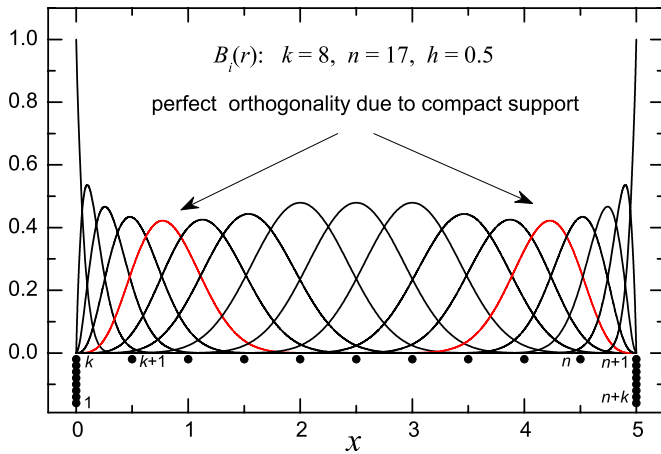


Figure 1. An example of a simple B -spline basis of order 8 with equidistant knots. Note that neighbouring splines have a finite overlap with each other, while distant splines are perfectly orthogonal. Also, multiple knots occur at the edges, with only one spline having a nonzero value on the boundary. The latter property makes it straightforward to impose appropriate boundary conditions.

$(N + 1)$ -electron terms and thus may reduce the pseudo-resonance problem.

Nonorthogonal orbital sets are often avoided, because the most time-consuming part of atomic structure calculations is connected with the angular integrations in constructing the Hamiltonian matrix elements. A key step in the development, namely the methodology for dealing with the additional complications associated with nonorthogonal orbitals in a general way, was described by Zatsarinny and Fischer [4].

The choice of B -splines as basis functions, introduced to atomic structure calculations in the 1980s, is advantageous due to their excellent numerical approximation properties [87]. B -splines are bell-shaped piecewise polynomial functions of order k_s (degree $k_s - 1$), defined by a given set of points in some finite radial interval [88]. An example of 17 splines of order 8, with knots located 0.5 units apart on the x -axis, is shown in figure 1. There is great flexibility in the choice of the radial grid in a B -spline basis, and machine accuracy may be achieved with simple Gaussian quadratures. Finite-difference algorithms are avoided and well-established linear algebra packages are used instead.

In contrast to perturbative approaches based on some variant of the Born series with usually only one or two terms of the series being evaluated due to practical constraints, the nonperturbative close-coupling method (see below) is, in principle, based on a complete expansion for the wavefunction of the system of interest. The method generally leads to a system of coupled integro-differential equations, which can be solved using a variety of methods. In recent years, the convergent close-coupling approach [89, 90], formulated for the transition (T) matrix elements in momentum space, has been highly successful for describing the valence electron(s) in H-like and He-like targets.

For more complex systems, and in a variety of applications for which results for a very large number of energies are required, the R -matrix technique [91] is often the method of choice, as illustrated in the compilation volume by Burke

and Berrington [92] and the recent comprehensive book by Burke [93]. The general R -matrix package RMATRX-I [94] remains the most general publicly available code. Other packages are around, such as the unpublished code RMATRX-II, with improved efficiency of the angular-momentum algebra, the parallelized version PRMAT [95], an alternative version of RMATRX-I [96] with the possibility of including radiative damping and the intermediate-energy R -matrix (IERM) method [97]. The basic idea of the R -matrix technique is to split the configuration space into two regimes separated by the box radius a , which is typically chosen such that exchange effects can be neglected outside the box. The solutions of a simpler system of coupled differential (rather than integro-differential) equations for the outer region are then matched at $r = a$ to the inner-region solution. The latter, in turn, is obtained via a basis-function expansion that can take full advantage of highly sophisticated and well-developed atomic (and molecular) structure methods.

One of the principal ingredients of the above-mentioned programs is a single set of orthogonal one-electron orbitals, with one subset being used to construct the target states in multi-configuration expansions and the other one being employed to represent the scattered projectile inside the box. This structure simplifies the calculations and allows for the development of efficient computer programs. On the other hand, it often leads to three major problems in the application to truly complex targets: (i) difficulties in describing all target states of interest to sufficient accuracy; (ii) the possible occurrence of unphysical structures, the so-called pseudo-resonances, when an attempt is made to address the former problem and (iii) numerical difficulties due to an ill-conditioned orthogonalization procedure and the need to modify the so-called Buttle correction.

Given the success of the R -matrix method for atomic collision processes, it is not surprising that many modifications of the general idea and improved algorithms have been suggested since the original programs were published. These include the eigenchannel formulation [98], a different choice of basis functions [99], particularly targeted to eliminate or reduce the sometimes problematic Buttle correction [100, 101], or specific recipes to avoid the pseudo-resonance problem [102, 103]. The use of B -splines as the R -matrix basis set was first outlined by van der Hart [104], who applied the method to e-H scattering and obtained excellent agreement with existing benchmark results.

The BSR method described in this topical review was developed with several key aspects in mind. These include the effective completeness and numerical stability that can be expected from a B -spline basis, whose numerical properties are well researched and for which a number of basic library routines could be developed in a straightforward way. Furthermore, since the primitive basis is already nonorthogonal (although with limited nonorthogonality involving only a few neighbouring splines and hence resulting in a banded overlap matrix), the first step towards employing nonorthogonal sets of *physical* orbitals was already taken. Hence, it was decided to allow for individually optimized, term-dependent one-electron orbitals *in general*, thereby

taking advantage of compact configuration expansions to generate accurate wavefunctions for the N -electron target and, if needed, the $(N + 1)$ -electron collision system and especially its resonances. Finally, the package was supposed to be general, allowing for structure calculations as well as the treatment of electron and photon-induced processes. Relativistic effects can be accounted for with the appropriate degree of accuracy, and the treatment of electron-impact ionization has become possible via the introduction of a large number of so-called pseudo-states that effectively provide a discretization of the target continuum via quasi-bound states. Finally, an extension towards the treatment of time-dependent processes was desirable as well.

This topical review is organized as follows. After this introduction, section 2 describes the basic ingredients of the BSR method. (Readers who are mainly interested in getting an impression of possible applications of the method could skip this section.) Starting with the close-coupling expansion, we introduce the R -matrix methodology in the inner region, the matching to the solution outside the box, the description of bound states rather than collisions and the calculation of radiative processes. This is followed by the treatment of relativistic effects, coupling to the ionization continuum, which allows for a fully nonperturbative description of ionization processes, and another extension dealing with time-dependent processes. Section 3 is devoted to illustration through a collection of sample results, ranging from atomic structure to electron scattering and photo-induced processes. While we will attempt to keep the review self-contained, space limitations require the omission of many details. The interested reader is thus referred to the original references listed in tables 1–4. We finish with a summary, conclusions, and an outlook in section 4.

Unless specified otherwise, atomic units (au) are used throughout this manuscript.

2. General theory

2.1. The close-coupling expansion

The problem of low-energy electron scattering from an N -electronic atomic target is reduced to solving the time-independent Schrödinger equation

$$(H_{N+1} - E)\Psi_\alpha(\Gamma X, x_{N+1}) = 0 \quad (1)$$

with appropriate boundary conditions. The collision wavefunction $\Psi_\alpha(\Gamma X, x_{N+1})$ represents a fully antisymmetrized wavefunction of the system ‘target atom + projectile electron’, where $X \equiv (x_1, x_2, \dots, x_N)$ and $x_i \equiv (\mathbf{r}_i, \sigma_i)$, with the spatial ($\mathbf{r}_i$) and spin (σ_i) coordinates of the i th electron. Furthermore, Γ is a complete set of quantum numbers of the $(N + 1)$ -electron system, and E is the total energy. The subscript α characterizes the initial conditions and usually denotes the incoming scattering channel.

The Hamiltonian H_{N+1} , which describes the scattering of an electron from an N -electron atomic target with nuclear charge Z , has the form

$$H_{N+1} = \sum_{i=1}^{N+1} \left(-\frac{1}{2} \nabla_i^2 - \frac{Z}{r_i} \right) + \sum_{i>j=1}^{N+1} \frac{1}{r_{ij}}, \quad (2)$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$, with $\mathbf{r}_i$ and $\mathbf{r}_j$ denoting the vector coordinates of electrons i and j . The origin of the coordinate frame is set at the target nucleus, which is assumed to have an infinite mass. For the time being we neglect all relativistic effects in order to simplify the notation. We introduce a set of target states, including possible pseudo-states Φ_i (see below), and their corresponding eigenenergies E_i by the equation

$$\langle \Phi_i | H_N | \Phi_j \rangle = E_i(Z, N) \delta_{ij}, \quad (3)$$

where the integration is carried out over all the space and spin coordinates of the target electrons. Then the total energy in a collision process is $E = E_i + k_i^2/2$, with E_i being the energy of the target in the state i , while $k_i^2/2$ represents the kinetic energy of the projectile electron. The target states are expanded in terms of single-configuration basis states φ_j by

$$\Phi_i(x_1, \dots, x_N) = \sum_j \varphi_j(x_1, \dots, x_N) c_{ij}, \quad (4)$$

where the coefficients c_{ij} are determined by diagonalizing the target Hamiltonian. The configurations φ_j are constructed from a one-electron bound orbital basis, usually consisting of physical self-consistent field orbitals plus possibly additional pseudo-orbitals. The latter are included to represent correlation effects. Note that we do not assume the one-electron basis to be orthogonal, as is often imposed in scattering calculations. This allows us to optimize the bound orbitals in independent calculations for each target state and thus to use term-dependent one-electron radial functions.

The solution of (1) has to satisfy the boundary conditions of an incoming wave packet in some scattering channel α and outgoing waves in this and all other channels. In the close-coupling approximation, the solution is expanded in terms of a set of N -electron target wavefunctions Φ_i . The corresponding expansion coefficients play effectively the role of wavefunctions for the incident electron. In practice, one uses

$$\begin{aligned} \Psi_\alpha^\Gamma(x_1, \dots, x_{N+1}) &= \mathcal{A} \sum_{i=1}^n \bar{\Phi}_i^\Gamma(x_1, \dots, x_N; \hat{\mathbf{r}}_{N+1}, \sigma_{N+1}) \frac{1}{r_{N+1}} F_{i\alpha}^\Gamma(r_{N+1}) \\ &\quad + \sum_{j=1}^m c_j \chi_j^\Gamma(x_1, \dots, x_{N+1}), \end{aligned} \quad (5)$$

where $\mathcal{A}$ is the antisymmetrization operator with respect to the exchange of any pair of electrons while $F_{i\alpha}^\Gamma(r)$ is the radial component of the scattered electron wavefunction when the target is in the i th state. The index i includes all quantum states of the system, and $\bar{\Phi}_i$ is a channel function that is obtained by coupling the target state with the spin-angle functions of the scattered electron. A major question, of course, concerns the completeness of the expansion, in particular when it comes to accounting for the ionization continuum. Traditionally, the expansion was cut off after including only a few discrete, low-lying physical target states, thereby resulting in the method being restricted to the low-energy near-threshold regime.

Next, let us denote the quantum numbers of the incident electron by $k_i l_i m_i m_{s_i}$. The Hamiltonian (2) is diagonal with respect to the total orbital momentum L , total spin S , their projections M_L and M_S onto the chosen axis and the parity π of

the total system. Therefore, in the expansion (5) it is convenient to use the coupled angular-momentum representation, in which

$$\Gamma \equiv \gamma L S M_L M_S \pi, \quad (6)$$

where γ denotes the set of all other quantum numbers. The channel functions Φ are defined according to the following coupling scheme:

$$\begin{aligned} \bar{\Phi}_i^\Gamma(x_1, \dots, x_N; \hat{\mathbf{r}}_{N+1}, \sigma_{N+1}) \\ = \sum_{M_L m_i} \sum_{M_S m_i} (L_i M_{L_i}, l_i m_{l_i} | L M_L) \left(S_i M_{S_i}, \frac{1}{2} m_i | S M_S \right) \\ \times \Phi_i(x_1, \dots, x_N) Y_{l_i m_{l_i}}(\hat{\mathbf{r}}_{N+1}) \chi_{\frac{1}{2} m_i}(\sigma_{N+1}). \end{aligned} \quad (7)$$

Here Y_{lm} is a spherical harmonic, $\chi(\sigma)$ is a spin function, and we use the standard notation for the Clebsch–Gordan coefficients. The function $F_{i\alpha}^\Gamma(x_{N+1})$ in (5) for the incident electron can describe both open and closed channels. If $(E - E_i)$ is positive, the channel is said to be ‘open’; otherwise, the channel is ‘closed’. Note, however, that the function $F_{i\alpha}^\Gamma(x_{N+1})$ is not quadratically integrable for open channels. Closed-channel radial functions must satisfy the same boundary conditions at $r = 0$ and the same orthogonality conditions as the open-channel functions, but the closed-channel functions are quadratically integrable.

The first term in expansion (5) should also include the integration over the continuous spectrum of the target, which corresponds to excitation into the ionization continuum. A direct inclusion of this term in (5) would tremendously complicate the computational problem since the channel index becomes a continuous variable and the number of channels is not countable. Very often, therefore, this term is omitted entirely. However, the continuum part of the close-coupling expansion was found to be very important at intermediate scattering energies from about the ionization threshold to a few times that threshold. It can be simulated, to some extent, by the inclusion of bound pseudo-states. Employing a large number of pseudo-states, therefore, can essentially eliminate the low-energy restriction of the close-coupling method. This will be discussed below.

The correlation functions χ_i are quadratically integrable functions, usually constructed from the same set of one-electron orbitals as the target states Φ_i . The correlation functions ensure that the expansion (5) is complete in the bound orbital basis, even when the continuum radial functions are chosen to be orthogonal to the bound orbitals.

In the central field approximation, the atomic orbitals are represented in the form

$$\varphi_j(x) = Y_{l_j m_j}(\hat{\mathbf{r}}) \chi(m_s | \sigma) \frac{1}{r} P_{n_j l_j}(r). \quad (8)$$

Then it is usually demanded that for $l_j = l_i$ the orthogonality condition

$$\int_0^\infty P_{n_j l_j}(r) F_{i\alpha}(r) dr = 0 \quad (9)$$

is satisfied. This condition does not follow from general principles and is introduced only to simplify numerical calculations. The introduction of the correlation functions

$\chi_j(x_1, \dots, x_{N+1})$ means that, in spite of implying conditions (9), the second sum in (5) permits us to account for the virtual capture of electrons into an unfilled subshell.

In our implementation of the BSR method and the corresponding computer code [1], the conditions (9) are optional. The use of nonorthogonal continuum orbitals allows us to avoid the introduction of the correlation functions $\chi_j(x_1, \dots, x_{N+1})$, or to use them directly to describe short-range correlations when the convergence of the expansion becomes too slow. In our case, they can also be generated independently from the target states.

Coupled equations for the radial components of the functions $F_i(r)$ and the coefficients c_j are obtained by substituting expansion (5) into the Schrödinger equation and projecting onto the target functions Φ_i and the L^2 functions χ_i . After separating out the spin and angular variables and eliminating the coefficients c_j , we obtain the following set of coupled integro-differential (‘close-coupling’) equations:

$$\begin{aligned} \left(\frac{d^2}{dr^2} - \frac{l_i(l_i + 1)}{r^2} + \frac{2Z}{r} + k_i^2 \right) F_i(r) \\ = 2 \sum_j (V_{ij} + W_{ij} + X_{ij}) F_j(r). \end{aligned} \quad (10)$$

Here l_i is the orbital angular momentum of the scattered electron while V_{ij} , W_{ij} and X_{ij} are the partial-wave decompositions of the local direct, nonlocal exchange and nonlocal correlation potentials, respectively. These potentials are too complicated to write down explicitly except for the simplest atoms. Instead they are constructed by general computer programs. The equations (10) can also contain terms that arise from the orthogonality constraints on the scattering radial functions $F_i(r)$.

Over the past decades, many computational methods have been developed for solving equations (10) to yield the scattering matrices and amplitudes, which can be combined to calculate observable quantities for comparison with experiment. These methods form the basis for a number of computer program packages, some of which are widely used. As examples we mention the linear algebraic equation method [105], the noniterative integral equation method [106], the *R*-matrix method (see below) and again the convergent close-coupling approach [90]. All these methods seek the solution of equations (10) in either configuration or momentum space for each collision energy.

A promising approach for the direct solution of the close-coupling equations (10), based on the *B*-spline basis and notable for its simplicity, which is a key to a successful computational implementation, was put forward by Fischer and Idrees [107]. The method determines the required solution within a finite boundary, with no assumed boundary conditions. This is not a limitation, provided that the asymptotic region is reached, so that the solutions can be matched to a linear combination of true asymptotic solutions. The core of the algorithm involves the evaluation of the Hamiltonian and overlap matrix elements in the *B*-spline basis,

$$H_{ij} = \langle \Psi_i, H \Psi_j \rangle, \quad S_{ij} = \langle \Psi_i, \Psi_j \rangle, \quad (11)$$

and the extraction of the eigenvectors relative to the minimum modulus eigenvalues of the nonhermitian, energy-dependent matrix

$$A(E) = H - ES \quad (12)$$

at each prefixed energy E .

2.2. The R -matrix method

As mentioned earlier, the R -matrix method is one of many for solving the close-coupling equations (10). A detailed description of the method and its many applications can be found in the recent book by Burke [93]. Here we only summarize the basic ideas.

The important difference from the straightforward close-coupling formalism is a separate treatment of two regions: an inner region, in which all the electrons are closely interacting with each other and possible external fields, and an outer region, in which the continuum electron only feels a local potential. Here the coupled equations (with simple long-range potentials) are solved for each collision energy and matched, at the boundary $r = a$, to the solution in the inner region. However, instead of solving a set of coupled integro-differential equations in the internal region for each collision energy, the $(N+1)$ -electron wavefunction is expanded in terms of an energy-independent basis set and treated similarly to electrons in atomic bound states. Consequently, general computer codes written for bound-state atomic structure problems can be used, with only minor modifications, to generate the scattering algebra.

In the internal region, the $(N+1)$ -electron wavefunction at energy E is expanded in terms of an *energy-independent* basis set, Ψ_k , as

$$\Psi_E^\Gamma = \sum_k A_{Ek}^\Gamma \Psi_k^\Gamma. \quad (13)$$

The basis states for a given total angular momenta are constructed as

$$\begin{aligned} \Psi_k^\Gamma(x_1, \dots, x_{N+1}) &= \mathcal{A} \sum_{i,j} \bar{\Phi}_i^\Gamma(x_1, \dots, x_N; \hat{\mathbf{r}}_{N+1}, \sigma_{N+1}) \frac{1}{r_{N+1}} u_j(r_{N+1}) c_{ijk}^\Gamma \\ &+ \sum_i \chi_i^\Gamma(x_1, \dots, x_{N+1}) d_{ik}^\Gamma. \end{aligned} \quad (14)$$

The u_i in equation (14) are radial continuum basis functions describing the motion of the scattering electron. They are nonzero on the boundary of the internal region and thus provide the link between the solution in the internal and external regions. The quadratically integrable functions χ_i have the same meaning as in equation (5) and are assumed to be fully confined to the internal region. The coefficients c_{ijk}^Γ and d_{ik}^Γ are determined below.

Consider now the solution of the Schrödinger equation in the internal region $[0, a]$. We note that the Hamiltonian H_{N+1} is not Hermitian in this region due to the surface terms at $r = a$ that arise from the kinetic energy operator. These surface terms can be cancelled by introducing the Bloch operator

$$L_{N+1} = \sum_{i=1}^{N+1} \frac{1}{2} \delta(r_i - a) \left(\frac{d}{dr_i} - \frac{b-1}{r_i} \right), \quad (15)$$

where b is an arbitrary constant (often chosen as $b = 0$). Note that $H_{N+1} + L_{N+1}$ is Hermitian for functions satisfying arbitrary boundary conditions at $r = a$. We then rewrite the Schrödinger equation in the inner region as

$$(H_{N+1} + L_{N+1} - E)\Psi = L_{N+1}\Psi. \quad (16)$$

This equation can be formally solved in terms of the R -matrix basis functions Ψ_k , which are obtained through

$$\langle \Psi_i^\Gamma | H_{N+1} + L_{N+1} | \Psi_j^\Gamma \rangle_{\text{int}} = E_i^\Gamma \langle \Psi_i^\Gamma | \Psi_j^\Gamma \rangle_{\text{int}}, \quad (17)$$

where the integration over the radial variables is restricted to the internal region. This generalized eigenvalue problem determines the coefficients c_{ijk}^Γ and d_{ik}^Γ in (13).

The formal solution of equation (17) can be expanded as

$$|\Psi^\Gamma\rangle = \sum_k |\Psi_k^\Gamma\rangle \frac{1}{E_k^\Gamma - E} \langle \Psi_k^\Gamma | L_{N+1} | \Psi^\Gamma \rangle_{\text{int}}. \quad (18)$$

Projecting this equation onto the channel functions $\bar{\Phi}$ and evaluating on the boundary of the internal region yields

$$F_i^\Gamma(a) = \sum_j R_{ij}^\Gamma(E) \left(a \frac{dF_j^\Gamma}{dr} - b F_j^\Gamma \right)_{r_{N+1}=a}, \quad (19)$$

where we have introduced the R -matrix with elements

$$R_{ij}^\Gamma(E) = \frac{1}{2a} \sum_k \frac{w_{ik}^\Gamma w_{jk}^\Gamma}{E_k^\Gamma - E}, \quad (20)$$

the reduced radial wavefunctions

$$F_i^\Gamma(r_{N+1}) = r_{N+1} \langle \bar{\Phi}_i^\Gamma | \Psi_k^\Gamma \rangle' \quad (21)$$

and the surface amplitudes

$$w_{ik}^\Gamma = a \langle \bar{\Phi}_i^\Gamma | \Psi_k^\Gamma \rangle'_{r_{N+1}=a}. \quad (22)$$

The primes on the brackets in equations (21) and (22) indicate that the integration is carried out over all the electronic space and spin coordinates, except for the radial coordinate of the scattered electron. Equations (19) and (20) describe the scattering of electrons from atoms or ions in the internal region. Together with the following relations for the coefficients A_{Ek} in (13),

$$\begin{aligned} A_{Ek}^\Gamma &= \frac{1}{2a} (E_k - E)^{-1} \sum_i w_{ik}(a) \left(a \frac{dF_i^\Gamma}{dr} - b F_i^\Gamma \right)_{r=a} \\ &= \frac{1}{2a} (E_k - E)^{-1} \mathbf{w}^T \mathbf{R}^{-1} \mathbf{F}^\Gamma, \end{aligned} \quad (23)$$

they allow us to establish the wavefunction Ψ_E in the inner region for any value of the total energy E given the values of the scattering orbitals on the boundary. The R -matrix with elements given by equation (20) is obtained at *all energies* by diagonalizing $H_{N+1} + L_{N+1}$ for each set of conserved quantum numbers Γ to determine the basis functions Ψ_k and the corresponding eigenenergies E_k . The logarithmic derivatives of the continuum radial wavefunctions $F_i(r)$ on the boundary of the internal region are then given by equation (19).

An important point in the R -matrix method is the choice of the radial continuum basis functions u_j in equation (14). Although members of any complete set of functions satisfying arbitrary boundary conditions can be used, a careful choice will speed up the convergence of the expansion (13). In the standard R -matrix approach developed by the Belfast group

[92, 93], numerical basis functions satisfying homogeneous boundary conditions at $r = a$ were adopted. This approach yields accurate results provided that corrections proposed by Buttle [108], to allow for the omitted high-lying poles in the R -matrix expansion, are included. The shortcoming of this approach is that all basis functions have the same (usually zero) logarithmic derivative at the boundary of the internal region. This yields a discontinuity in the slope of the resulting continuum orbitals, which is also addressed by including the Buttle correction. In the standard approach, the basis functions u_i are constructed to be orthogonal to the bound orbitals P_{nl} used for the construction of the target wavefunctions. To compensate for the resulting restrictions on the total wavefunctions, the basis states Ψ_k must contain the correlation functions χ_i as in equation (14). In the case of complex atoms, when extensive multi-configuration expansions are used for accurate representations of the target wavefunctions, this may lead to a very large number of the correlation functions to be included in the close-coupling expansion in order to compensate for the orthogonality constraints.

A key point of the BSR approach is the use of B -splines as the one-electron basis functions $u_i(r)$ in the R -matrix representation (14) of the inner region. The B -splines possess properties as if they were especially created for the R -matrix theory. They form a complete basis on the finite interval $[0, a]$, are of universal nature without numerical bias and are very convenient in numerical calculations because they avoid finite-difference formulae. Here we should emphasize the need to distinguish between using B -splines as another basis to represent the one-electron orbitals and employing them to generate the complete pseudo-spectrum for some one-electron Hamiltonian, as is done in many atomic structure calculations. The BSR program [1] provides both options. In the first case, the coefficients c_{ijk} are found from the diagonalization (17) of the full Hamiltonian in the B -spline representation. Such an approach is well suited for bound-state calculations. In the second case, we first perform a preliminary diagonalization of the Hamiltonian blocks corresponding to one channel. This generates a complete set of one-electron orbitals for each channel. After transforming the Hamiltonian matrix to the new representation based on these one-electron orbitals, we reduce the dimension of the full interaction matrix by dropping some of the basis orbitals, depending on the problem under consideration.

The boundary conditions in the B -spline basis define only the first and the last basis functions, which are the only nonzero terms, respectively, for $r = 0$ and $r = a$. The boundary conditions for the scattering function at the origin are satisfied in the form $F(0) = 0$ by simply removing the first B -spline from the basis set. For the definition of the R -matrix at the boundary (20), the amplitudes of the wavefunctions at $r = a$ are required. These values are defined by the coefficients of the last spline at the boundary. The summation over the entire expansion (14) then yields the surface amplitudes.

2.3. The external region

The next step in the calculation is to solve the scattering problem in the external region and to match the solutions

on the boundary $r = a$ in order to obtain the K -matrices, S -matrices or phase shifts. Since the radius a is chosen such that electron exchange is negligible in this region, we can expand the total wavefunction in the form

$$\Psi^\Gamma(x_1, \dots, x_{N+1}) = \sum_j \bar{\Phi}_i^\Gamma(x_1, \dots, x_N; \hat{r}_{N+1}, \sigma_{N+1}) \frac{1}{r_{N+1}} F_i^\Gamma(r_{N+1}); \quad r_{N+1} > a. \quad (24)$$

Here the $\bar{\Phi}_i^\Gamma$ are the same set of channel functions as those retained in expansion (5), and the $F_i^\Gamma(r)$ are the analytic continuations for $r > a$ of the reduced radial wavefunctions defined in (21). The radial functions satisfy the set of coupled differential equations

$$\left(\frac{d^2}{dr^2} - \frac{l_i(l_i+1)}{r^2} + \frac{2(Z-N)}{r} + k_i^2 \right) F_i^\Gamma(r) = 2 \sum_{j=1}^n \sum_{\lambda=1}^{\Lambda} \frac{a_{ij}^{\lambda, \Gamma}}{r^{\lambda+1}} F_j^\Gamma(r), \quad i = 1, n \quad (r \geq a). \quad (25)$$

Here n is the number of channel functions retained in expansions (5) and (24) while l_i and k_i^2 are the channel angular momenta and energies. The interaction between channels is defined by the long-range potential with coefficients

$$a_{ij}^{\lambda, \Gamma} = \langle \bar{\Phi}_i^\Gamma(x_1, \dots, x_N; \hat{r}_{N+1}, \sigma_{N+1}) | \sum_k^N r_j^\lambda P_\lambda(\cos \theta_{kN+1}) | \bar{\Phi}_j^\Gamma(x_1, \dots, x_N; \hat{r}_{N+1}, \sigma_{N+1}) \rangle, \quad (26)$$

where $\cos \theta_{kN+1} = \hat{r}_k \cdot \hat{r}_{N+1}$ and $P_\lambda(x)$ is a Legendre polynomial. The integral in equation (26) is again carried out over all electronic space and spin coordinates except for the radial coordinate of the scattered electron. In practical calculations, the long-range potential coefficients are the by-product of generating the interaction matrix (17) and are defined by the coefficients of the relevant Slater integrals R^k , describing the direct interaction between channels in the internal region.

Equations (25) can be integrated outwards from $r = a$ and fitted to an asymptotic expansion at large r as described in [109]. If all n scattering channels are open, then the asymptotic form of the radial wavefunctions $F_i(r)$ may be expressed in the form

$$F(r) \underset{r \rightarrow \infty}{\sim} k^{-1/2} (F + GK), \quad (27)$$

where we have written the channel momenta k as a diagonal matrix. The diagonal matrices F and G correspond to regular and irregular Coulomb (or Riccati–Bessel) functions in each scattering channel. The asymptotic expression (27) defines the K -matrix, which is appropriate for standing-wave boundary conditions. It is related to the scattering S -matrix and the transition T -matrix by the equation

$$S = 1 + T = \frac{1 + iK}{1 - iK}. \quad (28)$$

These matrices are used to calculate cross sections and other scattering observables.

The long-range coefficients, together with the target energies and the definition of the structure of the close-coupling equations, constitute the information needed to solve the scattering problem in the external region. This problem is well developed, and there exist several general computer codes for its solution. Note that, in addition to the K -matrix, we also need the outer region solutions at the boundary $r = a$ for the calculation of photoionization cross sections (see below).

2.4. Bound-state calculations

Electron collision theory is concerned with states of an $(N+1)$ -electron system for which N electrons are bound in an atom or atomic ion and one electron can escape to infinity. Such states may be represented accurately with the close-coupling expansion (5). Not surprisingly, however, this expansion is also very suitable for representing states with one electron highly excited and the other N electrons more tightly bound. When the close-coupling approximation is used to calculate bound states of atomic systems, it is referred to as the frozen-cores (FCS) approximation, and the Φ_i^c are usually labelled ‘core’ rather than ‘target’ functions.

The FCS method has several advantages. It can readily be extended to highly excited states. The multichannel form of (5) allows us to include explicitly the interaction between different Rydberg series, as well as the interaction of the Rydberg series with perturbers that may be represented in the second part of the expansion. The energies and wavefunctions can be computed efficiently with an accuracy comparable to that obtainable with the best alternative methods. Note that the same expansion can be used for close-coupling collision calculations as for FCS bound-state calculations. This consistency is very important for the ionization methodology discussed later. Also, a comparison of the calculated bound-state energies with experimental energies may provide a check regarding the accuracy of the collision calculations. At the same time, one can take advantage of the extensive experience accumulated from close-coupling collision calculations and the codes developed for such calculations, and one can apply them in a straightforward way to the study of Rydberg series.

Highly accurate numerical results can be obtained by using a spline-based FCS method for Rydberg series, in which the wavefunctions of the outer electrons are expanded directly in B -splines in some finite region $r \leq a$, with a sufficiently large value of a . The appropriate boundary conditions are imposed by deleting from the expansion the first and the last splines, i.e., the only splines with a nonzero value at the boundary. In practice, we also delete the next to last spline, in order to ensure a zero derivative at the border for all bound solutions.

The choice of B -splines as basis functions has some advantages. The completeness of the B -spline basis ensures that, in principle, we can study the entire Rydberg series. The number of physical states obtained in a single diagonalization is defined by the box radius a , which can easily be varied in the B -spline representation. Not surprisingly, an exponential grid of knots is most suitable for bound states, which allows us to use a large radius with a relatively small number of B -splines.

For example, in order to obtain Rydberg states up to $n = 10$ in neutral atoms, it is often sufficient to choose the box radius as $300a_0$ (where $a_0 = 0.529 \times 10^{-10}$ m denotes the Bohr radius) and the number of splines as 45. If we aim to study the Rydberg series up to $n = 20$, we should increase the border radius to $1200a_0$. With an exponential grid, this increases the number of splines to just 51. The size of the interaction matrix, therefore, which is proportional to the number of splines, does not increase considerably. Of course, these numbers somewhat depend on the nuclear charge Z . For very high Z , it is advisable to add a few splines at very small radii to achieve an accurate representation of the orbitals near the nucleus.

The wavefunctions in this method are obtained for all radii and for all Rydberg states under consideration. There is no need to obtain an asymptotic solution and to match it to the inner-region solution as in the R -matrix method discussed above. This considerably simplifies the calculations and the associated computer codes. In practice, the method was found to be most efficient for the study of moderately excited states with principal quantum numbers in the range 10–30.

Our implementation of the spline method differs from a previous one [87] through the use of nonorthogonal orbitals, both for the construction of the target (core) wavefunctions and for the representation of the outer electron. It provides us with a great deal of flexibility in the choice of the core wavefunctions, which can be optimized for each atomic state separately, and in the introduction of different correlation corrections. For example, the core–core correlation may be taken into account by using extensive multi-configuration target states. The core–valence correlation can be handled in two ways, either by using a large set of excited target states in the close-coupling expansion or by introducing additional $(N+1)$ -electron states, specially designed for this purpose. The convergence of the close-coupling expansion can be very slow, and hence the first approach is much more time consuming. Nevertheless, our experience shows that this method provides a more accurate description of the core-polarization potential, and the $(N+1)$ -electron terms in (5) are better used only for the inclusion of the short-range correlation.

2.5. Radiative processes

Photoionization cross sections can be defined through the dipole matrix between the initial bound state Φ_0 and the R -matrix basis states Ψ_k . This is straightforward in R -matrix theory, provided that all radial orbitals of the initial state are well confined to the inner region. The total photoionization cross section for a given photon energy ω is

$$\sigma(\omega) = \frac{4}{3} \pi^2 a_0^2 \alpha \frac{\omega C}{2L_0 + 1} \sum_j |(\Psi_j^- || D || \Phi_0)|^2, \quad (29)$$

where $\alpha \approx 1/137$ is the fine-structure constant and D is the dipole operator. The latter is usually given in the length ($C = 1$) or the velocity ($C = 4/\omega^2$) form, with the photon energy in Rydberg. The index j runs over the possible final-state solutions. The solutions Ψ_j^- in (29) correspond to asymptotic conditions with a plane wave in the direction of the ejected electron momentum $\mathbf{k}$ and ingoing waves in all open channels.

The corresponding radial functions F^- are related to the $F(r)$ with the K -matrix asymptotic form (27) via

$$F^- = -iF(1 - iK)^{-1}. \quad (30)$$

Expanding Ψ_j^- in terms of the R -matrix states as in (13) and using the expressions (23), we find that

$$(\Psi_j^- || D || \Phi_0) = \frac{1}{a} \sum_k \frac{(\Psi_k || D || \Phi_0)}{E_k - E_0 - \omega} \mathbf{w}_k^T \mathbf{R}^{-1} \mathbf{F}_j^-(a), \quad (31)$$

where $(\Psi_k || D || \Phi_0)$ are reduced matrix elements between the initial state and the R -matrix basis functions.

In order to use the expression (31), we need the values of the solutions $F_j(a)$ at the R -matrix boundary. They can be obtained by matching the general asymptotic solutions of (25) to the solutions in the internal region at $r = a$. This can be done with (22), which in matrix form reads

$$\mathbf{F} = a\mathbf{R}\mathbf{F}' - b\mathbf{R}\mathbf{F} \quad (r \leq a). \quad (32)$$

Note that there are n_0 independent physical solutions in the outer region, where n_0 is the number of open channels determined by all the accessible target states at a given excitation energy. To relate the $(n \times n)$ -dimensional R -matrix to the $n_0 \times n_0$ K -matrix defined in (27), we introduce $n + n_0$ linearly independent solutions $s_{ij}(r)$ and $c_{ij}(r)$ of (25) that satisfy the boundary conditions

$$\left. \begin{matrix} s_{ij}(r) \\ c_{ij}(r) \end{matrix} \right\} r \rightarrow \infty \begin{cases} \sin \theta_i \delta_{ij}; & i = 1, n; j = 1, n_0; \\ \cos \theta_i \delta_{ij}; & i = 1, n; j = 1, n_0; \\ \exp(-\phi_i) \delta_{i, j-n_0}; & i = 1, n; j = n_0 + 1, n. \end{cases} \quad (33)$$

Here θ_i and ϕ_i define the asymptotic phases in the open and closed channels, respectively:

$$\begin{aligned} \theta_i &= k_i r - \frac{1}{2} l_i \pi - \frac{z}{k_i} \ln 2k_i r + \arg \Gamma \left(l_i + 1 + i \frac{z}{k_i} \right) \quad (\text{open}); \\ \phi_i &= |k_i| r - \frac{z}{|k_i|} \ln(2|k_i| r) \quad (\text{closed}). \end{aligned} \quad (34)$$

Now we can rewrite the asymptotic form of the scattering wavefunction in the general form

$$\mathbf{F} = \mathbf{s} + \mathbf{c}\mathbf{K} \quad (r \geq a). \quad (35)$$

Substituting this into (32) and solving for $\mathbf{K}$, we obtain

$$\mathbf{K} = \mathbf{B}^{-1} \mathbf{A}, \quad (36)$$

where

$$\mathbf{A} = -\mathbf{s} + a\mathbf{R} \left(\mathbf{s}' - \frac{b}{a} \mathbf{s} \right) \text{ and } \mathbf{B} = +\mathbf{c} - a\mathbf{R} \left(\mathbf{c}' - \frac{b}{a} \mathbf{c} \right). \quad (37)$$

This completes the evaluation of the reactance matrix $\mathbf{K}$ and the vector $\mathbf{F}(a)$, provided that the asymptotic solutions $s_{ij}(r)$ and $c_{ij}(r)$ are known. As mentioned above, a number of computer packages are available for obtaining these solutions.

The initial state Φ_0 can be obtained either in an independent MCHF calculation, or in the framework of B -spline bound-state calculations discussed above. In general, it requires the evaluation of dipole matrix elements between states with nonorthogonal orbitals. Details for this case are presented in sections 9 and 10 of [1].

2.6. Inclusion of relativistic effects

With increasing nuclear charge Z , relativistic effects become important both in the target description and the scattering wavefunctions, even for low-energy electron scattering. In our BSR complex, relativistic corrections can be included either via the Breit–Pauli (BP) or the Dirac–Coulomb (DC) Hamiltonian. The former is generally appropriate for intermediate nuclear charges ($Z \lesssim 30$), while the latter should be used, if possible, for heavier targets. Note, however, that the computational effort is significantly larger in the DC case due to at least the doubling of the basis functions. In practice, the situation is even worse, since more splines are needed if the finite size of the nucleus is to be accounted for properly. Both of these methods are described below.

2.6.1. The Breit–Pauli approach. The BP Hamiltonian [110] can be considered as a first-order correction to the nonrelativistic atomic Hamiltonian. It includes all relativistic terms up to order αZ^2 . We recognize three parts of the BP Hamiltonian, namely

$$H_{\text{BP}} = H_{\text{NR}} + H_{\text{RS}} + H_{\text{FS}}. \quad (38)$$

Here H_{NR} is the ordinary nonrelativistic many-electron Hamiltonian

$$H_{\text{NR}} = -\frac{1}{2} \sum_{i=1}^N \nabla_i^2 - Z \sum_{i=1}^N \frac{1}{r_i} + \sum_{i < j}^N \frac{1}{r_{ij}}. \quad (39)$$

The *relativistic shift* operator H_{RS} commutes with the operators $\mathbf{L}$ for the total orbital angular momentum and $\mathbf{S}$ for the total spin. It can be written as

$$H_{\text{RS}} = H_{\text{MC}} + H_{\text{D1}} + H_{\text{D2}} + H_{\text{OO}} + H_{\text{SSC}}, \quad (40)$$

where H_{MC} is the *mass correction* term

$$H_{\text{MC}} = -\frac{\alpha^2}{8} \sum_{i=1}^N (\nabla_i^2)^+ \cdot \nabla_i^2, \quad (41)$$

while H_{D1} and H_{D2} are the one-body and two-body Darwin terms, i.e., the relativistic correction to the potential energy,

$$H_{\text{D1}} = -\frac{\alpha^2 Z}{8} \sum_{i=1}^N \nabla_i^2 \left(\frac{1}{r_i} \right) \quad (42)$$

and

$$H_{\text{D2}} = \frac{\alpha^2}{4} \sum_{i < j} \nabla_i^2 \left(\frac{1}{r_{ij}} \right). \quad (43)$$

Next, H_{SSC} is the *spin–spin contact* term

$$H_{\text{SSC}} = -\frac{8\pi\alpha^2}{8} \sum_{i < j} (\mathbf{s}_i \cdot \mathbf{s}_j) \delta(\mathbf{r}_i \cdot \mathbf{r}_j) \quad (44)$$

and, finally, H_{OO} is the *orbit–orbit* term

$$H_{\text{OO}} = -\frac{\alpha^2}{2} \sum_{i < j} \frac{1}{r_{ij}^3} \left\{ \frac{(\mathbf{p}_i \cdot \mathbf{p}_j)}{r_{ij}} + \frac{\mathbf{r}_{ij}(\mathbf{r}_{ij} \cdot \mathbf{p}_i) \cdot \mathbf{p}_j}{r_{ij}^3} \right\}. \quad (45)$$

The *fine-structure* operator H_{FS} describes interactions between the spin and the orbital angular momenta of the electrons. It does not commute with $\mathbf{L}$ and $\mathbf{S}$ individually but only with

the total electronic angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$. The fine-structure operator itself consists of three terms,

$$H_{\text{FS}} = H_{\text{SO}} + H_{\text{SOO}} + H_{\text{SS}}. \quad (46)$$

Here H_{SO} is the *nuclear spin-orbit* term

$$H_{\text{SO}} = \frac{\alpha^2 Z}{2} \sum_{i=1}^N \frac{1}{r_{ij}^3} (\mathbf{l}_i \cdot \mathbf{s}_i), \quad (47)$$

H_{SOO} is the *spin-other-orbit* term

$$H_{\text{SOO}} = -\frac{\alpha^2}{2} \sum_{i \neq j}^N \frac{\mathbf{r}_{ij} \times \mathbf{p}_i}{r_{ij}^3} \cdot (\mathbf{s}_i + 2\mathbf{s}_j) \quad (48)$$

and H_{SS} is the *spin-spin* term

$$H_{\text{SS}} = \alpha^2 \sum_{i < j} \frac{1}{r_{ij}^3} \left\{ (\mathbf{s}_i \cdot \mathbf{s}_j) - 3 \frac{(\mathbf{s}_i \cdot \mathbf{r}_{ij})(\mathbf{s}_j \cdot \mathbf{r}_{ij})}{r_{ij}^2} \right\}. \quad (49)$$

For most applications, it is sufficient to include only the one-electron terms and sometimes the two-electron spin-other-orbit interaction. Once again, all necessary integrals can be carried out efficiently and accurately in the B -spline basis. Finally, when we include the fine-structure interactions, a relativistic coupling scheme must be defined. Depending on the particular application and the strength of the spin-orbit interaction, we choose LSJ , jK , or jj , respectively.

2.6.2. The Dirac-Coulomb approach. With increasing nuclear charge, the BP approach will ultimately no longer be sufficient and a fully relativistic formulation is desirable. A Dirac scheme was already implemented in the relativistic ‘Dirac atomic R -matrix code’ developed by Norrington and Grant [111, 112]. Although highly successful in many applications of photon and electron collisions with atoms, ions and molecules (see, for example, [92, 113, 114]), the method has limitations, especially when used for very complex targets.

In our Dirac B -spline R -matrix (DBSR) approach [62], we also use the DC Hamiltonian to describe the N -electron target and the $(N+1)$ -electron collision systems, respectively. In atomic units, the DC Hamiltonian for N electrons in a central field of a nucleus with charge Z is given by

$$H^{\text{DC}} = \sum_{i=1}^N \left(c \boldsymbol{\alpha} \cdot \mathbf{p}_i + \beta c^2 - \frac{Z}{r_i} \right) + \sum_{i > j}^N \frac{1}{r_{ij}}, \quad (50)$$

where the components of the vector $\boldsymbol{\alpha}$ and β are the Dirac matrices, $\mathbf{p}_i$ is the momentum operator of electron ‘ i ’, and $c \approx 137$ is the speed of light. For each partial-wave symmetry J^π , with J denoting the total electronic angular momentum in a jj -coupling scheme and π indicating the parity, the total wavefunction is constructed from Dirac four-component spinors

$$\phi_{nkm} = \frac{1}{r} \begin{pmatrix} P_{n\kappa}(r) \chi_{km}(\vartheta, \varphi) \\ iQ_{n\kappa}(r) \chi_{-\kappa m}(\vartheta, \varphi) \end{pmatrix}, \quad (51)$$

where the real and imaginary radial Pauli spinors are the large and small components, respectively. Furthermore, χ_{km} is the spinor spherical harmonic and κ is the relativistic angular momentum quantum number.

In contrast to the nonrelativistic case, however, the direct implementation of B -splines for the solution of the Dirac

equations encounters a problem related to the occurrence of spurious states. A practically feasible solution was proposed by Fischer and Zatsarinny [3]. They noticed that employing B -splines of *different order* for the large and small component of the spinors provides a numerically stable basis that avoids the occurrence of spurious solutions. At the same time, this basis retains the simplicity and effectiveness of the original B -spline basis and provides the same accuracy as the kinetic balance bases considered by Igarashi [115] and Shabaev *et al* [116].

Consequently, we expand the radial functions for the large and small components $P(r)$ and $Q(r)$ in separate B -spline bases as

$$P(r) = \sum_{i=1}^{n_p} p_i B_i^{k_p}(r); \quad (52)$$

$$Q(r) = \sum_{i=1}^{n_q} q_i B_i^{k_q}(r). \quad (53)$$

Both B -spline bases are defined on the same grid, with the same number of intervals. Only in this case the calculations of various matrix elements and integrals of interest can be performed with the same routines and at the same level of required computational resources as in the case of a single B -spline basis.

The coefficients p_i and q_i are again found by solving the generalized eigenvalue problem for the total $(N+1)$ -electron Hamiltonian inside the R -matrix box. The R -matrix basis functions for the continuum electron are chosen to satisfy the boundary conditions [112]

$$\frac{Q_i(a)}{P_i(a)} = \frac{b + \kappa}{2ac} = \mu, \quad (54)$$

where b is an arbitrary constant, again usually chosen as $b = 0$.

To address the nonhermiticity of the Hamiltonian in the fully relativistic case, we use the Bloch operator suggested in [117]:

$$\mathcal{L} = c \delta(r - a) \begin{pmatrix} -\mu\eta & \eta \\ (\eta - 1) & (1 - \eta)/\mu \end{pmatrix}, \quad (55)$$

where μ defines the boundary conditions (54) and η is another arbitrary constant. After adding the Bloch operator to the Hamiltonian, the interaction matrix is reduced to the symmetric form and can be diagonalized readily to obtain the desired set of solutions. From this finite set of solutions, an R -matrix relation can be derived that connects the solutions in the inner and outer regions. For a given energy E , this relation has the form

$$P_i(a) = \sum_j R_{ij}(E) [2acQ_j(a) - (b + \kappa)P_j(a)], \quad (56)$$

where the relativistic R -matrix is defined as

$$R_{ij}(E) = \frac{1}{2a} \sum_k \frac{P_{ik}(a)P_{jk}(a)}{E_k^{N+1} - E}. \quad (57)$$

We note that a more rigorous expression for the R -matrix contains the correction $-(b + \kappa)/[(b + \kappa)^2 + (2ac)^2]$ [117]. This correction is due to the fact that the set of relativistic basis functions (P_i, Q_i) is incomplete on the surface $r = a$. In most realistic cases, however, it is small and thus usually omitted.

As mentioned previously, the reactance matrix in the R -matrix method is defined via the matching of the external and internal solutions at $r = a$. In the external region, exchange between the scattered electron and the target electrons is neglected. Relativistic effects are also small in this regime, and thus the scattered electron can be well described in a nonrelativistic framework. Consequently, we follow [111, 118] and use the nonrelativistic limit of the Dirac radial equations for the scattered electron in the asymptotic region. In this case, the matching procedure is identical to that used in the semirelativistic BSR code [1].

2.7. Pseudo-states

The use of ‘pseudo-orbitals’ and ‘pseudo-states’ constructed with at least one of these orbitals being in a dominant configuration has a long history in atomic structure and collision calculations. Generally speaking, their role is twofold: they can be used to account for (i) the term dependence of the physical orbitals and (ii) the coupling to the ionization continuum. Consider for example, the 1s and 2s orbitals in helium in the three states $(1s^2)^1S$ and $(1s2s)^3,^1S$. Independent Hartree–Fock calculations would yield different 1s and 2s one-electron orbitals for each of these states. If the computational method, however, is restricted to a single set of one-electron orbitals, then the accuracy of the *simultaneous* description of all these states can be improved by introducing a $\bar{3}s$ pseudo-orbital. That orbital would be optimized based on the criteria whose details depend on the chosen set of 1s and 2s orbitals and the weights given to the various states. As a result, the $\bar{3}s$ orbital would have a much shorter range than the physical 3s orbital, and hence it is termed a pseudo-orbital. With it, additional states such as $(1s\bar{3}s)^3,^1S$, $(2s\bar{3}s)^3,^1S$ and $(\bar{3}s^2)^1S$ can be constructed. These pseudo-states have unphysical thresholds whose effects need to be handled with care. Depending of the complexity of the situation, many more such pseudo-orbitals may be needed in order to achieve the same accuracy that can otherwise be readily obtained through the individual optimization of the orbitals for each state of interest.

While some of these pseudo-states are likely located above the ionization threshold already, the systematic construction of a large number of such states to map out the ionization continuum has become very popular in recent years. The principal reason is the success of the convergence close-coupling (CCC) method, first demonstrated by Bray and Stelbovics in the Temkin–Poet S-wave model problem of e–H collisions [119] and shortly thereafter for the full e–H case [120]. They used a Laguerre basis to generate the one-electron pseudo-orbitals, but other approaches such as box-based CCC [121] can be used as well. In the latter, the orbitals are forced to vanish at a boundary like the R -matrix radius. Hence, it is not surprising that the R -matrix basis functions themselves may also be used, as is done in the IERM approach [97], or Sturmian functions as in the well-known R -matrix with pseudo-states (RMPS) method [100], and, of course, B -splines. The basic idea is always the same: the target Hamiltonian is diagonalized in a basis of finite range, whether it be a box

with a hard wall like the R -matrix radius or a soft wall as in a Laguerre or Sturmian basis. States whose orbitals effectively fit in the box are excellent approximations of the physical target states. Then there will be some whose physical orbitals do not quite fit but whose energies are still below the ionization threshold, and then there are states above that threshold. The former set approximate the effect of the high-lying Rydberg states while the latter simulate the ionization continuum.

2.8. Electron-impact ionization

We consider the ionization of an atom by electron impact, schematically written as

$$e_0(k_0\mu_0) + A(L_0S_0) \rightarrow e_1(k_1\mu_1) + e_2(k_2\mu_2) + A^+(L_fS_f), \quad (58)$$

where k_i and μ_i ($i = 0, 1, 2$) are the linear momenta and spin components of the incident, scattered and ejected electrons, respectively. L_0, S_0 and L_f, S_f are the orbital and spin angular momenta of the initial $(N+1)$ -electron atom and the residual N -electron ion.

For a complete description of this process, we need the ionization amplitude

$$f(L_0M_0S_0M_{S_0}, k_0\mu_0 \rightarrow L_fM_fS_fM_{S_f}, k_1\mu_1, k_2\mu_2), \quad (59)$$

where we have introduced the magnetic quantum numbers of the atomic (M_0) and ionic (M_f) orbital angular momenta, as well as the corresponding spin components M_{S_0} and M_{S_f} .

To begin the discussion, we consider the first-order amplitude that can be written as

$$\begin{aligned} f(L_0M_0S_0M_{S_0}, k_0\mu_0 \rightarrow L_fM_fS_fM_{S_f}, k_1\mu_1, k_2\mu_2) \\ = \langle \varphi_{k_1\mu_1}^{(-)}(\mathbf{x}) \Psi_{L_fM_fS_fM_{S_f}}^{k_2\mu_2(-)}(\mathbf{X}) | \\ \times V(\mathbf{x}, \mathbf{X}) | \Phi_{L_0M_0S_0M_{S_0}}(\mathbf{X}) \varphi_{k_0\mu_0}^{(+)}(\mathbf{x}) \rangle. \end{aligned} \quad (60)$$

Here $\mathbf{X} = \{\mathbf{r}_1\sigma_1; \mathbf{r}_2\sigma_2; \dots; \mathbf{r}_{N+1}\sigma_{N+1}\}$ denotes a set of electronic spatial and spin coordinates in the $(N+1)$ -electron atom, while $\mathbf{x} = \{\mathbf{r}, \sigma\}$ represents the corresponding coordinates for the colliding electron. The Coulomb potential

$$V(\mathbf{x}, \mathbf{X}) = \sum_{n=1}^{N+1} \frac{1}{|\mathbf{r} - \mathbf{r}_n|} - \frac{Z}{r} \quad (61)$$

describes the interaction between the projectile and the atomic electrons as well as the nucleus of charge Z . Although the two outgoing electrons are, in principle, indistinguishable, we will refer to the faster (slower) one of these as the scattered (ejected) electron, respectively. This notation makes sense for sufficiently high incident energies (many times the ionization threshold) and strongly asymmetric energy sharing between the two outgoing electrons.

The functions $\varphi_{k_0\mu_0}^{(+)}(\mathbf{x})$ and $\varphi_{k_1\mu_1}^{(-)}(\mathbf{x})$ represent the incident and outgoing projectile with linear momenta k_0 and k_1 , respectively. Furthermore, the function $\Psi_{L_fM_fS_fM_{S_f}}^{k_2\mu_2(-)}(\mathbf{X})$ in (61) describes the scattering of the ejected electron by the residual ion. Post-collision interaction (PCI) as well as exchange effects between the two final continuum electrons are not accounted for in (60) above.

Neglecting the presence of the projectile, a rigorous treatment for the ejected-electron–residual-ion part of the problem is, once again, the close-coupling expansion

$$\begin{aligned} \Psi_{L_f M_f S_f M_{S_f}}^{k_2 \mu_2 (-)}(\mathbf{X}) &= \frac{1}{\sqrt{k_2}} \sum_{l_2 m_2} i^{l_2} \exp(-i\sigma_{l_2}) Y_{l_2 m_2}^*(\hat{\mathbf{k}}_2) \\ &\times \sum_{\substack{L, M \\ S, M_S}} (L_f M_f, l_2 m_2 | LM) \left(S_f M_{S_f}, \frac{1}{2} \mu_2 | SM_S \right) \\ &\times \psi_{L M S M_S}^{L_f M_f S_f M_{S_f}, k_2 (-)}. \end{aligned} \quad (62)$$

Here $\psi_{L M S M_S}^{L_f M_f S_f M_{S_f}, k_2 (-)}$ is a residual-ion–ejected-electron basis state of total orbital angular momentum L and spin S with appropriate boundary conditions. These channel functions are again being constructed by coupling the $\mathbf{X}$ coordinates of the target with the $\hat{\mathbf{r}}, \sigma$ coordinates of the free electron. The expansion (62) allows us to obtain wavefunctions for well-defined orbital and spin angular momenta L_f and S_f for each residual ionic state and the ejected electron. As will be shown below, having these functions available for projection is a critical part in developing an entirely nonperturbative methodology for describing electron-impact ionization processes.

Using the expansion (62) in (60) together with distorted waves for a ‘fast’ projectile is the basis of the hybrid approach developed by Bartschat and Burke [122] and later extended by Reid *et al* [123, 124] and Fang and Bartschat [125] to account, approximately, for second-order effects in the projectile–target interaction. This method is appropriate in highly asymmetric kinematical situations, where the faster of the two outgoing electrons has most of the excess energy and is detected at small (a few degrees) deflection angles from the incident-beam direction. Not surprisingly, therefore, it was very successful for such cases [126, 127], but then broke down with increasing detection angle of the fast electron, decreasing excess energy and comparable sharing of that excess energy [70, 128, 129].

An alternative to the fully or at least partially perturbative methods is the continuum pseudo-state approach, which does *not* start with the separation of the projectile from the rest of the problem as in (60). This fully nonperturbative approach was introduced by Bray and Fursa [130] in their extension of the CCC formalism from elastic scattering and electron-impact excitation processes in helium [131] to ionizing collisions. Specifically, one begins by replacing the true continuum orbitals in the wavefunction $\Psi_{L_f M_f S_f M_{S_f}}^{k_2 \mu_2 (-)}(\mathbf{X})$ by a square-integrable representation obtained by diagonalizing the atomic Hamiltonian in an appropriate basis. As mentioned above, popular choices in the CCC calculations have been the Laguerre basis or a box basis of finite range [121]. The final answers should be the same, except for numerical issues, but one basis may be advantageous over another in the practical aspects of a particular problem. The diagonalization procedure results in a number of discrete states, with the negative-energy (with respect to the ionization threshold) states representing a few low-lying physical bound states and an approximation for the infinite Rydberg spectrum, while the positive-energy states represent the ionization continuum.

The calculation then proceeds in two steps. First, the electron-impact *excitation* problem for both the physical

and the pseudo-states needs to be treated. This can be done, for example, by solving the corresponding coupled Lippmann–Schwinger equations (as in the momentum-space CCC method) for the transition matrix elements or by using the *R*-matrix approach formulated in coordinate space. In a second step, excitation of the positive-energy pseudo-states is then interpreted as ionization.

A significant challenge, however, concerns the procedure of extracting the information from the wavefunction obtained for the atom left in an excited pseudo-state. The total angle-integrated ionization cross section can be obtained directly as the sum of the excitation cross sections for all the pseudo-states above the ionization threshold. To obtain the more detailed differential cross sections (DCSs) (with regard to energy sharing and/or well-defined detection angles), on the other hand, one needs to relate the discrete finite-range pseudo-state functions to the true continuum functions at the proper ejected electron energy and construct the appropriate ionization amplitude (59). Even the extraction of the angle-integrated ionization cross section for a specific final ionic state is by no means trivial, due to the fact that CI effects make it difficult to uniquely assign the contribution of an individual pseudo-state to the result for a particular physical ionic state.

In our BSR approach [71], we proceed as follows. For a given energy of the incident electron, we obtain the scattering amplitudes for excitation of all energy-accessible atomic pseudo-states $\Phi^P(nln'l', LS)$ as

$$\begin{aligned} f^P(L_0 M_0 S_0 M_{S_0}, \mathbf{k}_0 \mu_0 \rightarrow L M S M_S, \mathbf{k}_1 \mu_1) \\ = \sqrt{\frac{\pi}{k_0 k_1}} \sum_{l_0, l_1, L_T, S_T, \Pi_T, M_{L_T}, M_{S_T}} i^{(l_0 - l_1)} \sqrt{(2l_0 + 1)} \\ \times (L_0 M_0, l_0 0 | L_T M_{L_T}) (LM, l_1 m_1 | L_T M_{L_T}) \\ \times \left(S_0 M_{S_0}, \frac{1}{2} \mu_0 | S_T M_{S_T} \right) \left(SM_S, \frac{1}{2} \mu_1 | S_T M_{S_T} \right) \\ \times T_{l_0 l_1}^{L_T S_T \Pi_T}(\alpha_0 L_0 S_0 \rightarrow \alpha LS) Y_{l_1 m_1}(\theta_1, \varphi_1), \end{aligned} \quad (63)$$

where $T_{l_0 l_1}^{L_T S_T \Pi_T}(\alpha_0 L_0 S_0 \rightarrow \alpha L_1 S_1)$ is an element of the *T*-matrix for a given L_T , total spin S_T and parity Π_T of the $(N+2)$ -electron system. Choosing the *z*-axis along the direction of the incident beam simplifies the formula to $m_0 = 0$.

We then obtain the ionization amplitude (59) by projecting the excitation amplitudes (63) to the true continuum functions $\Psi_{L_f M_f S_f M_{S_f}}^{k_2 \mu_2 (-)}$ and summing over all energetically accessible pseudo-states using the ansatz

$$\begin{aligned} f(L_0 M_0 S_0 M_{S_0}, \mathbf{k}_0 \mu_0 \rightarrow L_f M_f S_f M_{S_f}, \mathbf{k}_1 \mu_1, \mathbf{k}_2 \mu_2) \\ = \sum_p \langle \Psi_{L_f M_f S_f M_{S_f}}^{k_2 \mu_2 (-)} | \Phi^P(nln'l', LS) \rangle \\ \times f^P(L_0 M_0 S_0 M_{S_0}, \mathbf{k}_0 \mu_0 \rightarrow L M S M_S, \mathbf{k}_1 \mu_1). \end{aligned} \quad (64)$$

This requires the determination of the overlap factors $\langle \Psi_{L_f M_f S_f M_{S_f}}^{k_2 \mu_2 (-)} | \Phi^P(nln'l', LS) \rangle$ between the true continuum states and the corresponding pseudo-states. The continuum states $\Psi_{L_f M_f S_f M_{S_f}}^{k_2 \mu_2 (-)}$ are obtained using the *R*-matrix method with the *same* close-coupling expansion that is employed for the pseudo-states. Computationally, the only difference is the use of *R*-matrix boundary conditions by adding the corresponding Bloch operator. Both pseudo-states and continuum solutions

in the B -spline basis can be considered as vectors b_p and b_c whose length depends on the number of open channels.

The computational advantage of the present approach is that the needed overlap factors are obtained in a straightforward way as $\langle b_p | S | b_c \rangle$ using the already calculated overlap matrix. Note that the one-electron pseudo-state and the continuum orbitals are not orthogonal. Our nonorthogonal orbital technique takes this nonorthogonality into account to full extent. In order to obtain the required solutions with ingoing-wave boundary conditions, we renormalize the R -matrix solutions by multiplying through with the matrix $[1 + iK]^{-1}$.

It is worth pointing out an important subtlety of the approach. Note that $\Psi_f^{k_2}$ and $\Phi^p(LS)$ have *different* energies for the continuum electron represented by k_2 and the electron in the pseudo-state. Due to energy conservation, the excitation of $\Phi^p(LS)$ leads to $k_{1p} \neq k_1$ for the projectile. Having noted numerical instabilities in some cases, Bray and Fursa [130] suggested interpolating the transition-matrix elements as an alternative. While this interpolation, indeed, worked very well for the single-channel case, our direct projection method is necessary to maintain the crucial channel information in multichannel situations. This makes equation (64), which is the generalization of equation (15) of [130] for multichannel cases, a suitable approximation for the true ionization amplitude, provided the spectrum of the pseudo-states is sufficiently dense. Finally, the triple-DCS for ionization with the residual ion remaining in the final state f (this may include simultaneous ionization-excitation) is then given by

$$\frac{d\sigma_f}{d\Omega_1 d\Omega_2 dE} = \frac{k_1 k_2}{k_0} \frac{1}{2(2L_0 + 1)(2S_0 + 1)} \sum_{\substack{M_0, M_f, M_{S_0}, M_{S_f} \\ \mu_0, \mu_1, \mu_2}} |f(L_0 M_0 S_0 M_{S_0}, k_0 \mu_0 \rightarrow L_f M_f S_f M_{S_f}, k_1 \mu_1, k_2 \mu_2)|^2. \quad (65)$$

Another issue worth mentioning in this context is the size of the R -matrix box. As mentioned previously, the separation of configuration space was motivated by the ability to neglect electron exchange outside the box and the use of an energy-independent basis expansion inside in order to allow for an efficient calculation of results for many collision energies. In order to span a large energy range with relatively few basis functions, the box was made as small as possible. For ionization, on the other hand, the size of the box and the related range of the pseudo-states determine the region where correlation effects between the two outgoing electrons are still properly accounted for. Consequently, one often uses a much larger box than needed to simply ignore exchange between the projectile and the bound target electrons. Finally, due to the fundamental asymmetry in the treatment of the two outgoing electrons, the proper symmetry needs to be restored in the final results. Details can be found, for example, in [132, 133].

2.9. Time-dependent processes

The rapid progress in the development of ultra-short and ultra-intense light sources is providing a window to study the details of electron interactions in atoms, molecules, plasmas and

solids on an attosecond time scale. These capabilities promise a revolution in our microscopic knowledge and understanding of matter [134]. There is no doubt that highly challenging experiments such as those reported in [135, 136] will benefit tremendously from the theoretical support in order to reap the maximum profits from the enormous resources being invested in the experimental facilities.

The ingredients of an appropriate theoretical and computational formulation require an accurate and efficient generation of the Hamiltonian and the electron–field interaction matrix elements, as well as an optimal approach to propagate the TDSE in real time. There have been numerous calculations for two-electron systems such as He and H₂. While these investigations emphasize the important role of two-electron systems in studying electron–electron correlation in the presence of a strong laser field, in its presumably purest form, experiments with He atoms are difficult and other noble gases, such as Ne and Ar, are often favoured by the experimental community.

Fully *ab initio* theoretical approaches, which are applicable to complex targets beyond (quasi) two-electron systems, are still rare. For (infinitely) long interaction times, the R -matrix Floquet ansatz [137] has been highly successful. A critical ingredient of this method is the general atomic R -matrix method developed over many years by Burke and collaborators in Belfast. A modification of the method, allowing for relatively long though finite-length pulses, was described by Plummer and Noble [138]. A time-dependent formulation for short-pulse laser interactions with complex atoms was described in detail by Lysaght *et al* [139].

We start with the time-dependent Schrödinger equation

$$i \frac{\partial}{\partial t} \Psi(\mathbf{r}_1, \dots, \mathbf{r}_N; t) = [H_0(\mathbf{r}_1, \dots, \mathbf{r}_N) + V(\mathbf{r}_1, \dots, \mathbf{r}_N; t)] \Psi(\mathbf{r}_1, \dots, \mathbf{r}_N; t) \quad (66)$$

for the N -electron wavefunction $\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N; t)$, where $H_0(\mathbf{r}_1, \dots, \mathbf{r}_N)$ is the field-free Hamiltonian containing the kinetic energy of the N electrons, their potential energy in the field of the nucleus and their mutual Coulomb repulsion, while

$$V(\mathbf{r}_1, \dots, \mathbf{r}_N; t) = \sum_{i=1}^N \mathbf{E}(t) \cdot \mathbf{r}_i \quad (67)$$

represents the interaction of the electrons with the laser field $\mathbf{E}(t)$ in the dipole length form. The tasks to be carried out in order to computationally solve the TDSE and to extract the physical information of interest are:

- (1) Generate a representation of the field-free Hamiltonian and its eigenstates; these include the initial bound state, other bound states, autoionizing states, as well as single-continuum and double-continuum states to represent electron scattering from the residual ion.
- (2) Generate the dipole matrices to represent the coupling to the laser field.
- (3) Propagate the initial bound state until some time after the laser field is turned off.
- (4) Extract the physically relevant information from the final state.

Of particular interest in the experiments mentioned above are processes, in which one, two, or even more electrons undergo significant changes in their quantum state in the presence of an atomic core. These include excitation, single and double ionization (DI), ionization plus simultaneous excitation or inner-shell ionization with subsequent rearrangement in the hollow ion. The latter processes, in particular, can only be investigated in systems beyond the frequently studied two-electron helium atom or H_2 molecule. While the generalization to two electrons outside a multi-electron core is far from trivial, the flexibility of the BSR method is highly advantageous for tasks 1 and 2. Our method is formulated in a sufficiently general way to be applicable to *complex* atoms, such as inert gases other than helium and even open-shell systems with nonvanishing spin and orbital angular momenta. In reality, of course, the size of the problems that can be handled is also determined by the available computational resources.

The solution of the TDSE requires an accurate and efficient generation of the Hamiltonian and electron–field interaction matrix elements. In order to achieve this goal, we approximate the time-dependent wavefunction as

$$\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N; t) \approx \sum_q C_q(t) \Phi_q(\mathbf{r}_1, \dots, \mathbf{r}_N). \quad (68)$$

The $\Phi_q(\mathbf{r}_1, \dots, \mathbf{r}_N)$ are a set of time-independent N -electron states formed from appropriately symmetrized products of atomic orbitals. They are expanded as

$$\Phi_q(\mathbf{r}_1, \dots, \mathbf{r}_N) = \mathcal{A} \sum_{c,i,j} a_{ijcq} \Theta_c \times (x_1, \dots, x_{N-2}; \hat{\mathbf{r}}_{N-1} \sigma_{N-1}; \hat{\mathbf{r}}_N \sigma_N) R_i(r_{N-1}) R_j(r_N). \quad (69)$$

Here $\mathcal{A}$ is the anti-symmetrization operator, the $\Theta_c(x_1, \dots, x_{N-2}; \hat{\mathbf{r}}_{N-1} \sigma_{N-1}; \hat{\mathbf{r}}_N \sigma_N)$ are channel functions involving the space and spin coordinates (x_i) of $N - 2$ core electrons coupled to the angular ($\hat{\mathbf{r}}$) and spin (σ) coordinates of the two outer electrons, $R_i(r)$ is a radial basis function, and the a_{ijcq} are expansion coefficients. Although resembling a close-coupling ansatz with two continuum electrons, the expansion (69) contains bound states and singly ionized states as well. In general, the atomic orbitals, $R_i(r)$, are not orthogonal to one another or to the orbitals used to describe the atomic core. If orthogonality constraints are imposed on these functions, additional terms would need to be added to the expansion to relax the constraints.

Once again, a significant advantage of not forcing such orthogonality conditions is the flexibility gained by being able to tailor the optimization procedures to the individual neutral, ionic and continuum orbitals. In the BSR code, the outer orbitals (i.e., the R functions above) are expanded in B -splines. Factors that depend on angular and spin momenta are separated from the radial degrees of freedom through the construction of the channel functions. Since many Hamiltonian matrix elements share common features, this enables the production of a ‘formula tape’, resulting in an efficient procedure to generate the required matrix elements. When the expansion (68) is inserted into the Schrödinger equation, we obtain

$$i\mathbf{S} \frac{\partial}{\partial t} \mathbf{C}(t) = [\mathbf{H}_0 + E(t)\mathbf{D}] \cdot \mathbf{C}(t), \quad (70)$$

where $\mathbf{S}$ is the overlap matrix of the basis functions, $\mathbf{H}_0$ and $\mathbf{D}$ are matrix representations of the field-free Hamiltonian and the dipole coupling matrices, and $\mathbf{C}(t)$ is the time-dependent coefficient vector in (68).

The price to pay for the flexibility in the BSR approach, at least initially, is the representation of the field-free Hamiltonian and the dipole matrices in a nonorthogonal basis. While there are other possibilities that are described in the original papers, a straightforward way to address this problem is to solve the field-free generalized eigenvalue problem

$$\mathbf{H}_0 \mathbf{v}^L = \mathbf{E} \mathbf{S} \mathbf{v}^L \quad (71)$$

for each symmetry with total orbital angular momentum L . To simplify the notation, we apply the dipole selection rules and assume the most common practical case, namely a ^1S initial state of even parity. Hence, we only need to include the symmetries ^1S , $^1\text{P}^0$, ^1D , $^1\text{F}^0$, $\dots$. This gives us the eigenbasis of column vectors $\mathbf{v}_n^L$, $n = 1, 2, \dots, n_r^L$ with eigenvalues E_n^L , where n_r^L is the rank of the matrix for this particular symmetry.

Having obtained this eigenbasis, we define the transformation matrix

$$\mathbf{A}^L \equiv (\mathbf{v}_1^L, \mathbf{v}_2^L, \dots, \mathbf{v}_{n_r^L}^L), \quad (72)$$

in which we drop the eigenvectors corresponding to eigenvalues $E_n^L > E_c$. We then transform the field-free Hamiltonian and the dipole matrices coupling L and L' according to

$$\tilde{\mathbf{H}}_0^L \equiv (\mathbf{A}^L)^T \mathbf{H}_0^L \mathbf{A}^L, \quad (73)$$

$$\tilde{\mathbf{D}}^{LL'} \equiv (\mathbf{A}^L)^T \mathbf{D}^{LL'} \mathbf{A}^{L'}. \quad (74)$$

Here the superscript ‘ T ’ indicates the transposed matrix. As a result, the field-free Hamiltonian is now diagonal with the energy eigenvalues $E_n^L < E_c$ as the diagonal elements, while the dipole matrices are represented in the truncated eigenbasis. The initial state in this representation is the column vector $(1, 0, 0, \dots, 0)$, and the extraction of survival, excitation and ionization probabilities is straightforward. After the time propagation, the first element of the vector squares yields the survival probability of the ground state, the sum of the squares of all other coefficients corresponding to eigenstates below the ionization threshold represents excitation, and the sum of the squares of the remaining coefficients corresponds to ionization. Since the problem is orthogonal in the truncated eigenbasis after the transformation, we can use standard algorithms to propagate the initial state in time. Details were given in several papers by Guan *et al* [5, 84–86].

Finally, special care needs to be taken in the extraction of information for DI. So far we have addressed the helium case [84] with the bare He^{2+} ion left after DI. There we projected the solution to a properly symmetrized product of Coulomb functions for the two outgoing electrons. Since this is not an eigenfunction of the problem, however, the stability of the solution with respect to changing the size of the R -matrix box must be analysed. For details, also concerning the proper definition of effective interaction times and generalized cross sections, we refer again to the above papers by Guan *et al*.

Table 5. Binding energies (NIST [140]) and energy differences (computed—observed) in eV for some low-lying levels in Ne, Ar, Kr and Xe. The values in bold are average differences for the states in the respective configuration obtained when the core–valence correlation is omitted.

Ne	NIST	Diff.	Ar	NIST	Diff.	Kr	NIST	Diff.	Xe	NIST	Diff.
2p ⁶	21.565	0.061	3p ⁶	15.760	0.044	4p ⁶	14.000	0.024	5p ⁶	12.565	0.020
3s[3/2] ₂	4.945	0.012	4s[3/2] ₂	4.211	0.102	5s[3/2] ₂	4.084	0.128	6s[3/2] ₂	4.250	0.100
3s[3/2] ₁	4.894	0.015	4s[3/2] ₁	4.136	0.100	5s[3/2] ₁	3.967	0.115	6s[3/2] ₁	4.129	0.094
		0.200			0.310			0.350			0.300
3p[1/2] ₁	3.183	0.007	4p[1/2] ₁	2.853	0.033	5p[1/2] ₁	2.696	0.033	6p[1/2] ₁	2.985	0.071
3p[5/2] ₃	3.009	0.009	4p[5/2] ₃	2.684	0.024	5p[5/2] ₃	2.557	0.036	6p[5/2] ₂	2.880	0.068
3p[5/2] ₂	2.989	0.007	4p[5/2] ₂	2.665	0.028	5p[5/2] ₂	2.555	0.046	6p[5/2] ₃	2.845	0.055
3p[3/2] ₁	2.952	0.009	4p[3/2] ₁	2.606	0.022	5p[3/2] ₁	2.473	0.031	6p[3/2] ₁	2.776	0.043
3p[3/2] ₂	2.928	0.008	4p[3/2] ₂	2.588	0.029	5p[3/2] ₂	2.454	0.036	6p[3/2] ₂	2.744	0.057
3p[1/2] ₀	2.853	0.008	4p[1/2] ₀	2.487	0.024	5p[1/2] ₀	2.334	0.052	6p[1/2] ₀	2.632	0.064
		0.070			0.130			0.140			0.130
3d[1/2] ₀	1.540	0.004	3d[1/2] ₀	1.915	0.116	4d[1/2] ₀	2.001	0.116	5d[1/2] ₀	2.675	0.014
3d[1/2] ₁	1.538	0.004	3d[1/2] ₁	1.896	0.113	4d[1/2] ₁	1.963	0.112	5d[1/2] ₁	2.648	0.066
3d[7/2] ₄	1.530	0.001	3d[3/2] ₂	1.856	0.105	4d[3/2] ₂	1.888	0.101	5d[7/2] ₄	2.622	0.168
3d[7/2] ₃	1.530	0.006	3d[7/2] ₄	1.780	0.081	4d[7/2] ₄	1.874	0.094	5d[3/2] ₂	2.607	0.042
3d[3/2] ₂	1.528	0.005	3d[7/2] ₃	1.747	0.072	4d[7/2] ₃	1.821	0.083	5d[7/2] ₃	2.526	0.144
3d[3/2] ₁	1.524	0.005	3d[5/2] ₂	1.697	0.064	4d[5/2] ₂	1.742	0.072	5d[5/2] ₂	2.408	0.109
3d[5/2] ₂	1.516	0.007	3d[5/2] ₃	1.661	0.048	4d[5/2] ₃	1.715	0.068	5d[5/2] ₃	2.345	0.122
3d[5/2] ₃	1.516	0.006	3d[3/2] ₁	1.607	0.047	4d[3/2] ₁	1.645	0.058	5d[3/2] ₁	2.164	0.011
		0.015			0.180			0.200			0.250

3. Illustrative results

Due to space limitations, we can only present selected results here. An up-to-date list of all our calculations using the BSR method is provided in tables 1–4. Details can be found in the respective publications. We will start with some results for atomic structure, followed by electron collisions, weak-field steady-state photoionization and finally time-dependent processes in short laser pulses.

3.1. Structure calculations

3.1.1. Energy levels in heavy noble gases. Table 5 compares the binding energies for some low-lying levels in Ne, Ar, Kr and Xe. As can be seen from the table, accounting for core–valence correlation is important for accurate results to be obtained. Our *ab initio* models with open core states account for up to about 90% of the core–valence correlation in Ne and up to 60%–70% in Kr and Xe. Details can be found in [7].

3.1.2. Oscillator strengths in xenon. Table 6 shows the oscillator strengths for some transitions in Xe from the ground state. As one might expect, the fully relativistic results are close to the BP values for most transitions, but for some the changes are crucial and lead to much better agreement with experiment. The most prominent example is excitation to the first 5d level, where the *f*-value is reduced by a factor of 4, mainly due to inner-core correlation. It is interesting to note that the largest changes occur for the *d*-series with quantum number $\kappa = 1/2$.

3.1.3. Lifetimes of sulfur states. As another illustration of a BSR structure calculation for a complex open-shell system, table 7 presents atomic lifetimes for selected levels of neutral sulfur. These lifetimes offer an alternative way of comparing our results [8] with experiment and other theoretical

Table 6. Oscillator strengths for excitation from the ground state in Xe, as obtained in the length (f_L) and velocity (f_V) forms of the electric dipole operator. The experimental data are from Chan *et al* [141] and the superscripts refer to the results of Avgoustoglou and Beck [142] (a) and Dong *et al* [143] (b).

Upper level	Breit-Pauli		Dirac		Experiment
	f_L	f_V	f_L	f_V	
6s[3/2] ₁	0.278	0.224	0.260	0.258	0.273(14)
	0.249 ^a	0.256 ^a	0.271 ^b	0.263 ^b	
6s'[1/2] ₁	0.186	0.157	0.188	0.189	0.186(9)
			0.158 ^b	0.154 ^b	
5d[1/2] ₁	0.0399	0.0345	0.0083	0.0071	0.0105(5)
5d[3/2] ₁	0.380	0.303	0.303	0.327	0.379(19)
7s[3/2] ₁	0.0785	0.0633	0.0791	0.0783	0.0859(43)
6d[1/2] ₁	< 0.0001		0.0003	0.0005	< 0.001
6d[3/2] ₁	0.0939	0.0758	0.0987	0.0873	0.0835(84)
8s[3/2] ₁	0.0262	0.0211	0.0201	0.0192	0.0222(22)
7d[1/2] ₁	0.0146	0.0105	0.0395	0.0441	0.0227(23)
7d[3/2] ₁	0.0001	0.0002	0.0064	0.0081	< 0.001
9s[3/2] ₁	0.0088	0.0072	0.0002	0.0003	< 0.001
5d'[3/2] ₁	0.151	0.114	0.167	0.170	0.191(19)
8d[1/2] ₁	0.0123	0.0091	0.0068	0.0071	0.0088(9)
8d[3/2] ₁	0.119	0.091	0.0846	0.0835	0.0967(97)
10s[3/2] ₁	0.0139	0.0110	0.0139	0.0134	0.0288(29)

predictions. The BSR length and velocity results agree very well with each other, within a few per cent for most levels. Good agreement is obtained with the MCHF results of Fischer [144] for all but the $(3s^2 3p^3 3d)^3 D_J$ levels, where our lifetimes are about 30% larger. Good agreement, to within 10%, is also obtained with the CI results of Tayal [145], except for the $(3s^2 3p^3 4d)^3 D_J$ levels, where Tayal's results are considerably smaller than ours and also the experimental data. Significant disagreement for many levels was found with the predictions generated with Cowan's [146] code by Biémont *et al* [147]. This points to large correlation corrections in the present case, which can only be accounted for properly with extensive

Table 7. Atomic lifetimes (in nanoseconds) for selected levels of neutral sulfur. T_L and T_V denote the BSR results [8] obtained in the length and velocity forms of the electric dipole operator, respectively. The nl' notation represents the $3p^3(^2D)$ core while nl'' represents the $3p^3(^2P)$ core.

Level	Theory				Experiment			
	T_L	T_V	[147]	[145]	[144]	[147]	[148]	Others
4s 3S_1	1.93	1.98	1.5	1.76	2.04		1.875 (0.188)	1.5 (0.3) ^a , 1.4 ^b , 1.875 (0.188) ^f
5s 3S_1	6.87	6.97	4.2	7.27		7.1 (0.5)	6.9 (1.4)	<6.4 ^b
6s 3S_1	17.0	17.2	9.7			17.7 (1.1)	16.9 (3.4)	<5 ^b
7s 3S_1	34.5	34.9	18.8			35.6 (4.0)	31.0 (8.4)	
8s 3S_1	62.5	63.3	33.1			40 (5)	73.3 (44.0)	
4s 5S_2	17 615	19 771			39 037			9200 ^e , 11900 ^b
4s' 1D_2	2.23	2.24	1.7	2.42	2.30			1.5 (0.3) ^a , <1.7 ^b
4s'' 3P_0	2.22	2.25	3.0					
4s'' 3P_1	2.21	2.24	2.9				2.1 (0.2)	2.8 (0.3) ^d , <1.2 ^b , 2.034 (0.204) ^f
4s'' 3P_2	2.20	2.22	2.8				2.1 (0.2)	2.146 (0.256) ^f
4p 3P_0	46.3	45.4	36.1		49.5	46.1 (1.0)		33 (12) ^c
4p 3P_1	46.3	45.4	36.1		49.4	46.1 (1.0)		33 (12) ^c
4p 3P_2	46.3	45.4	36.0		49.3	46.1 (1.0)		33 (12) ^c
4p' 3P_2	53.6	53.8	63.0					53.5 (4.0) ^e
5p 3P_2	184	186	131.9			188(13)		185 (15) ^e
5p 5P_3	158	159	130.1					615 (50) ^e , <97 (30) ^c
6p 5P_3	352	350	289.9					265 (20) ^e
7p 5P_3	631	624	549.8					415 (25) ^e
3d 3D_1	3.69	3.77	1.9	4.07	2.62			
3D_2	3.68	3.77	1.9	4.05	2.61		3.0 (0.6)	2.1 (0.3) ^a , <1.5 ^b
3D_3	3.67	3.75	1.9	4.02	2.59			
4d 3D_1	12.1	12.6	6.0	4.34		12.6(1.3)		11.4 (1.5) ^d , <8.5 ^b
3D_2	12.1	12.6	6.0	4.29		12.5 (0.7)	9.6 (1.9)	11.4 (1.5) ^d , <8.5 ^b
3D_3	12.0	12.5	5.9	4.37		12.9 (1.1)		11.4 (1.5) ^d , <8.5 ^b
5d 3D_1	36.0	37.8	10.1			41(3)		
3D_2	36.3	38.1	10.0				37 (7.5)	
3D_3	36.6	38.4	9.9			40 (4)		
6d 5D_4	165	163	593.1					260 (20) ^e
3D_1	108	115	16.0					
3D_2	112	118	15.9			75(7)	199 (90)	
3D_3	116	123	15.8					
7d 5D_4	245	243	994.7					505 (40) ^e
8d 3D_1	511	514	31.9					
3D_2	522	523	31.9				>328	
3D_3	501	514	31.9					

The references for ‘others’ are: ^a [149]; ^b [150]; ^c [151]; ^d [152]; ^e [153]; ^f [154].

configuration expansions. Special consideration is required for the $(4s)^5S_2$ quasi-metastable state. Its lifetime is determined by the intercombination transition to the ground state, which is very sensitive to the spin–orbit mixing in the expansion. For a detailed discussion of these results, we refer to our original paper [8].

3.1.4. Polarizabilities. Our last example in the area of structure calculations concerns atomic polarizabilities and the long-range dispersion factors that can be calculated from them for molecules. Table 8 reports the static dipole (α_d), quadrupole (α_q) and octupole (α_o) polarizabilities for the ground state. Due to its $^2P^o$ symmetry, the ground state also has a tensor polarizability (α_2) for the dipole excitation. For a more detailed discussion, also including comparison with a number of other calculations for α_d and α_2 , we refer the reader to the original paper [12].

Table 8. Static dipole (α_d), tensor (α_2), quadrupole (α_q) and octupole (α_o) polarizabilities for the ground state of fluorine. When available, the BSR results [12] are compared with predictions obtained from a coupled electron-pair approximation using pseudo-natural orbitals (CEPA-PNO) [155, 156].

Calculation type	Polarizability type	Value
BSR	α_d	3.490
CEPA-PNO	α_d	3.759
BSR	α_2	0.303
CEPA-NO	α_2	0.293
BSR	α_q	12.24
CEPA-NO	α_q	12.69
BSR	α_o	88.43

As explained in [12], it is possible to combine the results for a number of reduced matrix elements, which also enter the calculation of the above polarizabilities, to generate long-range dispersion coefficients for molecules such as HF and F₂. A few results are listed in table 9.

Table 9. Dispersion coefficients for the HF and F₂ ground states. The results for the reduced matrix elements were slightly rescaled in order to fulfil the appropriate sum rules for the oscillator strengths. For HF, we also include the results of Zemke *et al* [157]. The index γ is positive (negative) when the wavefunction stays invariant (changes sign) upon the interchange of the magnetic quantum numbers between the two fluorine atoms.

System	γ	C_5 (au)	C_6 (au)	C_8 (au)	C_{10} (au)
HF					
Σ			6.589	123.5	2977
Σ [157])			7.766	145.25	3455
Π			7.369	138.2	3396
F ₂					
Σ_1	-1	0.0	9.836	170.0	3652
Σ_2	+1	0.0	9.250	160.3	3406
Σ_3	+1	-3.372	8.860	150.7	3135
Σ_{23}	-1	0.0	0.890	16.00	429.4
Π_1	+1	2.248	8.827	153.1	3.209
Π_2	-1	0.0	8.932	154.1	3.232
Δ	+1	-0.562	9.889	171.1	3.687

3.2. Electron collisions

Most likely, the BSR approach has had its biggest impact to date in the description of electron collisions, in particular for complex, open-shell and heavy targets. Given the well-known strength of the *R*-matrix approach in efficiently generating results for many collision energies once the diagonalization step for the inner region has been completed, it is not surprising that the first applications concentrated on the low-energy near-threshold regime, which is often dominated by a wealth of resonance structure. A few examples will be shown in the next subsection.

Following in the footsteps of the Belfast *R*-matrix work, the method has already been used to generate a large amount of atomic data for plasma applications. The validation of such data, of course, is a crucial aspect in determining whether or not they can reliably be used in the modelling. We will discuss one case and the lessons that we learned in that study. In fact, they led to a significant further development of the method, namely, the BSR with the pseudo-states approach. We will see that the inclusion of a large number of pseudo-states can first of all improve the results obtained for electron-induced transitions between discrete bound target states, especially in the ‘intermediate energy regime’ from about the ionization threshold to a few times that energy. Secondly, however, it is then possible to generate accurate results for ionization processes. Due to the generality of the BSR suite of computer codes, we are not restricted to quasi-one-electron and quasi-two-electron targets and effectively a single active target electron. Hence, in addition to direct ionization of He(1s²) with He⁺(1s) as the residual ion, we will show an example for the highly correlated four-body Coulomb problem of ionization with simultaneous excitation of He(1s²) to He⁺(2s,2p), as well as an application to more complex targets such as neon.

Finally, we demonstrate how BSR calculations have been used to predict more than just total and angle-DCSs. Specifically, we will show results for angle-integrated and angle-differential Stokes parameters that characterize the

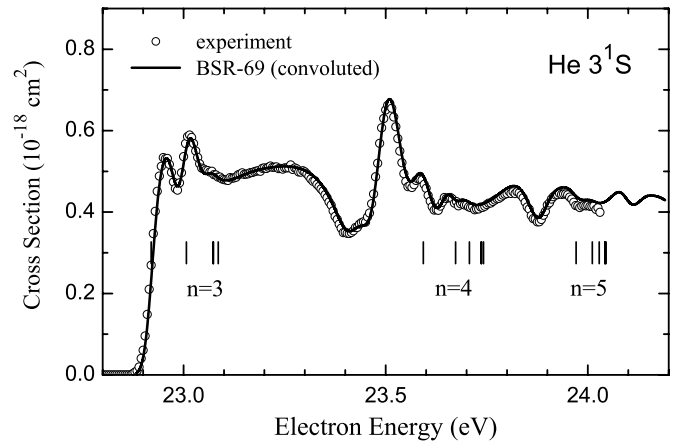


Figure 2. Angle-ICSs for the electron-impact excitation of the 3¹S state in He. The theoretical results were convoluted with a Gaussian of 37 meV (FWHM). The vertical bars represent the thresholds for the He target states. (Adapted from [20].)

polarization properties of the light emitted after electron-impact excitation, and for the spin-asymmetry function that determines the left-right asymmetry in the DCS if the incident electron beam is spin-polarized.

3.2.1. Near-threshold resonances. As our first example, angle-integrated cross sections (ICSs) for the electron-impact excitation of the 3¹S states of helium are shown in figure 2 [20]. For a direct comparison with the experimental data, the 69-state *B*-spline *R*-matrix (BSR-69) results, originally generated on a very fine energy mesh with a stepsize of 10⁻⁵ Ry, were convoluted with a Gaussian function of width 37 meV (FWHM), corresponding to the energy resolution in the experiment. The experimental data were normalized to the BSR-69 results at 23.20 eV, because the cross sections exhibit only a smooth energy dependence around this energy and hence the energy resolution does not play a role. Also, the cascade contribution was found to be negligible (less than 1%) for the 3¹S state while the effects of cascades needed to be included for the 3³S state [20].

The convoluted BSR-69 cross sections exhibit a remarkably good agreement with experiment, indicating that both the relative resonance contributions and their energies and shapes are well predicted by theory. All sophisticated theories, such as the BSR-69 as well as the RMPS [158] and RM29 [159] models, actually predict a much more complex and richer resonance structure than what would be expected from an inspection of the resonance features in figure 2. The features in the experimental cross sections are the combined result of the many resonances in the excitation cross sections. To illustrate the situation further, figure 3 exhibits the theoretical excitation cross sections without convolution with the experimental energy resolution. The resonances were analysed in detail in [20].

Moving on to a more complex system, figure 4 compares experimental data for the excitation of metastable states in noble gases [160] with predictions from earlier standard BP *R*-matrix calculations and the semirelativistic BSR results. The dashed and dash-dotted curves represent the most extensive

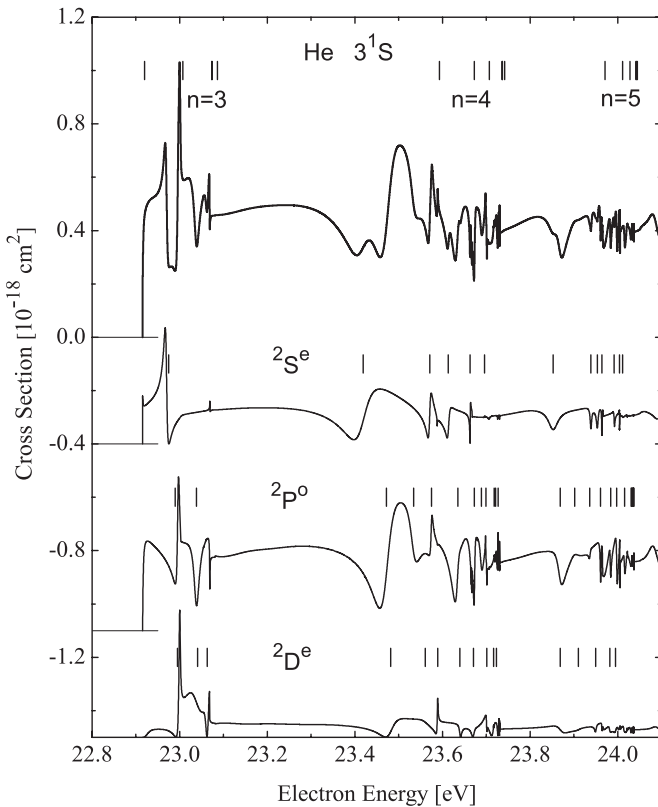


Figure 3. Angle-ICSs for the electron-impact excitation of the 3^1S state in He. The thick solid line at the top exhibits the predictions from the BSR-69 model summed over all partial waves, while the remaining thin solid lines (offset to increase the visibility) show the most important individual partial-wave contributions. The vertical bars represent the thresholds for the He target states at the top and prominent He^- resonances found for the partial waves.

and accurate results obtained with the standard *R*-matrix code [161–164], and there is at best qualitative agreement between theory and experiment. As seen from the figure, we obtain essentially perfect agreement with the shape of the experimental data for Ne as a function of energy, and we notice major improvements for all cases in comparison to previous calculations. Since we used the same scattering model (31 target states in the close-coupling expansion), the improvement is mainly due to employing more appropriate term-dependent target orbitals. For Ar we also see very satisfactory agreement with the available experimental data, with our calculation accurately reproducing all dominant resonance features.

The heavier the target, however, the more correlation we need to take into account. Not surprisingly, therefore, the agreement with experiment for Kr is still very good but no longer virtually perfect as was the case for Ne and Ar. Nevertheless, we again reproduce all the main features and achieve considerable improvement in comparison to the existing standard *R*-matrix calculations.

All the above results were obtained in the BP approximation regarding the inclusion of relativistic effects. This approximation becomes less and less accurate when we move to even heavier targets. As seen from the bottom right panel of figure 4, we still obtained significant improvement over previous work for electron scattering from Xe, but the

semirelativistic BSR model could not reproduce the resonance structure in all its details. Therefore, relativistic effects should be accounted for more accurately in this case, through a fully relativistic approach, not simply to first order as BP corrections calculated with effectively nonrelativistic orbitals.

The fully relativistic results, obtained with the DBSR code, are displayed in figure 5 and compared with the BP predictions and the experimental data. We see that the fully relativistic DBSR-31 model further improves the agreement between experiment and theory, thus indicating the importance of relativistic effects in the target orbitals, which are not accounted for to a sufficient extent in a perturbative treatment of these effects with nonrelativistic wavefunctions. Finally, including even more states in the DBSR-75 model further improves the results, now yielding very good though still not yet perfect agreement with experiment. It is possible that even more states and a further improved target description are necessary. Work in this direction is planned, but it will require both substantially increased resources and algorithm development, including the parallelization of the entire DBSR complex.

As our next example, figure 6 shows the DCS for the electron impact excitation of the first four excited states in Kr as a function of energy for two fixed scattering angles. The DCS is generally considered a more sensitive test of any theoretical method. We already reported very good agreement with experiment for our BSR-31 DCS calculations in Ne [32] and Ar [41], and recently we also compared our BSR-47 and DBSR-31 calculations for Kr [55] with early measurements by Phillips [165], who reported data for the excitation of the four levels in the $\text{Kr}(4p^55s)$ configuration at three angles (30° , 55° , 90°) from threshold up to about 14 eV. Our calculations showed encouraging but only qualitative agreement with respect to the position and energy dependence of the principal features up to incident energies of about 12.5 eV, and there was significant disagreement between the absolute values of [165] and the BSR predictions.

Fortunately, a new opportunity to test the DBSR program for such heavy targets as Kr opened up through recent measurements performed by Allan and published jointly with the BSR predictions in [56]. As seen from figure 6, our latest calculations in the even more extended DBSR-69 model yield very close agreement with Allan's experimental data, reproducing all principal resonances and other structures. Note that the experimental data are given on an absolute scale (independently normalized through comparison with He), and this aspect is very important for comparison with theory.

The close agreement between our predicted DCS and these latest measurements [56] gave us confidence in the accuracy and reliability of the DBSR calculations. The remaining discrepancies mostly occur for the very low-lying resonances around 10 eV, i.e., just above threshold. Such resonances can be very difficult for theory due to a sometimes slow convergence of the results with the number of states included in the close-coupling expansion. Also, their description requires a very good balance between correlation included in the *N*-electron target and the (*N*+1)-electron scattering systems. Finally, it should be noted that experiments

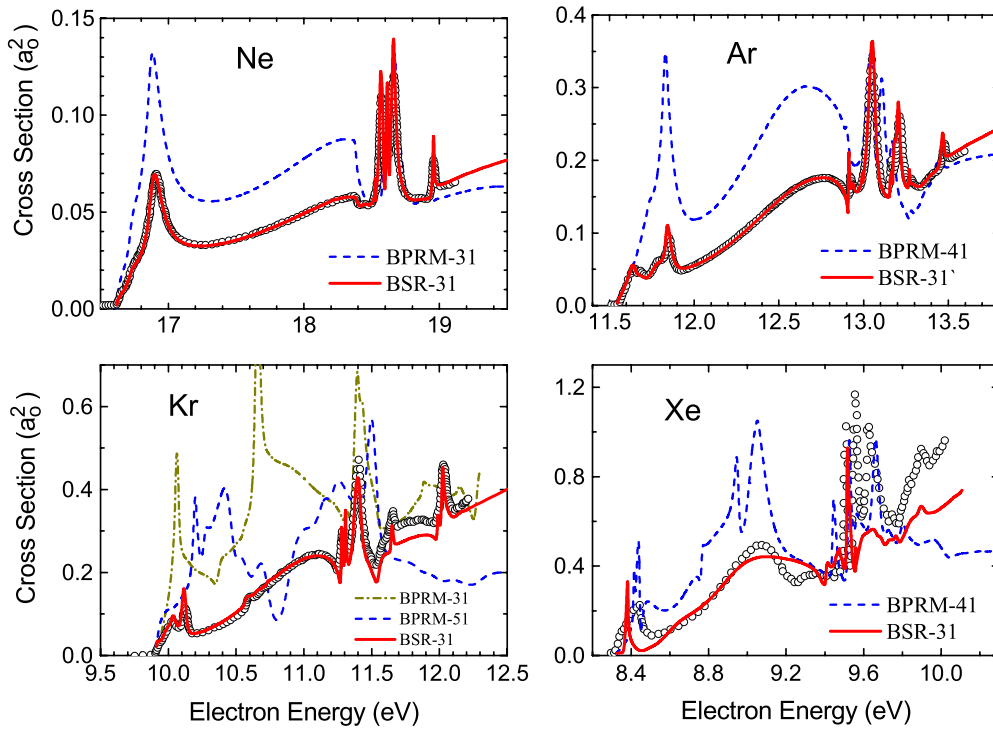


Figure 4. Cross sections for metastable production in the noble gases by electron impact. The BP *R*-matrix results [161–164], obtained with the standard Belfast code are shown with dashed and dash–dotted lines, while our results, obtained with the BSR code, are represented by solid lines. The circles are the experimental data of [160], which were given on an absolute scale for Ne, Ar and Kr, and on a relative scale for Xe. The curves shown here were all visually (re)normalized to the BSR-31 results.

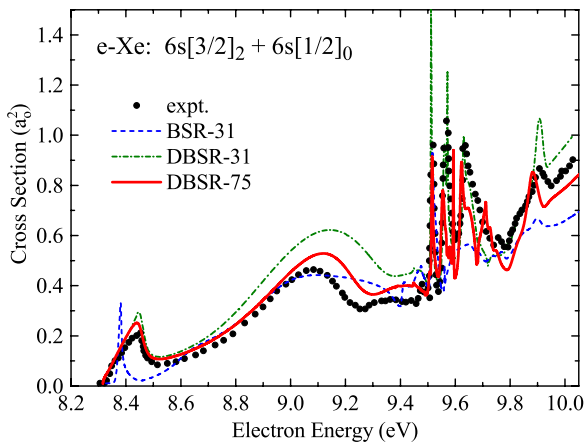


Figure 5. Metastable electron-impact excitation function of the $5p^5 6s(J = 0, 2)$ states in Xe. We compare the experimental data of Buckman *et al* [160] with the BSR-31, DBSR-31 and DBSR-75 results. The relative experimental data were visually normalized to the DBSR-75 results. The BSR and DBSR predictions shown here include cascade contributions from all higher lying states included in the respective models.

also become difficult in this regime, due to the detector response function that often varies considerably with energy for scattered electrons of just a few tens of eV.

Our next example concerns a series of studies for the electron-impact excitation of the lowest autoionizing states of the alkali atoms Na, K, Rb and Cs. These resonances, of the general configuration $(np^5[n+1]s^2)^2P_{3/2,1/2}$, are much more difficult to describe than many other observables

involving only the outer valence electron. Figure 7 compares the measured and calculated cross sections for the $(4p^5 5s^2)^2P_{3/2,1/2}$ states in potassium [58]. The relative experimental data were first cross-normalized to each other and then absolutely normalized to the theoretical predictions by visual fit to the $(4p^5 5s^2)^2P_{3/2}$ excitation function in the nonresonant energy region above 17 eV. As seen from figure 7, the BSR theory accurately reproduces the measured relative intensities of the $(4p^5 5s^2)^2P_J$ lines, in addition to yielding very good agreement in the shape of the excitation functions. Note that the ratio of about 4 at the peaks differs considerably from the statistical ratio of 2 found outside of the resonance region. The dominant maxima in the $(4p^5 5s^2)^2P_J$ excitation functions represent contributions from many individual negative-ion resonances. A detailed analysis can be found in [58].

Another target of interest for various applications is gold. A few representative results for the electron-impact excitation of the $(5d^{10} 6s)^2S_{1/2} \rightarrow (5d^{10} 6p)^2P_{3/2}$ and $(5d^{10} 6s)^2S_{1/2} \rightarrow (5d^9 6s^2)^2D_{3/2}$ transitions are shown in figures 8 and 9, respectively. For the $(5d^{10} 6p)^2P_{3/2}$ state, the experimental data are compared with predictions from relativistic distorted-wave (RDW), relativistic convergent close-coupling (RCCC) and DBSR models. We note overall good agreement between experiment and the various theoretical results. Even the first-order plane-wave Born approximation (FBA), calculated with the target descriptions used in the RCCC model, does quite well, with the typical problem of overestimating the cross section near the maximum. For energies of 20 eV and higher, the RDW (without cascade) and the DBSR results (with cascade) are very similar. The RCCC results, on the other hand,

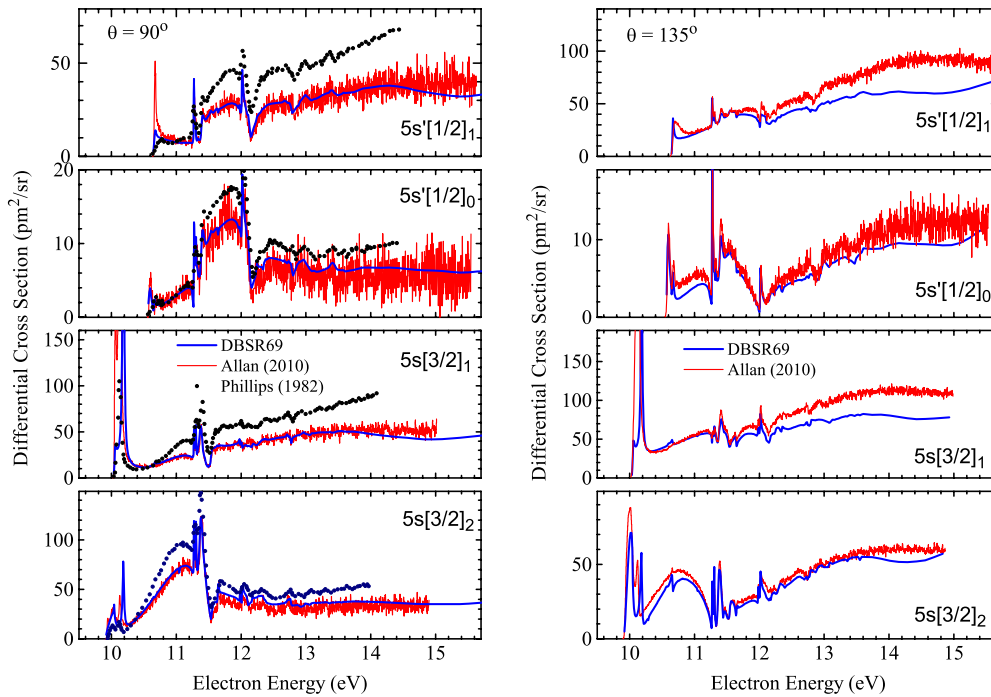


Figure 6. DCS for the electron-impact excitation of the individual states in the $4p^5 5s$ manifold of Kr as a function of the incident energy and fixed scattering angles of 90° and 135° . The experimental datasets are from Phillips [165] and Allan [56].

are generally lower than those from the other models. Looking at the RCCC with and without cascades suggests a maximum cascade contribution of about 10% near 20 eV. A similar size of the cascade effect was seen in the corresponding DBSR results (not shown). In addition, however, the DBSR method allows for an estimate of the cascade *loss* in the observed photon signal to the ground state due to transitions to the $(5d^9 6s^2)^2D_{5/2,3/2}$ levels.

Significantly more challenging than the excitation of the $(5d^{10} 6s)^2S_{1/2} \rightarrow (5d^{10} 6p)^2P_{3/2}$ optically allowed resonance transition is the optically forbidden transition $(5d^{10} 6s)^2S_{1/2} \rightarrow (5d^9 6s^2)^2D_{3/2}$. Only DBSR calculations are available for this case, due to the complexity associated with the open 5d subshell. As seen in figure 9, several DBSR models with a varying number of states were all unable to reproduce the large near-threshold maximum in this case. It is possible that the close-coupling expansion converges very slowly in this case, and thus new strategies to improve the calculations need to be designed. This will be further discussed in subsequent sections below. Nevertheless, the agreement between experiment and theory was judged sufficiently good to put the experimental data on an absolute scale by normalizing in the nonresonant regime. Albeit with an uncertainty of about 50%, the joint experimental and theoretical study resulted in a set of recommended *absolute* cross sections for this transition [64]. Note that these numbers are crucial for the understanding of the gold–vapour laser, since the $(5d^9 6s^2)^2D_{3/2}$ state of gold serves as the lower level for the 312.3 nm line.

Our final example in the area of near-threshold resonances is for FeII, a very complex open-shell target of significant importance for the interpretation of astrophysical observations. Table 10 shows the improvement of the BSR structure

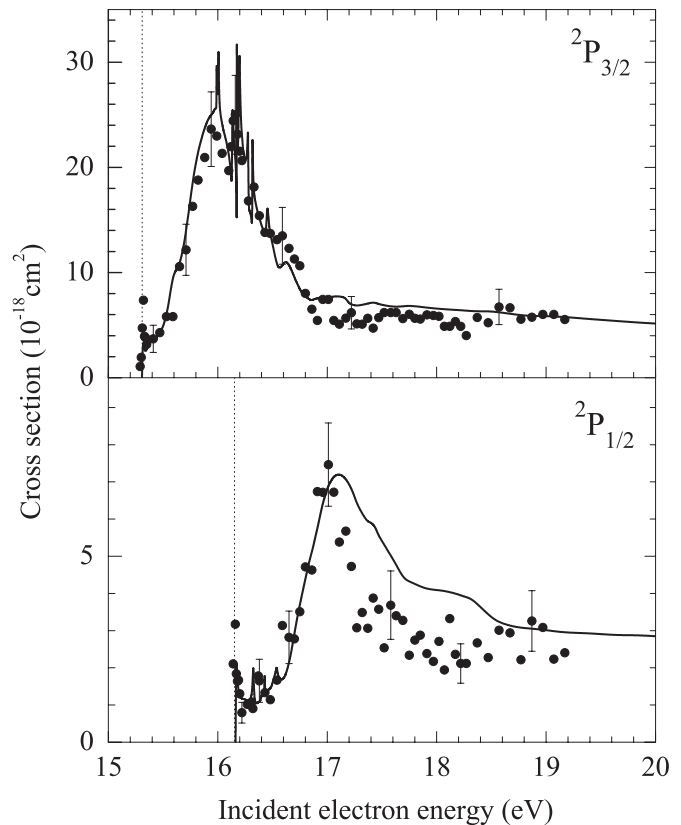


Figure 7. Experimental ejected-electron excitation functions (solid circles) and calculated angle-ICs (solid line) for the electron-impact excitation of the $(4p^5 5s^2)^2P_{3/2,1/2}$ states in rubidium. The experimental data were visually normalized to the BSR results for the $^2P_{3/2}$ state in the nonresonant regime above 17 eV.

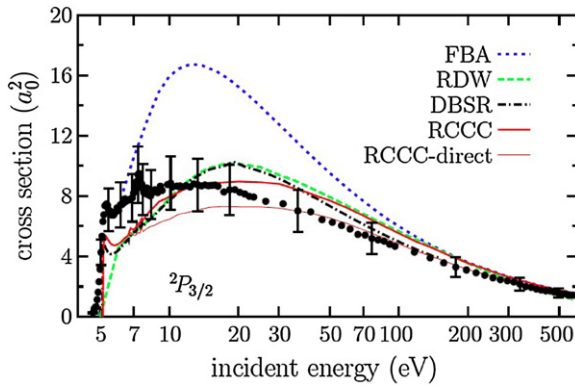


Figure 8. Cross section for electron-impact excitation of the $(5d^{10}6p)^2P_{3/2}$ state in gold. The present experimental data are compared with results obtained with RDW, RCCC, and DBSR models (see text). The curves labelled ‘RDW’ and ‘RCCC-direct’ show the direct cross sections, whereas all the other curves include cascade effects. Also shown are the results from a first-order plane-wave Born approximation (FBA) calculated with the target descriptions used in the RCCC model. (Adapted from [63].)

Table 10. Energy level separation (in 10^{-4} Ry) for the lowest 13 states of Fe^+ , and the difference from experiment in two calculations [14, 166].

State	NIST	Diff. [166]	Diff. BSR [14]
$3d^6(^5D)4s^6D$	0	0	0
$3d^7a^4F$	182	−53	16
$3d^6(^5D)4sa^4D$	720	5	−19
$3d^7a^4P$	1203	2	49
$3d^6(^3P)4sb^4P$	1914	106	−46
$3d^6(^3H)4sa^4H$	1918	411	8
$3d^6(^3F)4sb^4F$	2040	242	−2
$3d^54s^2a^6S$	2087	394	1
$3d^6(^3G)4sa^4G$	2310	387	1
$3d^6(^3D)4sb^4D$	2825	353	−23
$3d^6(^5D)4pz^6D^o$	3490	−229	−82
$3d^6(^5D)4pz^6F^o$	3805	−187	−82
$3d^6(^5D)4pz^6P^o$	3886	−162	−2

calculations over a standard *R*-matrix approach carried out at approximately the same time. Figure 10 compares two partial-wave contributions to the collision strength for the $a^6D \rightarrow a^4F$ transition. There is overall qualitative agreement between the BSR results [14] and those of [166], but we see different positions, heights and shapes of the resonances. Having two entirely independent calculations available is crucial to validate the predictions, in particular when no experimental data are available. For the effective collision strengths, which are obtained by performing the appropriate thermal averaging, the overall agreement between the predictions from the two calculations was satisfactory and indeed provided confidence in using these results in modelling applications.

3.2.2. Validation through a neon discharge. As seen in the previous subsections, the validation of individual theoretical predictions for energy levels, oscillator strengths and cross sections for state-to-state transitions by comparing with experimental benchmark data provides a detailed test of a

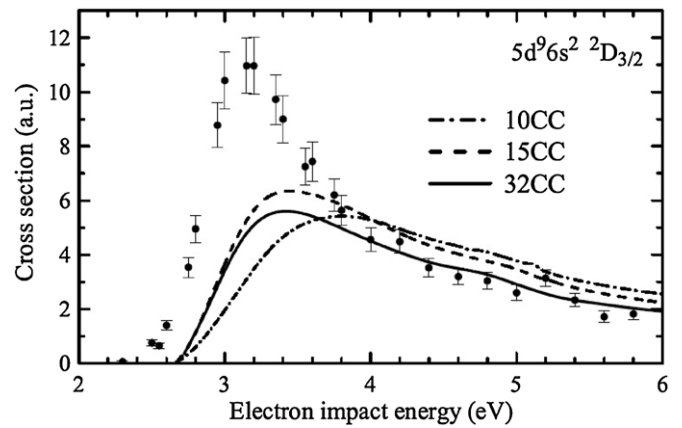


Figure 9. Cross section for the electron-impact excitation of the $(5d^9 6s^2)^2D_{3/2}$ state in gold. The experimental data were visually normalized to the 32 CC predictions in the energy range 4–6 eV. (Adapted from [64].)

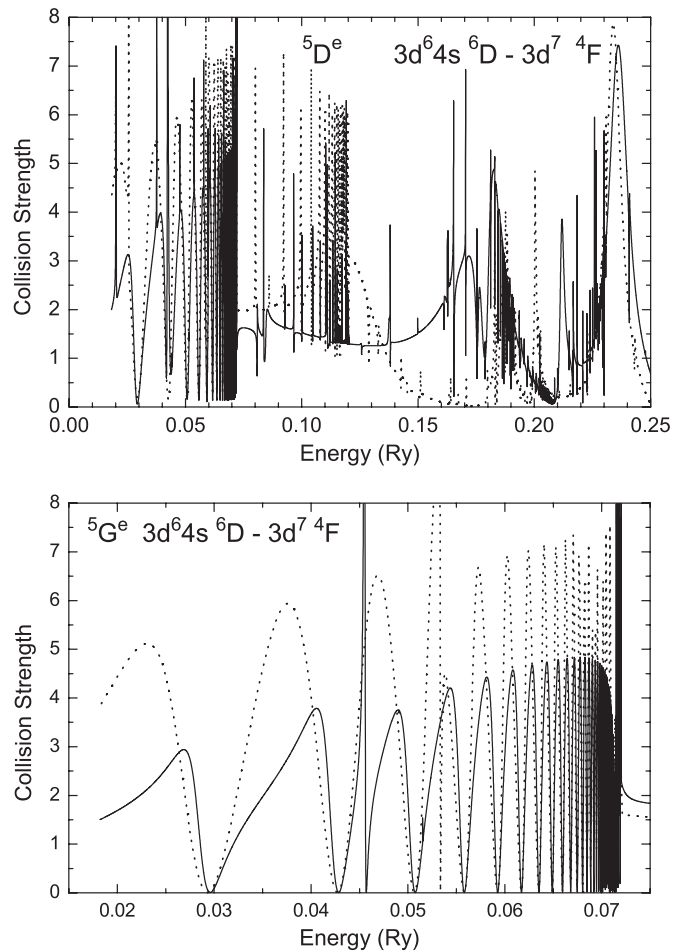


Figure 10. Selected partial-wave contributions to the collision strength for the $a^6D \rightarrow a^4F$ forbidden transition in Fe^+ . —, present results; ····, [166].

theoretical model. Such tests are limited, however, simply by the availability of suitable data. Furthermore, it is not immediately clear how potential inaccuracies in the individual predictions might affect the outcome of a collisional-radiative model (CRM) with a large number of parameters.

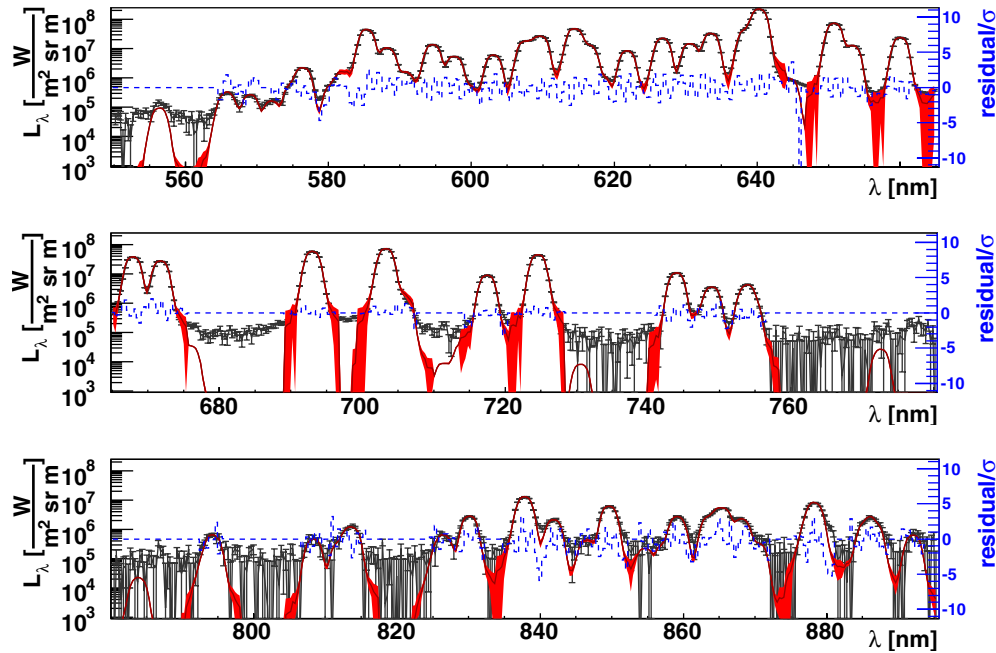


Figure 11. Result of the spectroscopic model of the discharge and the measurement. The three rows of the figure show the emission spectrum in the range of 550–900 nm. The logarithmic scale allows us to depict also the lines with low intensity. The dashed curve represents the difference between model and measurements in units of the standard deviation σ .

Figure 11 shows the result of a more global test of our predictions by comparing the results from a CRM of a Ne discharge, in which about 350 oscillator strengths and 150 rate coefficients for state-to-state transitions were used. Some of those numbers were taken from experiment, but most of them from the BP BSR-31 calculations. Other elementary processes, such as heavy-particle collisions, ionization and radiation transport had to be modelled as well. It was then assumed that the atomic input data were correct within a specified uncertainty, which was estimated based on our experience with the individual comparisons described above, and an attempt was made to model the results of a spectral measurement including many lines with these input data. Figure 11 reveals very good agreement between the model and the spectral measurement over three orders of magnitude of the emitted light intensity.

In the next step, the results were inverted and values for the input variables were extracted from the experimental data. In an ideal world, the exact same input numbers would come out again, but in practice there is an uncertainty range in the extracted results. If all ranges contain the respective input values, then the model is internally consistent. If not, however, such a finding strongly suggests a problem with the input data.

The result of the validation procedure for the Ne discharge is summarized in figure 12, which depicts the marginal distributions of the correction factors for the electronic excitation rate coefficients. The majority of the obtained correction factors show good agreement with a value of 1.0, thereby confirming that a description of the spectroscopic data can be obtained without considerable correction of the atomic input data.

Details of the validation procedure were given by Dodt *et al* [34]. Here we only note that it indeed identified

a small number of transitions, for which the BSR input predictions for the cross sections and, consequently, the rate coefficients seemed to overestimate the experimental result. These transitions were exactly those that had been identified by Ballance and Griffin [167] as possibly being affected by strong coupling to the ionization continuum, an effect that was left out in the BSR-31 calculation.

3.2.3. Extension to intermediate energies. The problem raised above was addressed in a recent extension of the BSR approach by the inclusion of a large number of pseudo-states, in order to account for coupling to the ionization continuum as well as the Rydberg spectrum of bound states. We will demonstrate the effect using two recent calculations on electron collisions with neon atoms, one performed in nonrelativistic framework [73] and one performed in the semirelativistic BP approach [74].

Figure 13 exhibits selected nonrelativistic results for the excitation of LS -coupled terms. They were chosen in order to discuss the convergence of the close-coupling expansion for e–Ne collisions. These results cannot be directly compared to experiment, since most Ne target states should be described at least in a semirelativistic intermediate-coupling scheme involving several terms. Nevertheless, we can compare our predictions with the RMPS results of Ballance and Griffin [167], who performed both a standard 61-term R -matrix calculation (RM-61) with only discrete terms included in the close-coupling expansion and a 243-term RMPS calculation (RMPS-243).

The results of these calculations for the excitation of the $(2p^53s)^3P$ and $(2p^53s)^1P$ terms are shown in the top panels of figure 13. The striking differences between the 61-term and the 243-term results clearly demonstrate the significance of

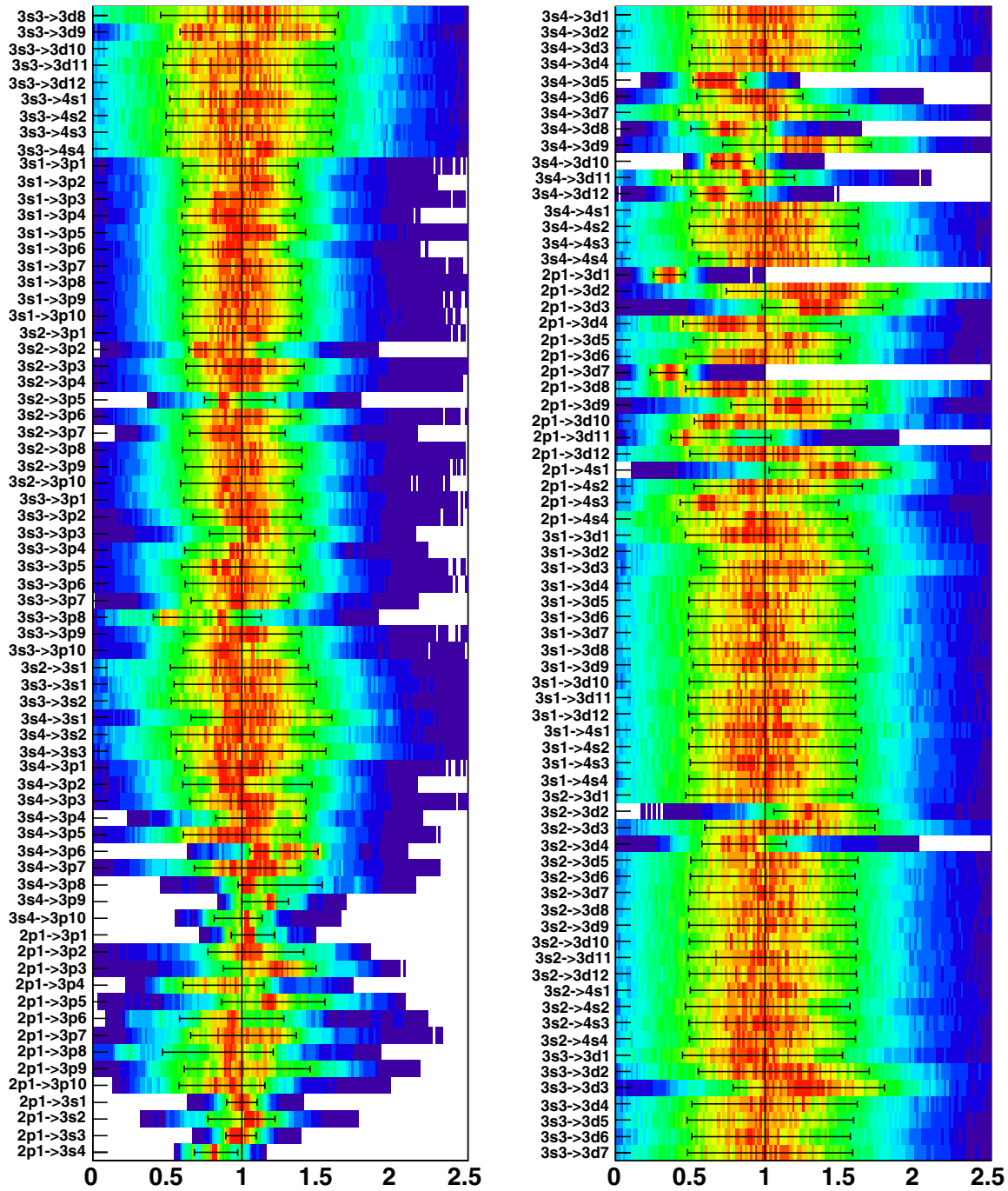


Figure 12. Marginal probability distributions for the rate-coefficient correction factors [34]. The error bars indicate one standard deviation.

continuum coupling effects. These effects have a pronounced influence on the theoretical cross sections, especially above the ionization limit, where the discrete-term-only RM-61 cross section is a factor of 2 larger than the RMPS-243 cross section at 30 eV for excitation of the $(2p^53s)^3P$ term and a factor of 1.5 larger at 32 eV for the excitation of the $(2p^53s)^1P$ term. Our even more extended BSR-679 calculation [73] produces results slightly below the RMPS-243 numbers. This may not just be due to additional channel coupling, but also due to differences in the atomic wavefunctions from the respective structure models. Overall, however, we conclude that the

pseudo-state expansion in the RMPS-243 model is already sufficiently complete to yield reliable cross sections.

The panels in the centre row of figure 13 show the corresponding comparisons for excitation from the ground state to the $(2p^53p)^3S$ and $(2p^53p)^3D$ terms, respectively. At 30 eV, continuum coupling reduces both cross sections by more than a factor of 2, and these effects persist even below the ionization limit. Our BSR-679 results suggest even smaller cross sections than the RMPS-243 model [167].

Finally, the bottom panels of figure 13 compare predictions for the excitation of the $(2p^53d)^3P$ and $(2p^53d)^1P$ terms from the ground state. The effects of continuum coupling

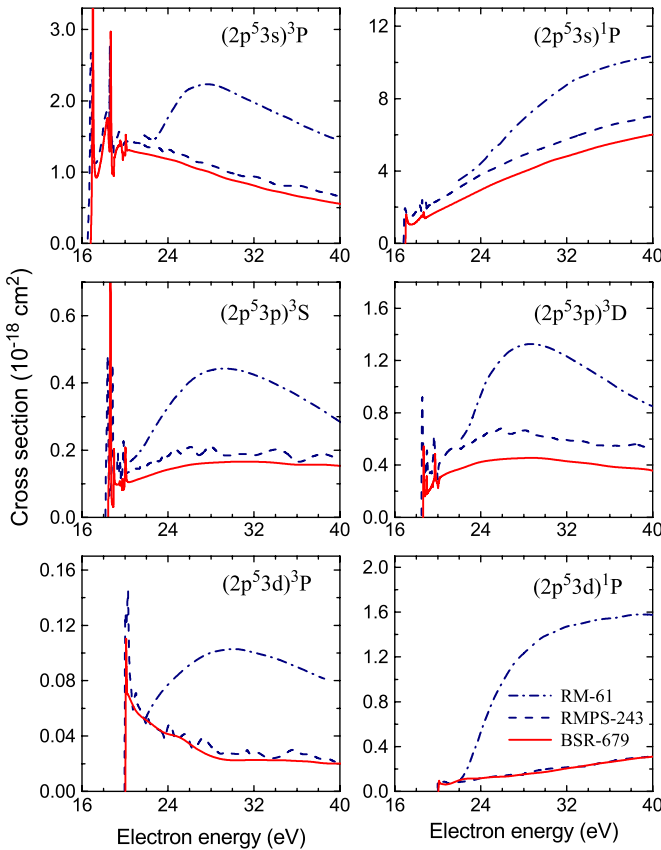


Figure 13. *LS*-coupling results for electron-impact excitation cross sections from the $(2p^6)^1S$ ground state of neon to the $(2p^5 3s)^3,1P$ (top row), $(2p^5 3p)^3S$ and $(2p^5 3p)^3D$ (centre row), and $(2p^5 3d)^3,1P$ (bottom row) terms. The BSR-679 predictions [73] are compared with the 61-state and 243-state results of Ballance and Griffin [167].

are very large for both cases. Most surprisingly, even for the relatively strong dipole-allowed $(2p^6)^1S \rightarrow (2p^5 3d)^1P$ transition, they cause a huge reduction of the calculated cross section by nearly a factor of 5 at 40 eV. The 679-state calculation completely supports the findings of Ballance and Griffin [167], and we see very close agreement between the BSR-679 and RMPS-243 results. Hence, we conclude that the remaining differences in the target wavefunctions are not very important here, and that both the RMPS and BSR expansions are sufficiently complete to describe these transitions. These findings are in line with the indirect evidence [34] mentioned above: discrete-state-only calculations such as our original BSR-31 model may significantly overestimate the cross sections for the excitation of states with dominant configuration $2p^5 3d$.

Recall that the intermediate-coupling nature of the Ne target states does not allow for a direct comparison with experimental data for the excitation of most individual states. However, the above results gave us confidence in extending the calculations to a semirelativistic BP description, and we will now present and discuss some of those results. Much more information can be found in [74].

The angle-ICS for the excitation of the four $2p^5 3s$ states are presented in figure 14 over an extended energy range of up to 300 eV. We compare the latest 457-state (BSR-457)

results [74] with those from the 31-state *B*-spline *R*-matrix calculation (BSR-31) [30] and with available experimental data. In order to demonstrate the convergence of the predicted cross sections, we also show five-state results (BSR-5), which are expected to be close to what one might obtain in a distorted-wave approximation with the same target description. The differences between the BSR-5 and BSR-31 results illustrate the effects of coupling to the higher lying bound states of Ne. This coupling considerably reduces the theoretical cross sections near the maximum and is responsible for the resonance structure at low energies. The effects of coupling to the continuum, represented by the differences between the BSR-31 and BSR-457 results, are even more significant. The pseudo-states in the expansion not only further reduce the predicted cross sections but also change their energy dependence. The discrete-state-only *R*-matrix cross sections rise rapidly just above the ionization limit, while the results obtained with the pseudo-states included do not. Although the most significant effects of continuum coupling occur above the ionization limit, they persist down into the bound-state energy region.

The agreement between the BSR results and various experimental data is rather scattered. Although we designate the $2p^5 3s$ levels in the *JK*-coupling scheme, they are still reasonably well described in the *LS*-coupling representation. This is particularly true for the metastable $3s[3/2]_2$ and $3s'[3/2]_0$ levels with dominant (nearly 100%) 3P_2 and 3P_0 character, respectively. Consequently, the excitation cross sections for these two states clearly exhibit the typical ‘exchange’ character, dropping quickly as $1/E^3$ at high energies. In contrast to the previous calculations, our BSR-457 results lie below the experimental values. The closest agreement is with the early measurements of Phillips *et al* [170], which were based on the optical method employing the laser-induced fluorescence technique. The principal uncertainty in such measurements originates from extracting the cascade contributions from the measured apparent excitation cross sections. Register *et al* [169] and Khakoo *et al* [168], on the other hand, measured DCS at fixed energies in crossed-beam setups. These DCSs were extrapolated to 0° and 180° and then integrated to yield absolute excitation cross sections as a function of incident electron energy. Uncertainties in this method necessarily arise from the unknown behaviour of the DCS at small and large scattering angles, especially if somewhat questionable theoretical predictions are used as a guide. These uncertainties are almost certainly responsible for the scatter in the energy dependence of the experimental ICSs. We will further comment on the Khakoo *et al* [168] data below.

As seen from figure 14, coupling to the continuum is even important for the strong dipole-allowed transition to the $3s'[1/2]_1(^1P_1)$ level, and its influence persists over a large range of electron energies. Here the experimental cross sections appear to be in good agreement with the BSR-31 predictions and hence exceed the new BSR-457 results by 20–40% at intermediate energies from 30 eV to 150 eV. New measurements, however, are currently in progress at the Sophia University [172], and preliminary results suggest smaller cross sections at 100 and 200 eV than the data shown in figure 14.

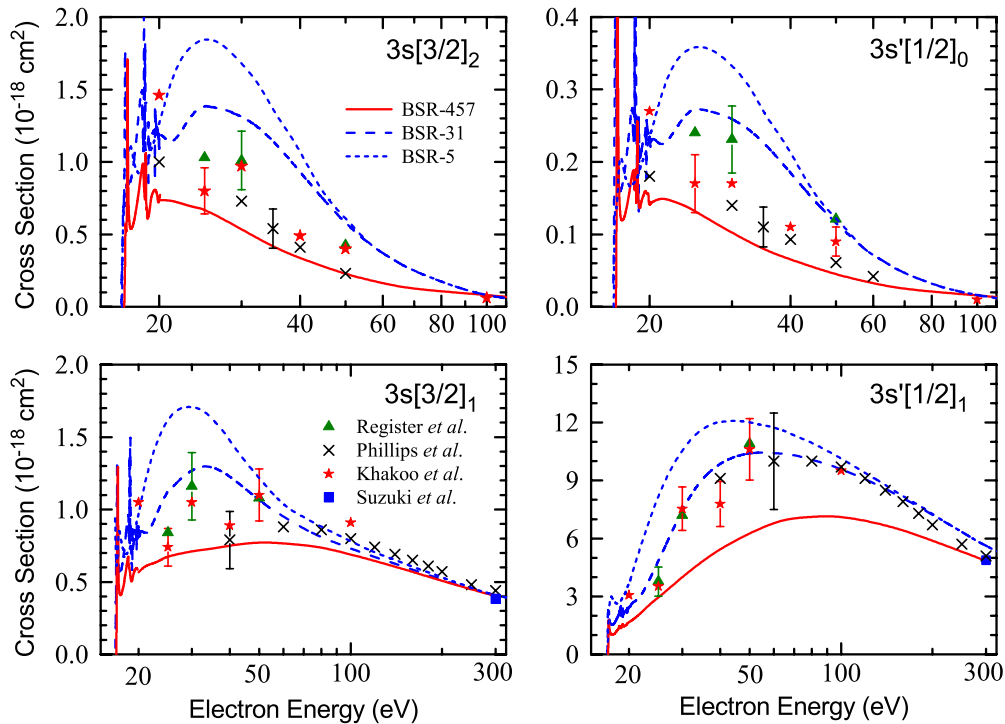


Figure 14. Angle-ICSs for the electron-impact excitation of the $2p^5 3s$ states in neon from the ground state $(2p^6)^1S_0$. The BSR-457 results [74] are compared with those from the BSR-5 and BSR-31 models, as well as the experimental data from Khakoo *et al* [168], Register *et al* [169], Phillips *et al* [170] and Suzuki *et al* [171].

All scattering models agree with the experimental datum of Suzuki *et al* [171] at 300 eV, where the coupling effects are expected to be small.

The best agreement with the experimental data is found for the excitation of the $3s[3/2]_1$ (3P_1) level. The cross section for this level is affected by a small mixing with the $3s'[1/2]_1$ (1P_1) level. At low energies, therefore, its variation with energy resembles that of a spin-forbidden transition, but then it transforms to the characteristics of a dipole-allowed transition at higher energies. For energies $E \geq 100$ eV, all scattering models provide very similar results and agree well with the available experimental data. At lower energies, we see good agreement between the BSR-457 results and experiment at 25 and 40 eV, while the experimental data of [168] and [169] at 30 and 50 eV are significantly larger than our predictions. On the other hand, there is no obvious physical reason for the scatter in these measured excitation cross sections at such high energies, and hence some of the experimental data are much further off from theory than others.

Figure 15 presents results for the angle-DCSs for the electron-impact excitation of the $2p^5 3s$ states in neon for an incident projectile energy of 30 eV. The BSR-457 results are compared with the experimental data of Khakoo *et al* [168], and also with our previous BSR-31 results [30]. Note that the experimental data were renormalized to yield good visual agreement with the BSR-457 results. The corresponding normalization coefficients indicated in the figures are the same for all $2p^5 3s$ states at a given scattering energy. In contrast to the BSR-31 predictions, we now see excellent agreement between the BSR-457 results and the experimental DCS as a function of the scattering angle. The new calculations

correctly reproduce all the minima and maxima for both the spin-forbidden and the dipole-allowed transitions. There is a striking improvement over the BSR-31 results, once again stressing the importance of coupling to the continuum. As expected, the latter is particularly important for the excitation of the metastable $3s[3/2]_2$ and $3s'[1/2]_0$ states. As discussed in detail by Khakoo *et al* [168], no previous calculation could reproduce the measured angular dependence of the DCS for the metastable states. The new results clearly confirm our earlier suggestion [30] that the results for these optically forbidden transitions are very sensitive to the inclusion of pseudo-states.

3.2.4. Ionization. The development of the BSRMPS approach not only improves the results for excitation processes, in particular at intermediate energies, but it also allows for the calculation of ionization processes. As described above, this is generally done in two steps: first we obtain scattering amplitudes for the excitation of all pseudo-states, and then we use a projection technique to map those amplitudes onto the true continuum for the ejected electron. In the case of total and single-differential (with respect to energy loss) cross sections, short-cuts can be taken by adding up the total cross sections for all states above the ionization threshold, or by projecting results for the individual cross sections rather than the amplitudes.

To illustrate the potential of the BSRMPS method, we first look at the frequently studied case of electron-impact ionization of $\text{He}(1s^2)^1S$ leaving the residual ion in the $\text{He}^+(1s)^2S$ ground state. Except for the different $1s$ orbitals in these two states (their overlap is still about 0.97), this is reducing the four-body Coulomb problem to an effective

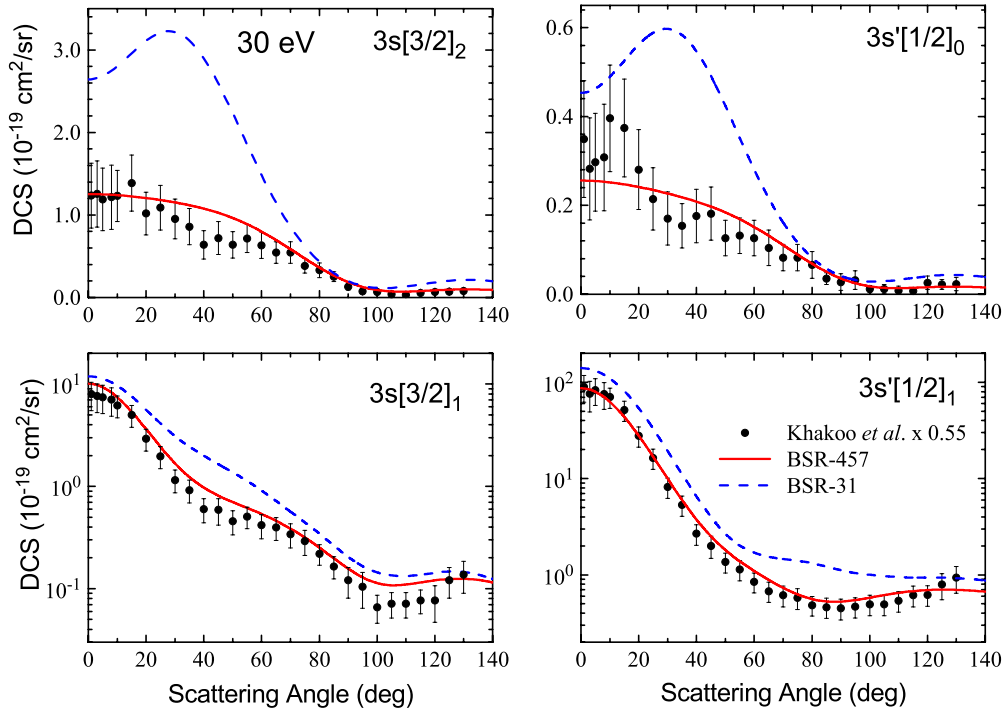


Figure 15. Angle-DCSs for the electron-impact excitation of the $2p^5 3s$ states in neon from the ground state $(2p^6)^1S_0$ at an incident projectile energy of 30 eV. The BSR-457 results [74] are compared with the BSR-31 predictions [30] and the experimental data of Khakoo *et al* [168]. The experimental data were renormalized by a factor of 0.55.

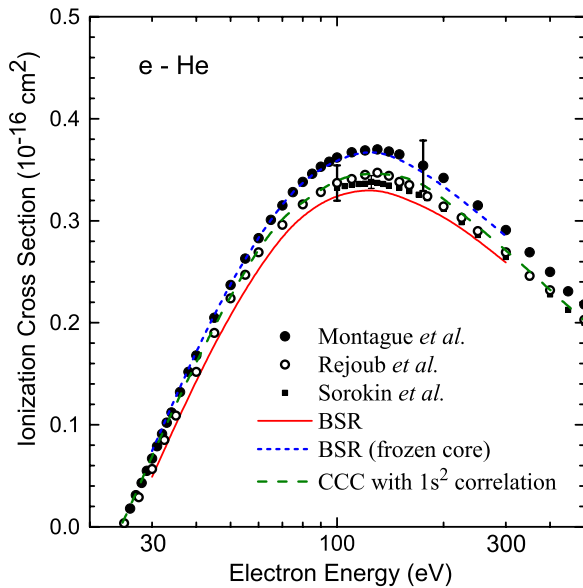


Figure 16. Total cross section for the electron-impact ionization of helium in its $(1s^2)^1S$ ground state as a function of the incident projectile energy. The experimental data of Montague *et al* [173], Rejoub *et al* [174] and Sorokin *et al* [175] are compared with BSR and CCC predictions with and without correlation terms in the description of the ground state. (Reproduced with permission from [71]. © 2012 by The American Physical Society.)

three-body case, since the remaining bound electron essentially remains a spectator.

We begin with the total ionization cross section depicted in figure 16. We show three sets of experimental data as well as predictions from three different theoretical models.

For a long time, the data obtained by Montague *et al* [173] had been considered the benchmark to check theory against. More recently, however, both Rejoub *et al* [174] and Sorokin *et al* [175] obtained significantly lower results. While large pseudo-state models that treat the initial state in the frozen-core approximation (including our 500-state BSR model with a frozen-core ground state) and several CCC calculations (presented in [176]) produce nearly perfect agreement with the Montague *et al* data, improving the description of the initial state by adding correlation terms (with $2\bar{p}^2$ being the most important one) drops the theoretical predictions by 10–15%, in agreement with the more recent experimental data. Since our BSR with the correlation model includes even more configurations than the CCC with the correlation approach, it is not surprising that the final BSR results are even lower than the improved CCC numbers. Both correlated datasets agree with the most recent data within the published uncertainties of the experiment. It should be noted that the frozen-core results are for ionization with $\text{He}^+(1s)$ as the only possible final state, while the correlated model also contains ionization with excitation to $\text{He}^+(2s)$ and $\text{He}^+(2p)$. Nevertheless, the former are larger than the latter owing to the different description of the initial state.

Next, we show the single DCS as a function of the energy loss in figure 17. Here the experimental data of Müller-Fiedler *et al* [177] are compared with our BSR predictions in the frozen-core model. The latter were obtained by either interpolation of the transition-matrix elements or by the direct projection approach. A thorough inspection of the figure reveals that the interpolation procedure, indeed, produces smoother results than the projection. The unphysical oscillations, however, deviate by less than 2% from the smooth

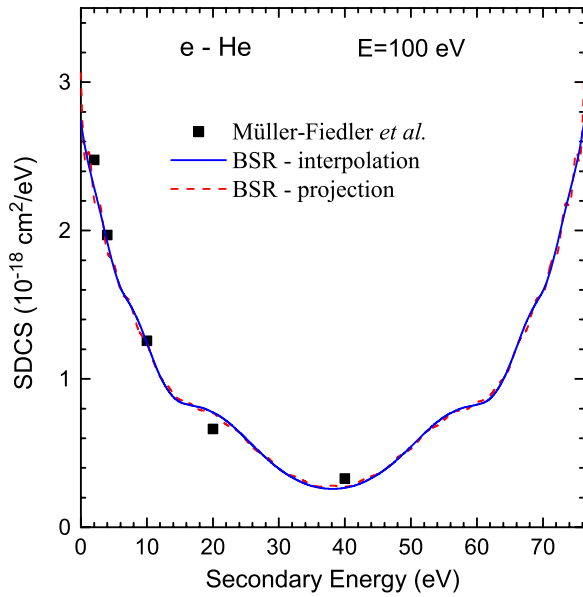


Figure 17. Single DCS for the electron-impact ionization of helium in its $(1s^2)^1S$ ground state for an incident projectile energy of 100 eV. The experimental data of Müller-Fiedler *et al* [177] (obtained by integrating their double-DCSs) are compared with BSR predictions (in the frozen-core model) generated by interpolation and direct projection. (Reproduced with permission from [71]. © 2012 by The American Physical Society.)

results and would not be detectable with current experimental techniques. As mentioned above, the projection approach is more generally applicable than the interpolation method. Hence, we use this example to illustrate the degree of validity.

The double-DCS as a function of the detection angle of one of the outgoing electrons with fixed energy is depicted in figure 18. There is excellent agreement between our BSR results and the experimental data of Müller-Fiedler *et al* [177] and also the CCC predictions [130]. For small energies of the detected electron (2 eV, 4 eV and 10 eV), the DDCS peaks in the backward direction, while it is the other way around for large detection energies. This is consistent with the expectation that the faster of the outgoing electron can be considered the ‘projectile’, which is not deflected very much. Consequently, the DDCS near the backward direction falls off fast with increasing detection angle of the faster of the two outgoing electrons. This poses significant challenges to both experiment and theory, resulting in limited experimental data and, particularly for 55 eV, in differences of up to a factor of 2 between the CCC and BSR predictions for large angles. Differences also occur for the case of near-symmetric energy sharing between the two electrons (35 eV), but once again in an angular regime where there are no experimental data for comparison. Finally, for a detection energy of 10 eV, the BSR predictions agree better with experiment than those from the CCC calculation at angles less than 60°, while it is the other way around for angles above 120°.

Figure 19 shows our results compared with the recent experimental benchmark data of Ren *et al* [178] for electron-impact ionization of helium in its $(1s^2)^1S$ ground with the residual ion left in the $\text{He}^+(1s)^2S$ state. The primary energy E_0 is 70.6 eV and the two final-state electrons have

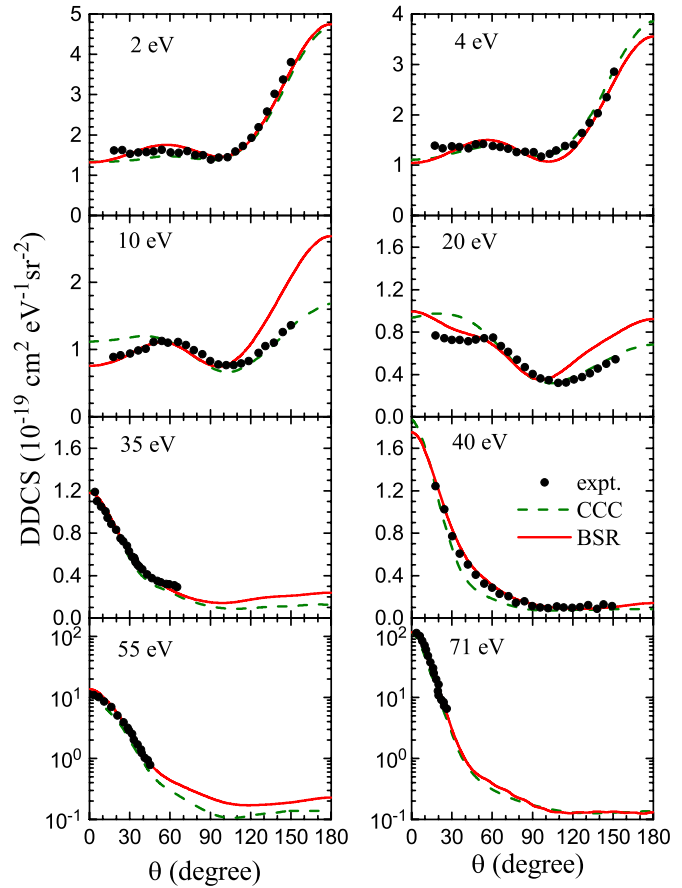


Figure 18. Double-DCS for the electron-impact ionization of helium in its $(1s^2)^1S$ ground state for an incident projectile energy of 100 eV and a fixed energy of the detected electron as the function of its detection angle. The experimental data of Müller-Fiedler *et al* [177] are compared with BSR and CCC [130] predictions. (Reproduced with permission from [71]. © 2012 by The American Physical Society.)

energies of 41.0 eV and 5.0 eV, respectively. These are examples of asymmetric energy sharing between the two outgoing electrons. The momentum vector of the incident electron and the faster outgoing electron with angle θ_1 form the scattering plane. Results are then presented for cases where the other electron with energy E_2 is detected at a variable angle θ_2 either in the same plane (the co-planar arrangement) or in the ‘perpendicular plane’ that also contains the incident momentum vector. Overall, we see excellent agreement with experiment as well as the CCC and TDCC predictions presented by Ren *et al* [178]. Interestingly, the largest differences between the theoretical predictions occur near a detection angle of 180°, where there are no experimental data available to decide between the three theories.

The next example is shown in figure 20, where we compare the experimental data of Nixon *et al* [179] to our BSR results. The data are for the doubly-symmetric perpendicular geometry (see figure 1 of [179] for details), where the momentum vectors of both final-state electrons lie in the plane perpendicular to the incident beam direction, both outgoing electrons have the same energy, and the angle between the two final-state momenta is varied. Since the experimental data are relative and not cross-normalized either, we compare the

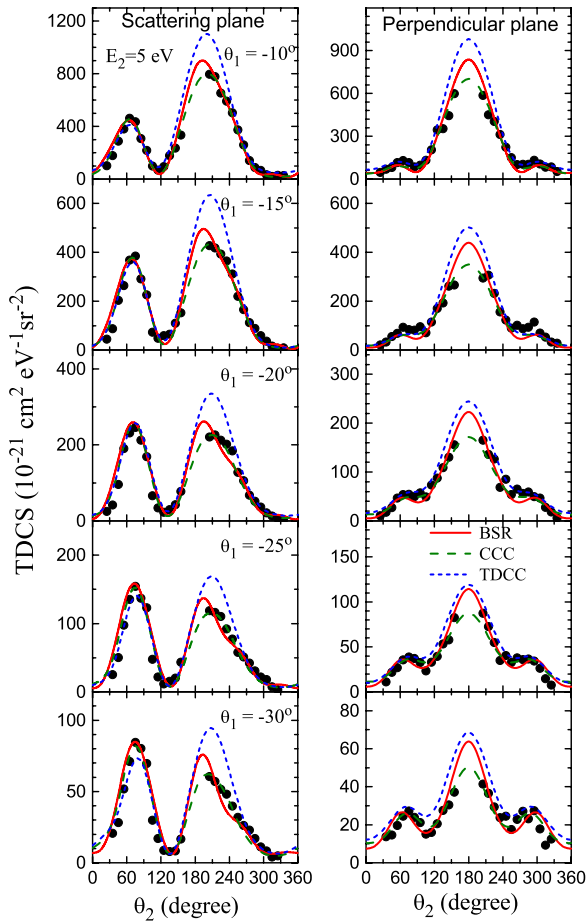


Figure 19. TDCS for the electron-impact ionization of helium in its $(1s^2)^1S$ ground with the residual ion left in the $\text{He}^+(1s)^2S$ state. The primary energy E_0 is 70.6 eV, and the final-state electrons have energies of 41.0 eV and 5.0 eV, respectively. The faster of the two electrons is detected at a fixed angle θ_1 indicated for each panel, while the detection angle θ_2 for the slower electron with energy E_2 is varied. The minus sign indicates the frequently used convention that the angles of the two electrons are counted clockwise for one and counterclockwise for the other. The experimental data for the co-planar and perpendicular geometries, as well as the CCC and TDCC predictions, are from Ren *et al* [178].

relative angular dependence. Here we visually normalize the experimental data to the absolute BSR numbers.

The overall agreement between experiment and the BSR results is very good, certainly regarding the shape. However, as discussed by Bray *et al* [90], the situation regarding absolute normalization is not so clear. The results for 180° in this geometry also appear in the coplanar geometry and hence, in principle, could be normalized to absolute data in that plane. Only a few other theoretical predictions are available (CCC at 10 eV [133], both CCC and TDCC at 20 eV [180]). For 20 eV in particular, there is a significant difference in the magnitudes predicted by CCC/TDCC and BSR, while the agreement between BSR and TDCC is better, though not perfect, for 10 eV. Clearly, both CCC and RMPS yield good agreement with the measured angular dependence, while the centre peak of the TDCC results for 20 eV seems too low relatively to the side peaks. We do not have an explanation for the discrepancies in magnitude at the present time. We note,

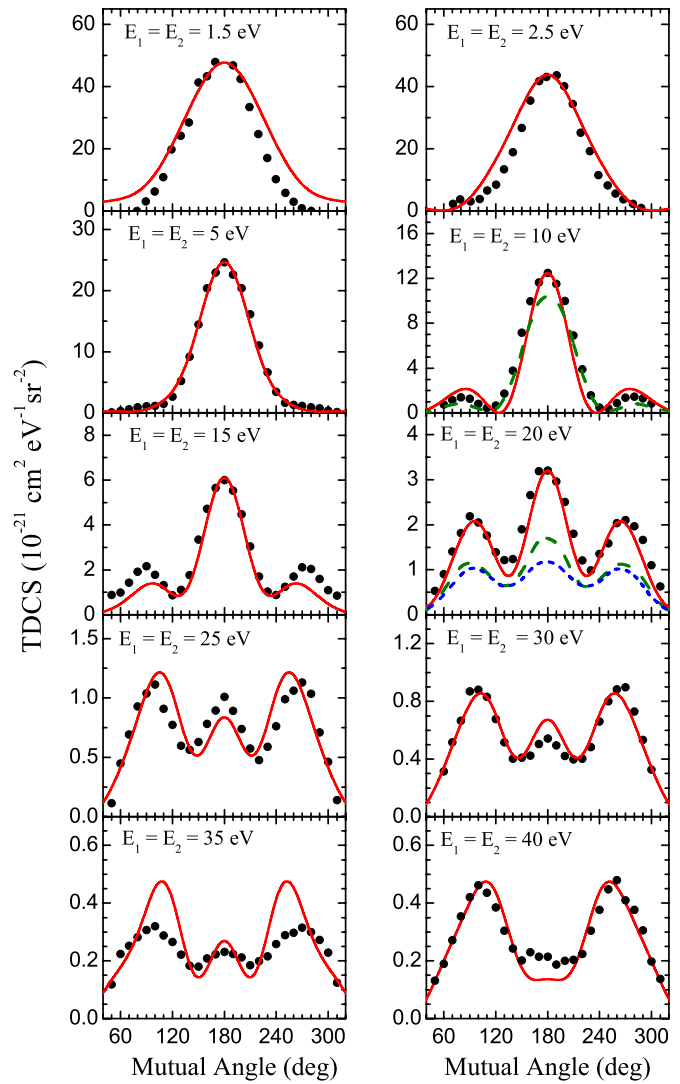


Figure 20. TDCS for the electron-impact ionization of helium in its $(1s^2)^1S$ ground with the residual ion left in the $\text{He}^+(1s)^2S$ state. The experimental data of Nixon *et al* [179] for the doubly-symmetric perpendicular geometry are compared with the present BSR predictions. Since the experimental data are relative and not cross-normalized, they were visually normalized to the BSR theory. Also available are CCC calculations for 10 eV [133] (long-dashed line), and both CCC and TDCC (short-dashed line) calculations for 20 eV [180].

however, that the TDCS values are generally smaller at 20 eV compared to 10 eV, and hence it is likely more difficult to calculate them accurately at 20 eV.

The apparently largest discrepancy between experiment and theory at 35 eV, which is also seen in the shape, can be attributed to the $2s^2$ and $2p^2$ doubly-excited autoionizing states of helium near that energy (see, for example, deHarak *et al* [181]). The theoretical cross sections then become very sensitive to the electron energies chosen in the projection step, as well as to the quality of the description of the autoionizing states.

As seen above, electron-impact ionization of $\text{He}(1s^2)^1S$ leaving the residual ion in the $\text{He}^+(1s)^2S$ ground state can also be handled well by CCC, TDCC, and in many cases also by perturbative methods subject to some restrictions regarding the

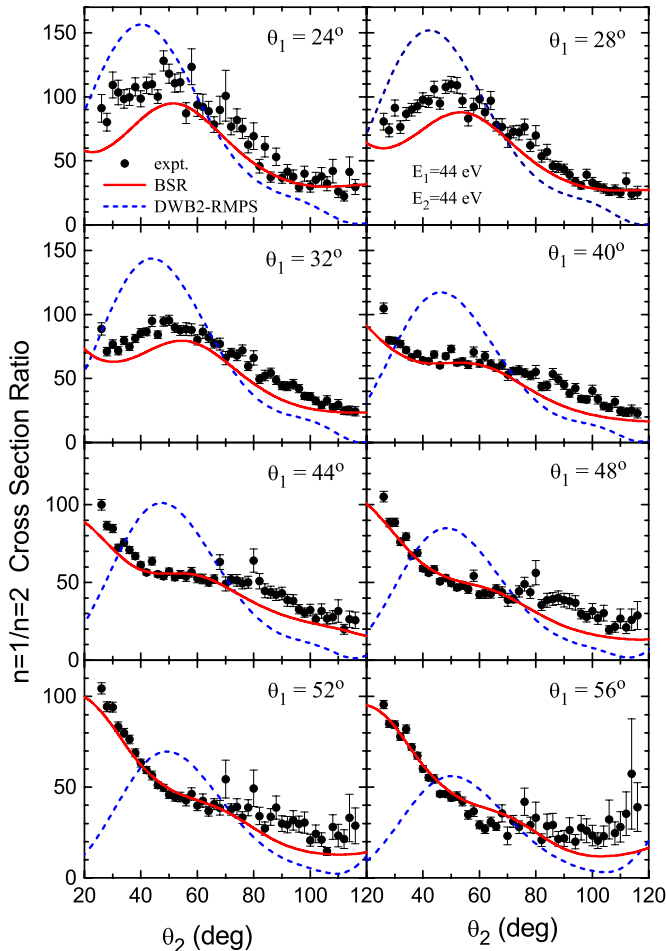


Figure 21. TDCS ratio for the electron-impact ionization of helium in its $(1s^2)^1S$ ground with the residual ion left in either the $\text{He}^+(1s)2S$ state or the excited $\text{He}^+(n=2)$ states. The primary energy E_0 is either 112.6 eV ($n=1$) or 153.4 eV ($n=2$) and both final-state electrons have energies of 44.0 eV. The experimental data of Bellm *et al* [128] are compared with BSR results and predictions from the hybrid DWB2-RMPS approach. (Reproduced with permission from [70]. © 2011 by The American Physical Society.)

kinematics. The most significant progress achieved by the BSR with pseudo-states method, therefore, was in the successful description of a much more complex process, namely the ionization of one $1s$ electron and simultaneous excitation of the other one [70]. This is a true, highly correlated four-body Coulomb process that posed major problems to theoretical attempts for many years. The hybrid approach mentioned previously had been somewhat successful [126–128], but was limited to highly asymmetric energy sharing and small scattering angles of the projectile.

Figure 21 compares the BSR and DWB2-RMPS predictions with the directly measured experimental cross-section ratios [128] for ionization without excitation (leaving the electron in He^+ in the $1s$ state) and ionization with excitation to $\text{He}^+(2s+2p)$. The agreement between the BSR results and the experimental data is excellent at all angles of the reference electron between 24° and 56° , and all detection angles of the other electron between 25° and 115° . As expected, the hybrid approach is inappropriate for large angles of the fixed electron.

The study of more complex targets (e.g., neon, argon, etc) impose additional problems in the description of the target structure. While a few attempts have been made to apply the CCC and TDCC methods, once again in the single-active-electron approach, to other targets (e.g., [182, 183]), it is not clear how accurate the results might be in the light of the necessary approximations made in the structure description. Until very recently, most theoretical work on such targets was still based on variants of the distorted-wave Born approximation (DWBA) pioneered by Madison *et al* [184] and the previously mentioned first-order (DWB1-RM) [122] and later second-order (DWB2-RM) [123, 124] hybrid distorted-wave + R -matrix approach. Despite some success, problems for the low-energy regime prevailed due to a less than ideal (if any) description of the PCI between the two outgoing electrons. Attempts to fix this problem include the use of asymptotically correct three-body wavefunctions in the 3DW approach [185] or the *ad hoc* introduction of some version of the Gamov factor [186]. Serious problems, however, are the facts that the 3DW wavefunction may not be sufficiently accurate at small distances between the projectile and the target (where the ionization occurs), while the Gamov factor is known to be very problematic regarding the overall normalization. Neither of these methods has been shown to yield a systematic and predictable improvement in the general case.

The first application of the BSR method to such systems was carried out for the ionization of $\text{Ar}(3p)$, where extensive experimental data were available for 200 eV [178] and 71 eV [187], respectively. While an unprecedented reproduction of the experimentally observed cross-section patterns as a function of the emission angle for the slow electron at fixed electron energies was observed, open questions remained concerning the dependence of the cross-section magnitude as a function of the projectile scattering angle. After a detailed analysis of the BSR predictions as well as the DWB2-RM hybrid results [77], the data evaluation procedure on the experimental side was revisited. It revealed that this issue, to a large extent, can be assigned to a missing solid-angle correction factor in processing the experimental raw data [188].

Very recently, the experiment was performed for the ionization of $\text{Ne}(2p)$ at an incident electron energy of 100 eV [189], with particular emphasis on the proper cross-normalization of the results. The experimental side of this work was carried out by employing an advanced reaction microscope [190, 191]). By means of homogeneous electric and magnetic fields the residual ion and the two final-state electrons e_1 and e_2 are projected onto time- and position-sensitive detectors. From the individually measured time-of-flight and position of each particle the three-dimensional momentum vector is calculated. Thus, full three-dimensional 3DCS are accessible. Since the complete experimentally accessible phase space is measured simultaneously, all relative 3DCS are cross normalized and only a single global factor is required in comparison of theory and experiment. This factor is used to put the 3DCS on an absolute scale.

Figure 22 exhibits a three-dimensional 3DCS pattern for a projectile scattering angle of -10° and 8 eV (central)

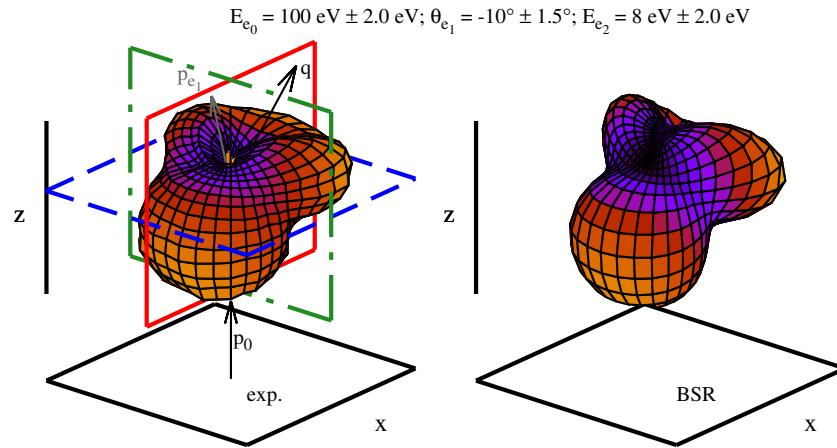


Figure 22. 3D triple-DCS for Ne(2p) ionization at an incident electron energy of 100 eV. The experimental pattern on the left is compared with the BSR predictions on the right. The scattering angle is $\theta_{e_1} = -10^\circ \pm 1.5^\circ$, and the ejection energy is $E_{e_2} = 8 \text{ eV} \pm 2 \text{ eV}$. (Reproduced with permission from [189]. © 2013 by The American Physical Society.)

energy of the ejected electron. The left panel represents the measured data while the BSR prediction is shown on the right. In this particular representation, the relative 3DCS is given by the radial distance from the origin to the surface of the plot. The projectile with momentum $\mathbf{p}_0$ enters the interaction region from below and after scattering has the momentum $\mathbf{p}_{e_1}$. The momentum transfer to the target system is indicated by the arrow labelled $\mathbf{q}$. Two general features are observable. Electrons emitted roughly in the direction of the momentum transfer form the binary peak, while those emitted in the opposite direction form the recoil peak.

Here, the binary peak is split by a dip (in theory) in the direction of $\mathbf{q}$. This is the result of the characteristic momentum profile of a p -orbital that has a node for vanishing momentum. Therefore, figure 22 exhibits a maximum cross section for electron emission on a cone around the $\mathbf{q}$ direction. Due to the low energy of the projectile, PCI is strong (as expected) and, consequently, the binary peak is suppressed near the forward direction. The qualitative comparison of experiment and theory shows very good overall agreement, with the exception that the binary peak is no longer split in the experimental data (see further discussion below).

Furthermore, three planes are indicated in figure 22, which will be used to compare experiment and theory quantitatively in figure 23. Those are the xz -plane or scattering plane (top), the yz -plane or perpendicular plane (centre) and the xy -plane or full-perpendicular plane (bottom). The global scaling factor used to normalize the experimental data to the BSR theory was found by achieving the best fit to the binary peak in the scattering plane. It was subsequently applied to all other (48 total) kinematics and planes [189], with overall impressive agreement between experiment and the BSR predictions, not only with regard to the angular dependence for each fixed set of final-state electron energies and detection angle of the projectile, but also the cross-normalization.

Figure 23 shows results for the three planes for one set of parameters, namely the incident electron energy of 100 eV, the projectile detection angle of -10° and an ejected electron energy of 8 eV. We see excellent agreement between experiment and the BSR predictions. For the coplanar case

(top panel), the DWB2-RM (which neglects PCI) shows a very pronounced split of the binary peak structure in the direction of $\mathbf{q}$. As seen already in figure 22, the peak towards smaller angles still exists in the BSR curve, but it is strongly suppressed by the PCI effects included in this model. The experimental data do not exhibit a small-angle peak at all, but it is not clear how much weight can be put on the ‘shape-determining’ point at 25° , i.e., at the very edge of the data range. The DWB2-RM results are relatively close to the BSR numbers, except for ejection angles where strong post-collision effects can be expected due to the two electrons leaving the target close to each other. This is never the case for the full-perpendicular geometry, and hence DWB2-RM does best in this case (bottom panel of figure 23).

Finally, we can close the circle by showing how these detailed results can be useful for applications. Figure 24 compares the total (elastic + excitation + ionization) cross section for electron collisions with Ne atoms in their ground state with a number of experimental data. We draw special attention to the higher energies, where ionization processes become increasingly important. The figure also shows the relative importance of the contributions from individual collision processes. Elastic scattering dominates over the entire energy regime shown in the figure. Excitation processes, on the other hand, only provide a relatively small contribution to the total cross section, while ionization becomes more and more important with increasing projectile energy and makes up nearly 30% of the total cross section at 200 eV. The overall excellent agreement with the available experimental data confirms the accuracy of the present approach. These results, including the important momentum-transfer cross sections, are now available through the publicly accessible LXCat database [196].

3.2.5. Electron-impact coherence parameters, spin polarizations and spin asymmetries. As our final examples for electron collisions, we show some recent BSR calculations to model experiments, in which the light emitted after electron-impact excitation is analysed or where the spin polarization of the electrons may play a role. We begin with a recent study

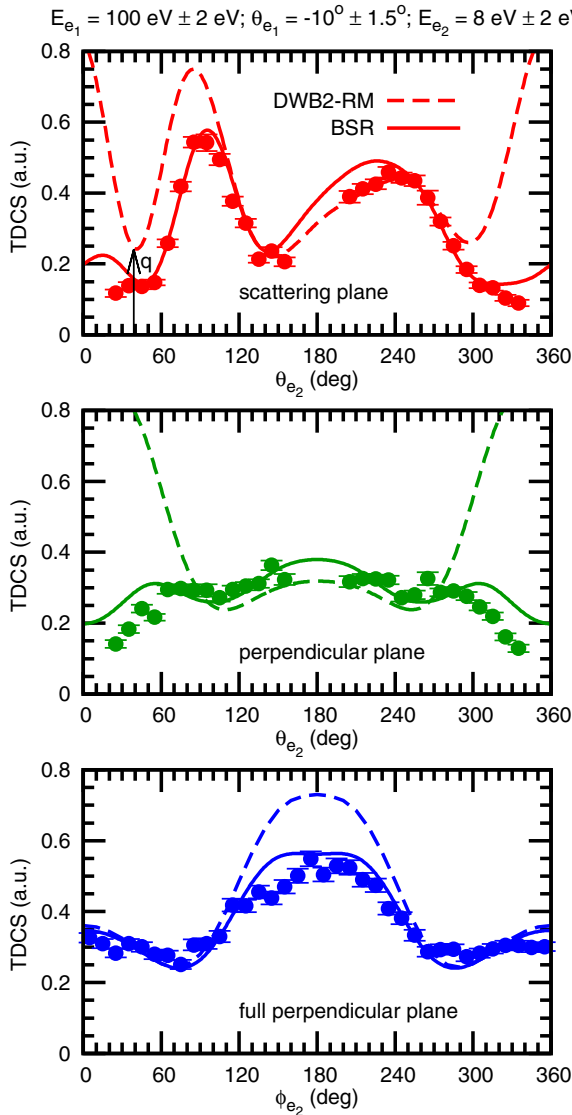


Figure 23. TDCS for the electron-impact ionization of Ne(2p) for an incident projectile energy of 100 eV, a scattering angle of $\theta_{e_1} = -10^\circ \pm 1.5^\circ$, and an ejection energy of $E_{e_2} = 8 \text{ eV} \pm 2 \text{ eV}$ for the co-planar (top), perpendicular (centre) and full-perpendicular (bottom) planes. The experimental data of Pflüger *et al* [189] are compared with predictions from the BSR model and DWB2-RM models.

[75], in which the light emitted after the excitation of Ne ($2p^5 3s$) 1P_1 state by electron impact was observed in coincidence with the scattered projectile. The photon detector was placed perpendicular to the scattering plane and the polarization of the light was analysed. Figure 25 exhibits the results for the circular polarization P_3 . This is a particularly interesting parameter that has drawn much attention over the years due to a propensity rule that predicts P_3 to start off with negative values at small scattering angles. This corresponds to a positive angular momentum transfer, which one would intuitively expect from a classical picture. A detailed discussion can be found, for example, in the book by Andersen and Bartschat [197].

The top part of figure 25 shows how the BSR predictions converge with the number of states in the close-coupling

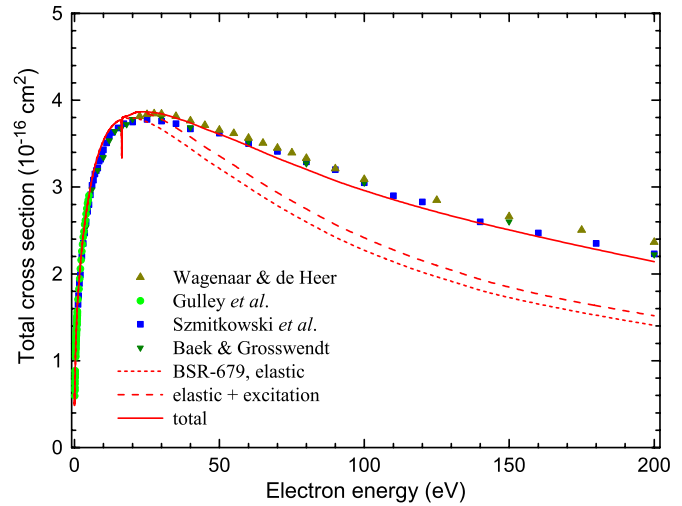


Figure 24. Total cross section for electron scattering from neon. The BSR-679 predictions [73] for elastic scattering alone, elastic scattering plus excitation, and Wagenaar and de Heer [192], Gulley *et al* [193], Szmitkowski *et al* [194] and Baek and Grosswendt [195].

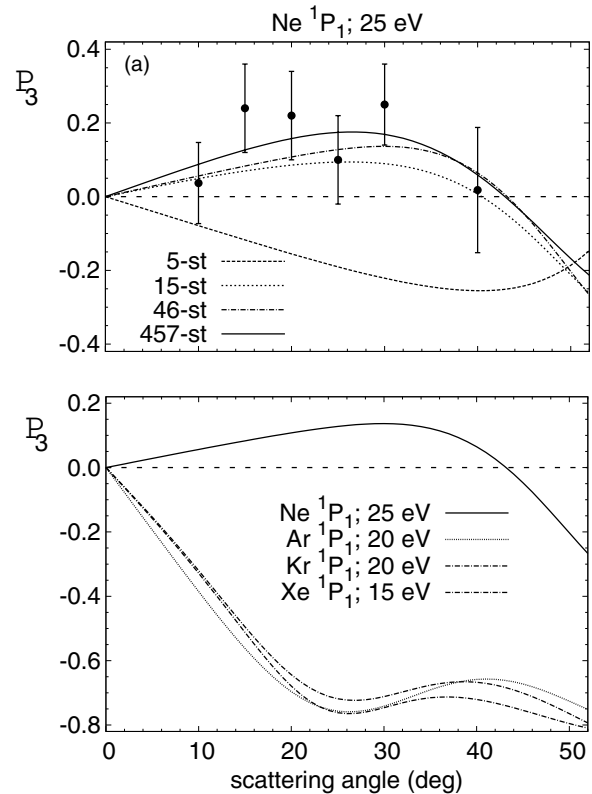


Figure 25. Top: circular light polarization P_3 for the Ne ($2p^5 3s$) 1P_1 state, as predicted by the BSR model with different numbers of coupled states. Bottom: circular light polarization P_3 for the corresponding lowest 1P_1 states in Ne, Ar, Kr and Xe, as predicted by BSR calculations with a sufficient number of states for convergence. (Reproduced with permission from [75]. © 2012 by The American Physical Society.)

expansion. A simple five-state model is completely inadequate for this case, while the apparently converged 457-state model, including a large number of pseudo-states, provides excellent agreement with the experimental data. Interestingly, the

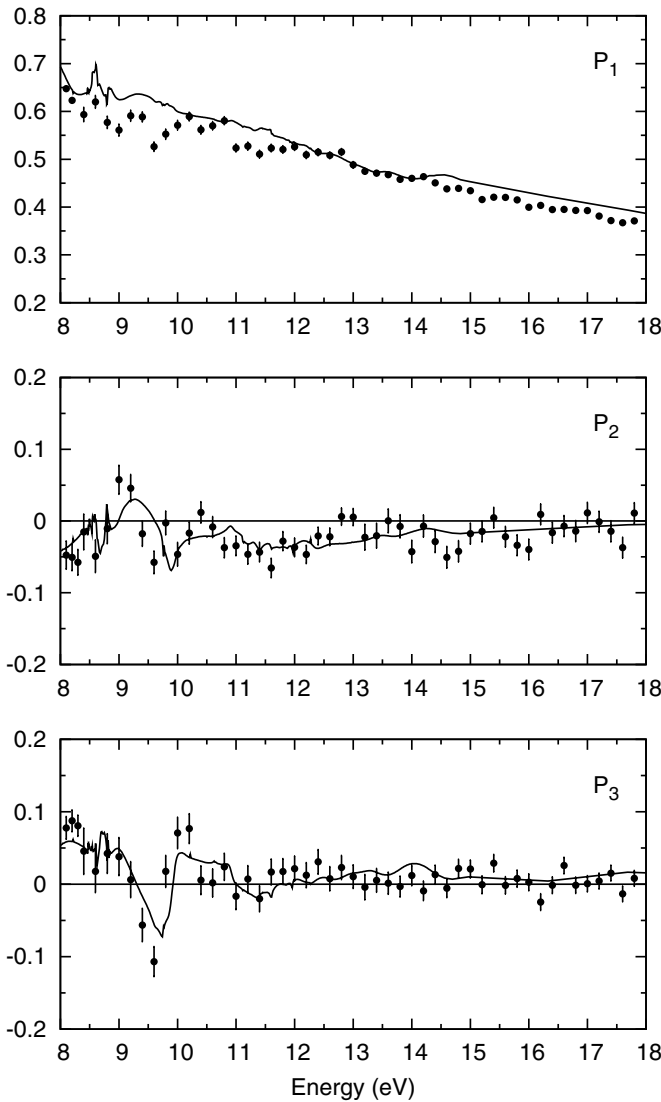


Figure 26. P_1 , P_2 and P_3 for the 185 nm line in mercury for incident energies between 8 and 18 eV. Solid circles: experimental results; solid line: DBSR-36 calculation. Since P_2 and P_3 are proportional to the electron polarization (P_1 does not depend on it), their values are normalized to 100% electron polarization. (Reproduced with permission from [66]. © 2009 by The American Physical Society.)

violation of the propensity rule seems to be limited to a small energy range around 25 eV and also to the Ne target. As seen in the bottom panel of figure 25, the other noble gases are behaving ‘normal’. It is worth mentioning that the unusual, but then confirmed, BSR predictions triggered the experimental investigation in this case, with further work currently being in progress.

Figure 26 shows results for the angle-integrated light polarizations P_1 , P_2 and P_3 for electron-impact excitation of the $\text{Hg}(6s6p)^1P_1$ state in the incident-energy range from 8 to 18 eV [66]. In this case the scattered electrons were not observed, but the incident electrons are transversely spin-polarized. The two linear polarizations $P_1(0^\circ, 90^\circ)$ and $P_2(45^\circ, 135^\circ)$ are defined with linear polarization filters set at the angles given in parentheses relative to the incident beam direction, while the photon detector is aligned parallel to the initial spin-

polarization vector. The energy range shown here is very challenging, both experimentally and theoretically, due to a wealth of resonances above the first ionization threshold of 10.4 eV in Hg, the vacuum ultraviolet nature of the emitted radiation and the relatively small values of P_2 and P_3 . Not surprisingly, not all details of the structure could be mapped one-to-one between experiment and the 36-state fully relativistic DBSR theory (DBSR-36), but the overall agreement is very satisfactory.

Another observable that is often studied when spin-polarized electrons are available is the spin-asymmetry function S_A , which determines a left–right asymmetry in the DCS when the incident electron beam is spin-polarized with a polarization vector perpendicular to the scattering plane [198]. Many experiments for a variety of targets were performed over the years in Münster in the groups of Kessler and Hanne, and a particularly challenging case is the lead atom. With a nuclear charge of $Z = 82$, it should be described in a fully relativistic framework, and the four electrons in the 6s and 6p sub-shells strongly interact with each other. While some theories were able to describe elastic scattering in a reasonable way, they were either inapplicable or simply inadequate for inelastic transitions.

Figure 27 shows the spin-asymmetry function S_A after the electron-impact excitation of the $(6p^2)^3P_0 \rightarrow (6p^2)^3P_2$ transition in Pb. We see a dramatic improvement in the agreement between the experimental data over Geesmann *et al* [199] achieved by two different fully-relativistic DBSR models [69] with 20 (DBSR-20) and 31 (DBSR-31) states, respectively, compared to a five-state semirelativistic BP calculation (BPRM-5) [200] performed at the time of the experiment over 20 years ago. It is truly remarkable that it has taken this long for theory to catch up with these benchmark experiments.

3.3. Photoionization and photodetachment

As mentioned above, the BSR method can also be used to calculate photo-induced processes. Except for a slight change in the boundary conditions (see the theory section), the problem effectively reduces to calculating the relevant dipole matrix elements between the initial bound state and the final continuum state involving the scattering of the ejected electron from the residual ion in photoionization or the neutral atom in photodetachment. Not surprisingly, the BSR approach with nonorthogonal orbitals can provide accurate descriptions of all the necessary ingredients.

3.3.1. Weak-field steady-state radiation. Figures 28 and 29 exhibit BSR results for the photoionization of potassium from its 4s ground state or the 4p excited state as a function of the photoelectron energy. For the ground state, there is excellent agreement between the results obtained in the length and velocity forms of the dipole operator, respectively, as well as with the experimental data of Marr and Creek [202]. The calculations [82] clearly favour the latter datasets over the former measurements performed by Hudson and Carter [203]. A number of other calculations were carried out for this problem as well. Details can be found in [82].

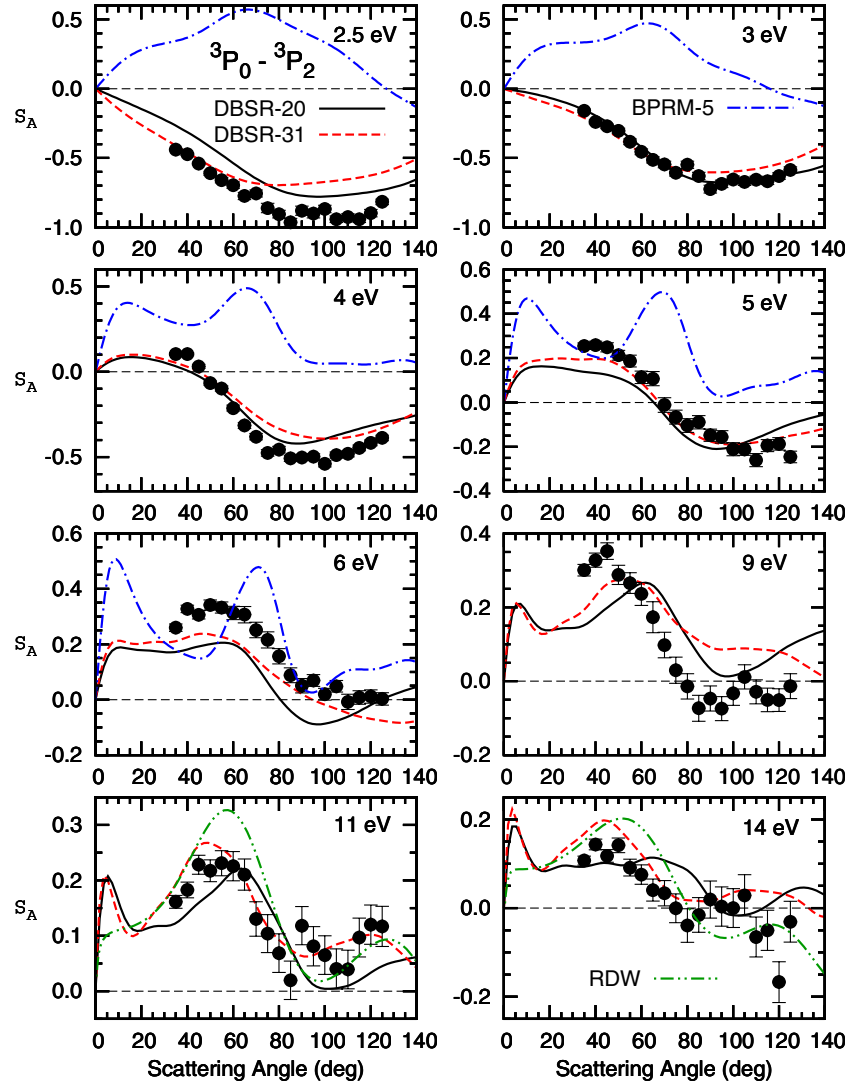


Figure 27. Spin-asymmetry function S_A for the electron-impact excitation of the $(6p^2)^3P_0 \rightarrow (6p^2)^3P_2$ transition in Pb for a variety of projectile energies. The DBSR-20 and DBSR-31 predictions [69] are compared with the experimental data of Geesmann *et al* [199]. Where available, we also show BPRM-5 [200] and RDW [201] results.

The photodetachment cross section of He^- from the metastable $(1s2s2p)^4P^o$ state is depicted in figure 30. This is one of the earlier applications of the BSR approach to a relatively simple problem, which can also be solved reliably using other approaches [206, 207]. Hence, it is not surprising that three different sets of theoretical predictions agree well with each other across the first maximum around a photon energy of 39 eV. A discussion about the potential reasons for the discrepancies between experiment and theory in this regime can be found in [78]. In the high-energy regime, on the other hand, the agreement between the experimental data and the BSR results is excellent.

3.3.2. Strong-field short-pulse radiation. As our first example from this area, figure 31 shows the response of the Ne atom to pulses with central photon energies of 11.6 and 7.3 eV, respectively. In these cases, at least two or three photons, need to be absorbed in order to ionize the system. This was our first attempt at such a problem, and hence we only used the

ground state $(1s^22s^22p^5)^2P$ of Ne^+ as the target state for the ‘half collision’ of the ejected electron with the residual ion. In order to ensure converged results for the above cases, we coupled LS symmetries up to a total orbital angular momentum $L_{\text{max}} = 6$ for the electron–ion collision system. Note that excitation rather than ionization appears as the dominating reaction process for $\omega = 0.27$ au and the laser parameters chosen here.

For the argon target, we performed a more sophisticated calculation [86], in which we coupled three states of the Ar^+ ion. In this project, we studied the effects of the finite pulse length and the intensity on various intermediate resonance states. Figure 32 depicts the excitation probability at different photon energies. Rabi oscillations occur when the photon energy matches the energy gap between the ground state and, in this case, the $(3p^54s)^1P$ excited state. This matching leads to oscillations in the excitation probability with a large amplitude and a long period. There are still oscillations when the photon energy is tuned away from the energy gap, but they have much smaller amplitudes and shorter periods.

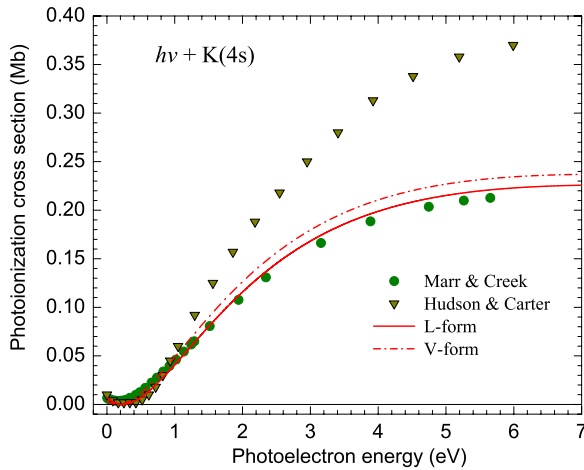


Figure 28. Photoionization cross sections for the 4s ground state of potassium as a function of photoelectron energy. The solid and dotted lines represent the length and velocity BSR results, respectively [82]. The experimental data are from Marr and Creek [202] (solid circles) and from Hudson and Carter [203] (solid triangles).

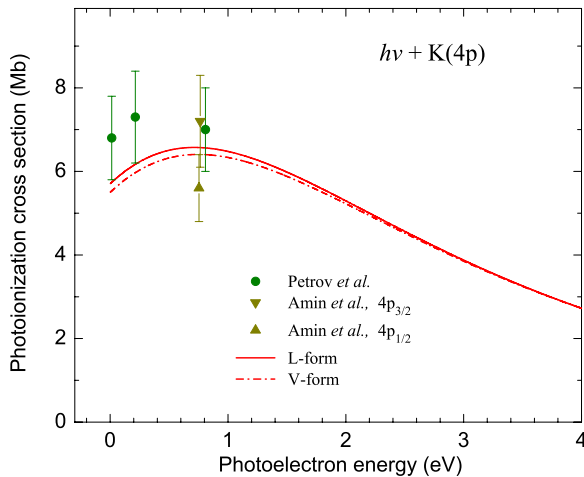


Figure 29. Photoionization cross sections for the $4p_{3/2}$ and $4p_{1/2}$ excited levels of potassium as a function of the photoelectron energy. The solid and dashed lines represent the BSR results (length form) for the $4p_{3/2}$ and $4p_{1/2}$ levels, respectively [82]. The experimental data are from Petrov *et al* [204] (solid circles) without resolved finestructure and from Amin *et al* [205] for the $4p_{3/2}$ (inverted solid triangles) and $4p_{1/2}$ (upward solid triangles) initial states.

As a result of coupling several ionic states, Rydberg-type resonances converging to different thresholds can be observed. However, both the finite length of the pulse and its intensity have an effect on the details of the observed structures. Figure 33 shows the effect of the laser intensity on the generalized two-photon cross section. While both peak intensities, 10^{12} and 10^{13} W cm $^{-2}$, still lie in the perturbative regime, the height of the first resonance peak is significantly diminished for the more intense laser field. Similar results have been obtained by another time-dependent *R*-matrix approach, which is independently being developed by the Belfast group [139].

Figure 34 exhibits two examples for the single ionization rates in argon [86] obtained in few-cycle laser pulses of

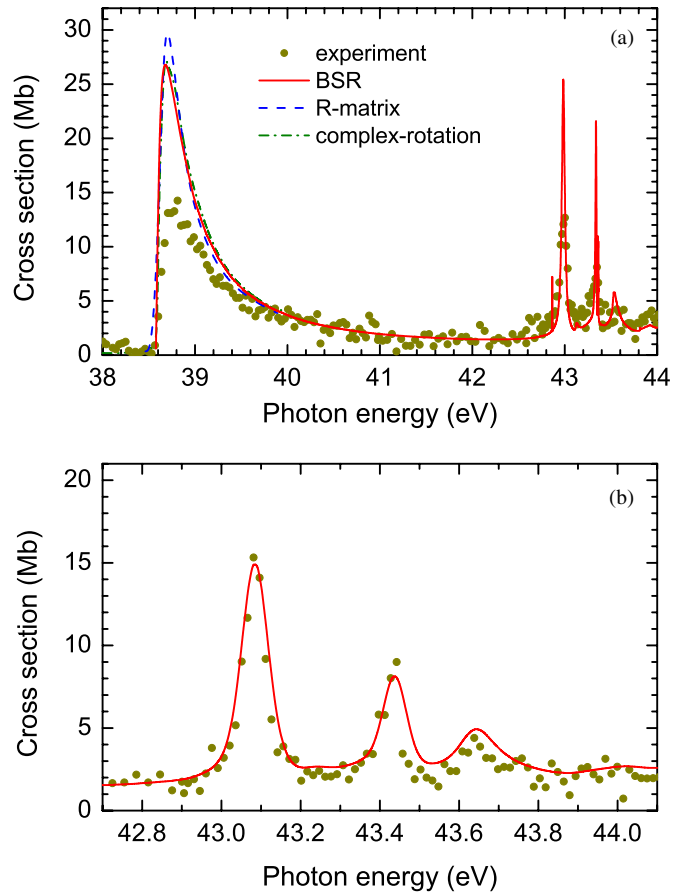


Figure 30. Total photodetachment cross section for He^- in the metastable $(1s2s2p)^4P^o$ state as a function of the photon energy [78]. The BSR results (solid line) are compared with the experimental data of Berrah *et al* [208] and near the first maximum with predictions from a standard *R*-matrix calculation [206] and a complex-rotation CI calculation [207]. (b) Comparison between the BSR predictions (convoluted with a 70 meV FWHM Gaussian) and high-resolution experimental data in the higher energy regime.

different lengths, intensities and central wavelengths. There is clearly a nontrivial dependence on the various laser parameters, and once again resonances appear when the photon wavelength is varied. These predictions are currently awaiting experimental tests, but they show the richness of effects that can be expected in complex targets where inter-shell correlation effects play an important role.

As our final example, we consider the two-photon DI of the helium atom by the absorption of two photons at different central photon energies. In other words, the target helium atom is exposed to the irradiation by two laser pulses, of potentially different frequencies and with a controllable time delay. Specifically, the following two-colour laser parameters were used for the results shown in figure 35: pulse 1 has a central photon energy of 35.3 eV and a peak intensity of 10^{14} W cm $^{-2}$, while the corresponding parameters for pulse 2 are 57.1 eV and 10^{13} W cm $^{-2}$, respectively. The length of each pulse is 10 optical cycles with a Gaussian envelope, thus corresponding to pulse lengths of about 1.2 and 0.7 fs, respectively. We are interested in the mechanism for the ejection of two electrons when the time delay between the two pulses is varied. We define this delay as the time distance

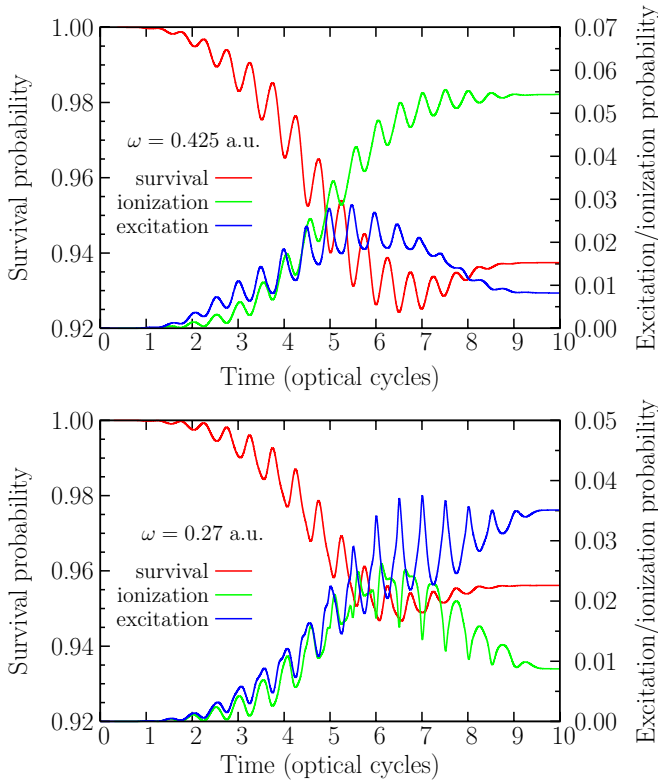


Figure 31. Ground-state survival (left scale) and total excitation and ionization probabilities (right scale) for Ne exposed to 10-cycle laser pulses of peak intensity $3.5 \times 10^{14} \text{ W cm}^{-2}$ with a Gaussian envelope. The central laser frequencies are 0.425 au (11.6 eV, top panel) and 0.27 au (7.3 eV, bottom panel).

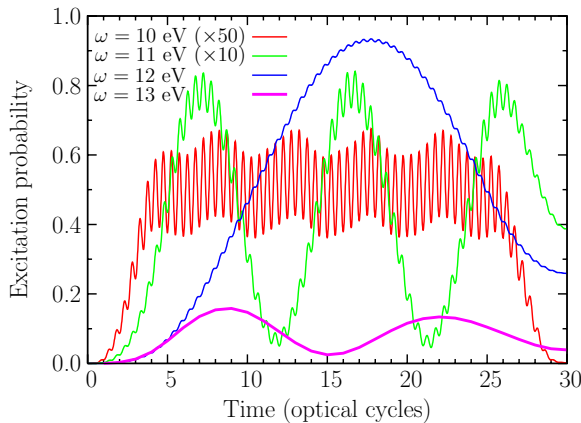


Figure 32. Excitation probability of argon for a 30-cycle laser pulse with the peak intensity of $2 \times 10^{13} \text{ W cm}^{-2}$ and photon energies of 10, 11, 12 and 13 eV. Note the different scales for the individual photon energies.

between the peak intensities. Consequently, there is no overlap at all between the two pulses for a delay of about 0.95 fs, corresponding to a little less than 40 au of time (1 au $\approx$ 24 attoseconds).

Figure 35 shows the results for a variety of delays, ranging from about -120 (i.e., the second photon with the higher energy comes first) to 600 attoseconds. Not surprisingly, the probability for DI is small in the first case, since the only chance for this process to happen is the two photons working

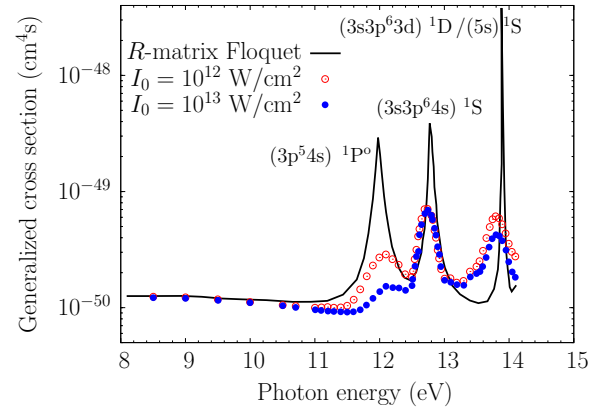


Figure 33. Effect of the laser peak intensity on the generalized two-photon cross section for a 30-cycle laser pulse interacting with argon. The Floquet-results are from McKenna and van der Hart [209].

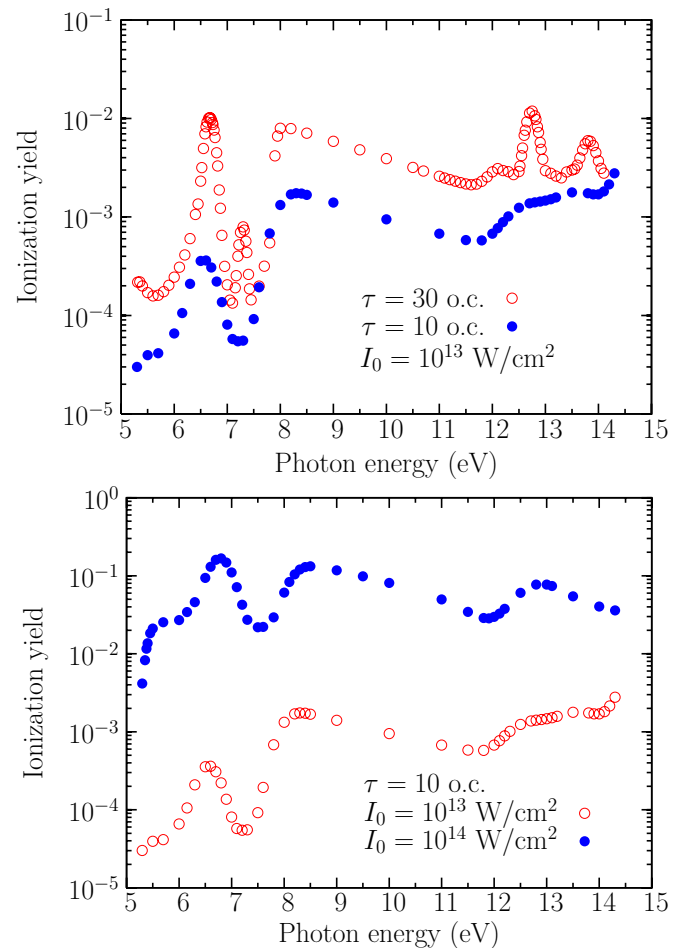


Figure 34. Ionization yield as a function of photon energy in 10-cycle and 30-cycle laser pulses of peak intensity $10^{13} \text{ W cm}^{-2}$ (top) and for 10-cycle laser pulses with peak intensities of 10^{13} and $10^{14} \text{ W cm}^{-2}$ (bottom).

together on the two electrons. Even when the two pulses come simultaneously, the probability for double ejection remains relatively small, and the peak for equal-energy sharing is a clear indication of the direct process. With increasing time delay, the peaks expected for the sequential process—one at

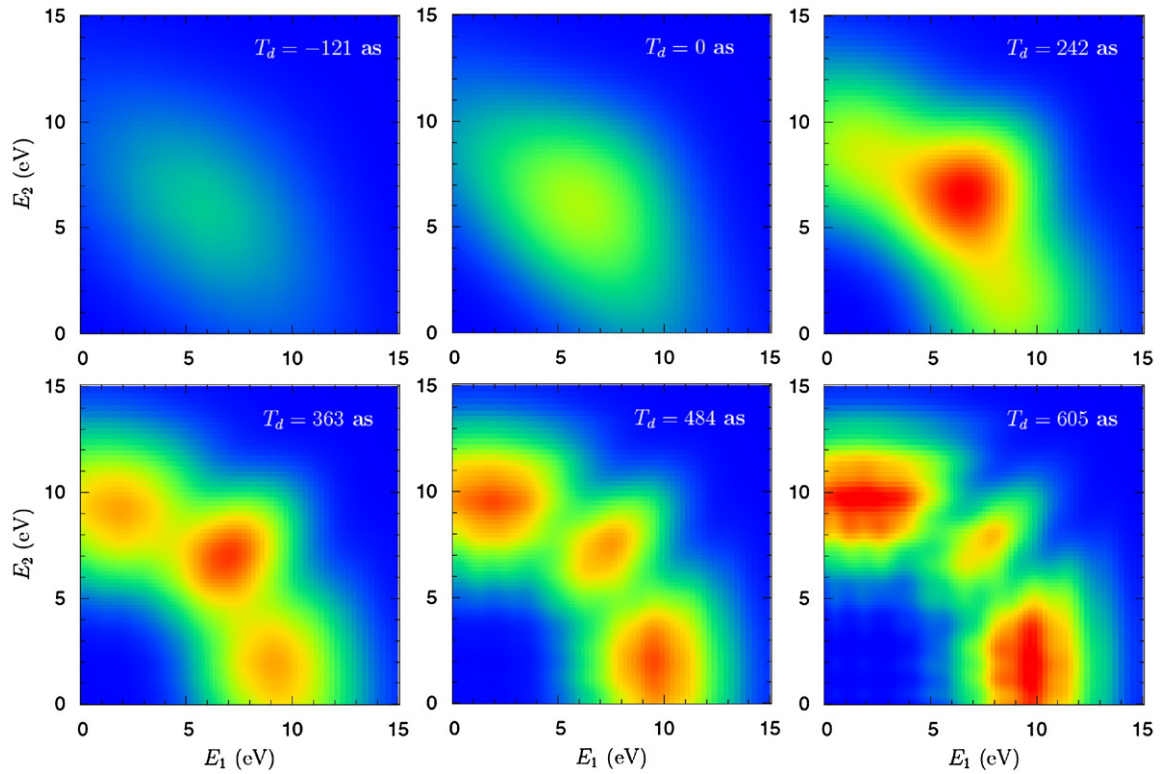


Figure 35. Energy distributions of the two escaping electrons in two-colour laser pulses of ten optical cycles each. The laser parameters are: $\omega_1 = 35.3$ eV at a peak intensity of 10^{14} W cm $^{-2}$ and $\omega_2 = 57.1$ eV at a peak intensity of 10^{13} W cm $^{-2}$. The delay between the two pulses is varied between -121 as and $+605$ as. See text for details.

10.7 eV and the other at 2.7 eV, corresponding to the 35.1 eV photon ionizing the neutral helium atom with an ionization potential of 24.6 eV and the other one ionizing the He $^+(1s)$ ion with an ionization potential of 54.4 eV—start to grow, but the two processes still have about equal weight for a time delay as large as 400 attoseconds.

From this example, it is clear that the time delay plays a decisive role in determining how the two electrons are ejected by two-colour XUV laser pulses. Depending on the details of the time delay, the electrons can be ejected in ways either similar to the sequential or the nonsequential process. Our findings qualitatively agree with those of Fomouuo *et al* [210] for the two-colour problem and Feist *et al* [211] in the single-colour problem. They serve as an independent confirmation of their predictions, and also give us confidence in our computer code.

4. Conclusions and outlook

We have presented the general ideas behind the B -spline R -matrix (BSR) approach with nonorthogonal orbitals and a variety of examples for applications of the method to problems in atomic structure, electron collisions and photon-driven processes. Allowing for nonorthogonal one-electron orbitals and hence eliminating orthogonality restrictions that are introduced in many other theoretical formulations merely for computational convenience, rather than physical necessity, often increases the accuracy that can be obtained in the target description while a subsequent collision calculation is still

possible. From a numerical point of view, key ingredients of the method are the B -splines that represent the orbitals of the active target electrons as well as the projectile, if necessary.

We presently have a suite of semirelativistic computer codes based on [1] that has been fully parallelized to run efficiently on large clusters such as Lonestar [212], Stampede [213] and Gordon [214]. As summarized in tables 1–4, results from this program, and its fully relativistic (DBSR) companion, have been used both to study fundamental physics and to generate a variety of atomic data for plasma applications. We are currently working on parallelizing the DBSR suite of codes. We also plan to further extend the capabilities of handling time-dependent processes, and we hope to move towards simple diatomic molecules in the foreseeable future.

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